US012405018B2

(12) United States Patent
Sinur et al.

(10) Patent No.: US 12,405,018 B2
(45) Date of Patent: Sep. 2, 2025

(54) **SYSTEM AND METHOD FOR CONTROLLING INDOOR AIR QUALITY**

(71) Applicant: **Broan-NuTone LLC**, Hartford, WI (US)

(72) Inventors: **Richard R. Sinur**, Hartford, WI (US); **Kyle Anderson**, Hartford, WI (US); **Jeremy Yingst**, Hartford, WI (US); **Eric Theriault**, Hartford, WI (US); **Loic Ares**, Hartford, WI (US); **Jason Asmus**, Hartford, WI (US); **Seddik Rougab**, Hartford, WI (US)

(73) Assignee: **Broan-NuTone LLC**, Hartford, WI (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 821 days.

(21) Appl. No.: **17/417,471**

(22) PCT Filed: **Jan. 7, 2020**

(86) PCT No.: **PCT/US2020/012487**

§ 371 (c)(1),
(2) Date: **Jun. 23, 2021**

(87) PCT Pub. No.: **WO2020/146315**

PCT Pub. Date: **Jul. 16, 2020**

(65) **Prior Publication Data**

US 2022/0154954 A1 May 19, 2022

Related U.S. Application Data

(60) Provisional application No. 62/789,501, filed on Jan. 7, 2019.

(51) **Int. Cl.**
F24F 11/00 (2018.01)
F24F 11/30 (2018.01)
(Continued)

(52) **U.S. Cl.**
CPC ***F24F 11/0001*** (2013.01); ***F24F 11/30*** (2018.01); ***F24F 11/65*** (2018.01);
(Continued)

(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,726,111	B2	4/2004	Weimer et al.
7,044,397	B2	5/2006	Bartlett et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CA	2297682	8/2001
CA	2600529	9/2007

(Continued)

OTHER PUBLICATIONS

English Translation of portions of Office Action for Chinese Patent Application No. 201980077675.7, mailed Jun. 10, 2023 (6 pages).

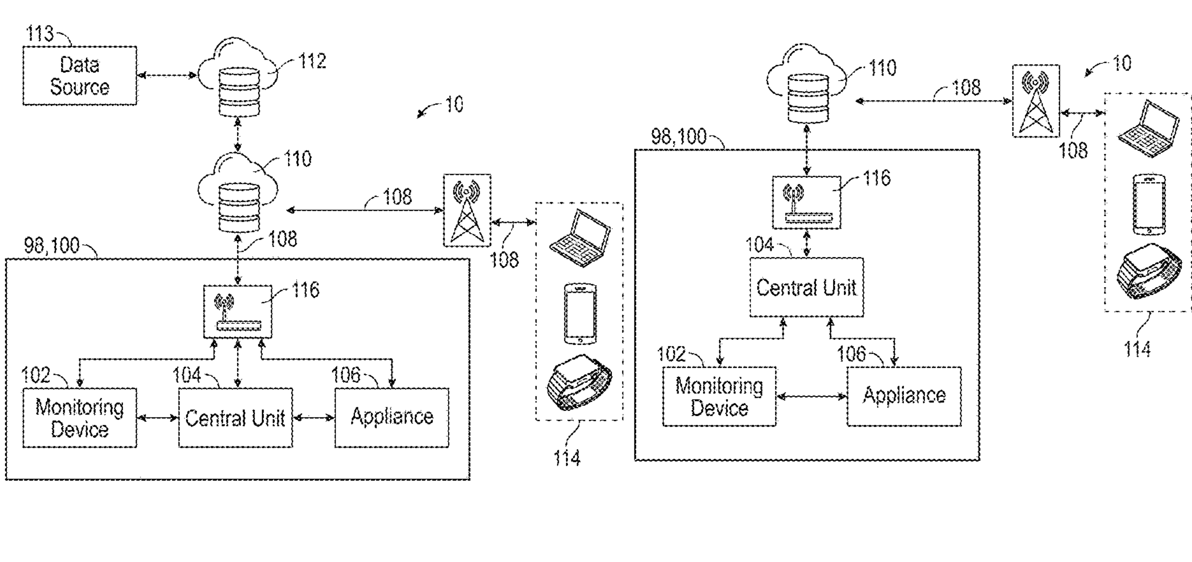
(Continued)

Primary Examiner — Tuan S Nguyen
(74) *Attorney, Agent, or Firm* — Barnes & Thornburg LLP

(57) **ABSTRACT**

A system and method for obtaining environmental data—namely air quality information—from various devices contained within a structure is disclosed herein. The various devices contain sensors that can obtain environmental data, which is then analyzed by the system to determine if any level of a component within the data is outside of a predefined threshold range. If the system determines that the level of the component is outside of the predefined threshold range for that given component, the system will carry out certain steps in order to bring the level within the predetermined threshold range. These steps include selecting the

(Continued)



appropriate appliance and the proper operating conditions to most efficiently bring the level back within the predetermined threshold range. Once the system has determined that the level is back within the predetermined threshold range, the system will instruct the selected appliance to turn OFF.

30 Claims, 61 Drawing Sheets

(51) **Int. Cl.**

F24F 11/65	(2018.01)
F24C 15/20	(2006.01)
F24F 7/06	(2006.01)
F24F 11/56	(2018.01)
F24F 11/58	(2018.01)
F24F 110/65	(2018.01)
F24F 110/70	(2018.01)
F24F 110/72	(2018.01)

(52) **U.S. Cl.**
CPC *F24C 15/2021* (2013.01); *F24F 7/06* (2013.01); *F24F 11/56* (2018.01); *F24F 11/58* (2018.01); *F24F 2110/65* (2018.01); *F24F 2110/70* (2018.01); *F24F 2110/72* (2018.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,475,828	B2	1/2009	Bartlett et al.	
7,632,178	B2	12/2009	Meneely, Jr.	
7,979,163	B2	7/2011	Terlson et al.	
8,100,746	B2	1/2012	Heidel et al.	
8,147,302	B2	4/2012	Desrochers et al.	
8,755,942	B2	6/2014	Bonilla et al.	
9,109,981	B2	8/2015	Sharp	
9,347,678	B2	5/2016	Stakutis et al.	
9,494,324	B2	11/2016	Livchak et al.	
9,618,222	B1	4/2017	Hussain et al.	
9,618,224	B2	4/2017	Emmons et al.	
9,618,948	B2	4/2017	Yamaguchi et al.	
9,677,772	B2	6/2017	Siegel et al.	
9,924,243	B2	3/2018	Lupien et al.	
9,946,670	B2	4/2018	Klein et al.	
10,082,299	B2	9/2018	Livchak et al.	
10,085,585	B2	10/2018	Siegel et al.	
10,126,009	B2	11/2018	Flaherty et al.	
10,234,158	B2	3/2019	Laughman et al.	
10,242,129	B2	3/2019	Flaherty et al.	
10,281,896	B1	5/2019	Goodman et al.	
10,305,323	B2	5/2019	Wester et al.	
10,309,668	B2	6/2019	Song et al.	
10,347,112	B2	7/2019	Myllymäki	
10,458,668	B2	10/2019	Emmons et al.	
10,462,184	B1 *	10/2019	Gu	H04W 4/33
10,490,055	B2	11/2019	Myllymaki	
10,514,177	B2	12/2019	Hussain et al.	
10,527,307	B2	1/2020	Saffre et al.	
10,529,215	B2	1/2020	Brown	
10,598,398	B2	3/2020	Wiley et al.	
10,655,884	B2	5/2020	Iura et al.	
10,712,038	B2	7/2020	Ribbich et al.	
10,837,665	B2	11/2020	Ajax et al.	
10,859,283	B1	12/2020	Goodman et al.	
10,867,254	B2	12/2020	Benazzouz et al.	
10,907,844	B2	2/2021	Ribbich et al.	
10,915,669	B2	2/2021	Flaherty et al.	
10,928,084	B2	2/2021	Ajax et al.	
11,009,248	B2	5/2021	Salem et al.	
2005/0260946	A1	11/2005	Kim	
2006/0278216	A1	12/2006	Gagas et al.	
2009/0057430	A1	3/2009	Cherewatti	
2013/0226849	A1	8/2013	Benazzouz et al.	
2015/0032264	A1	1/2015	Emmons	
2015/0096335	A1	4/2015	Haidak et al.	
2015/0096352	A1 *	4/2015	Peterson	H04M 1/72445 73/31.02
2015/0156031	A1 *	6/2015	Fadell	G08B 27/003 700/90
2015/0204558	A1	7/2015	Sartain	
2015/0369507	A1	12/2015	Flaherty et al.	
2016/0066068	A1	3/2016	Schultz et al.	
2016/0131376	A1	5/2016	Yoon	
2016/0231014	A1	8/2016	Ro et al.	
2016/0318368	A1	11/2016	Alger et al.	
2016/0361452	A1	12/2016	Blackley	
2017/0084156	A1	3/2017	Myllymäki	
2017/0168467	A1	6/2017	Liu et al.	
2017/0176030	A1	6/2017	Emmons et al.	
2017/0328591	A1	11/2017	Kelly et al.	
2017/0350611	A1	12/2017	Su et al.	
2018/0156483	A1	6/2018	Kim et al.	
2018/0156484	A1	6/2018	Kim et al.	
2018/0158315	A1	6/2018	Sloo et al.	
2018/0299150	A1	10/2018	Ajax et al.	
2018/0299151	A1	10/2018	Ajax et al.	
2018/0299159	A1	10/2018	Ajax et al.	
2019/0154285	A1	5/2019	Pham et al.	
2019/0221089	A1	7/2019	Wedig et al.	
2019/0288553	A1	9/2019	Wester et al.	
2019/0360717	A1	11/2019	Chae	
2019/0363576	A1	11/2019	Wester et al.	
2020/0011559	A1	1/2020	Wan	
2020/0072486	A1	3/2020	Matsubara et al.	
2020/0088438	A1	3/2020	Shiono et al.	
2020/0152040	A1	5/2020	Brown	
2020/0224915	A1	7/2020	Nourbakhsh et al.	
2020/0232670	A1	7/2020	Ritmanich et al.	
2020/0333753	A1	10/2020	Chenier et al.	
2020/0381939	A1	12/2020	Wester et al.	
2020/0393152	A1	12/2020	Ramirez et al.	
2020/0408434	A1	12/2020	Arentsen	
2021/0003308	A1	1/2021	Venne	
2021/0041118	A1	2/2021	Morgan et al.	
2021/0071891	A1	3/2021	Ferrere, Jr. et al.	
2021/0088241	A1	3/2021	Goodman et al.	
2021/0088248	A1	3/2021	Matsugi et al.	
2021/0103260	A1	4/2021	Khurana et al.	
2021/0123623	A1	4/2021	Notaro et al.	
2021/0131692	A1	5/2021	Martin	
2021/0156582	A1	5/2021	Stevenson et al.	
2021/0165926	A1	6/2021	Flaherty et al.	
2021/0173366	A1	6/2021	Turney et al.	
2021/0180815	A1	6/2021	Shamoon et al.	
2021/0199329	A9	7/2021	Mowris et al.	
2021/0207826	A1	7/2021	Sotokawa et al.	
2021/0208546	A1	7/2021	Locke et al.	
2021/0231325	A1	7/2021	Stevenson et al.	
2021/0239335	A1	8/2021	Morgan et al.	

FOREIGN PATENT DOCUMENTS

CA	2745432		6/2010
CA	2863117		11/2014
CA	2933630	A1	3/2017
CN	101208563	A	6/2008
CN	101208563	B	6/2008
CN	102301187	A	12/2011
CN	103557579	A	2/2014
CN	203571824	U	4/2014
CN	2014240733	A	12/2014
CN	103557579	B	2/2015
CN	105823172	A	8/2016
JP	H11287489	A	10/1999
JP	2004045016	A	2/2004
JP	2014240733		12/2014
JP	2018151114	A	9/2018
WO	WO2020055872	A1	3/2020

(56) **References Cited**

FOREIGN PATENT DOCUMENTS

WO	WO2020112981	A1	6/2020
WO	WO2020146315	A1	7/2020

OTHER PUBLICATIONS

Office Action issued to CN202080008234.4, mailed Jul. 26, 2022 (17 pages).
International Search Report and Written Opinion issued to PCT/US20/12487, mailed Apr. 2, 2020 (14 pages).
English Translation of portions of Office Action for Chinese Patent Application No. 202080008234.4, mailed Jul. 19, 2023 (3 pages).
Office Action for Chinese Patent Application No. 202080008234.4, mailed Jul. 19, 2023 (6 pages).
Search Report and Written Opinion for PCT Application No. PCT/US23/11580, mailed Apr. 14, 2023 (22 pages).
Office Action for Chinese Patent Application No. 202080008234.4, mailed Apr. 1, 2023 (8 pages).
Office Action issued to Chinese App. No. 202080008234.4 dated Jul. 26, 2022 (16 pages).
Canadian Application No. 3,125,578, Office Action mailed Apr. 4, 2024 (5 pages).
U.S. Appl. No. 18/817,338, Office Action mailed Jun. 4, 2025 (50 pages).
Investigation of a Novel Dew Point Indirect Evaporative Air Conditioning System for Buildings, Zhiyin Duan, University of Nottingham (188 pages).
A Critical Review on Air Quality Index, S. Kumari et al., Environmental Pollution: Select Proceedings of ICWEES-2016 (16 pages).

* cited by examiner

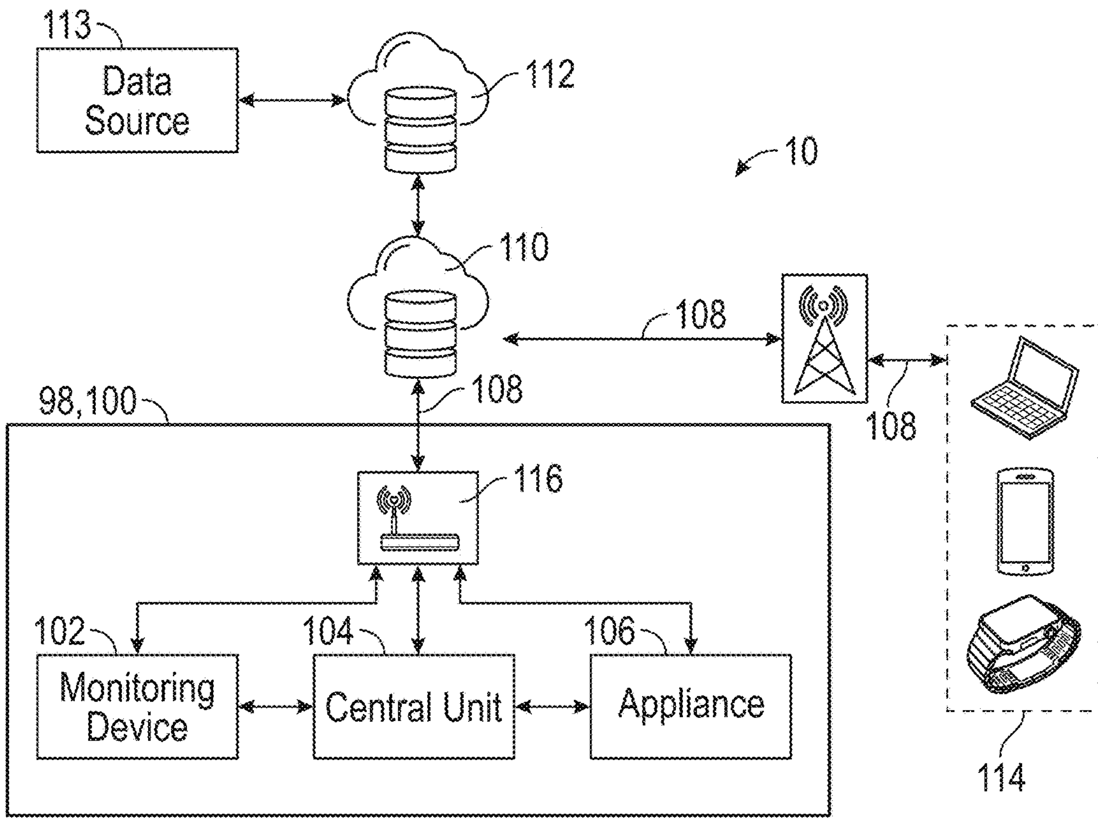

113
Data Source
112
10
110
108
98,100
108
116
108
102
104
106
Monitoring Device
Central Unit
Appliance
114

FIG. 1A

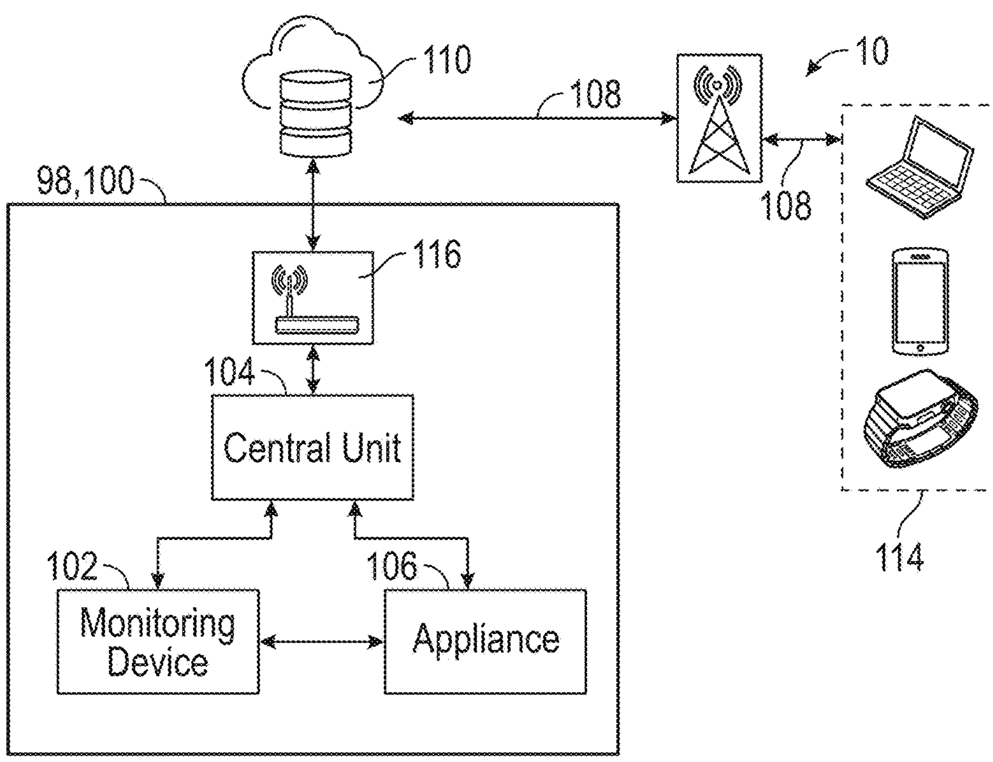

110
108
10
98,100
108
116
104
Central Unit
102
106
Monitoring Device
Appliance
114

FIG. 1B

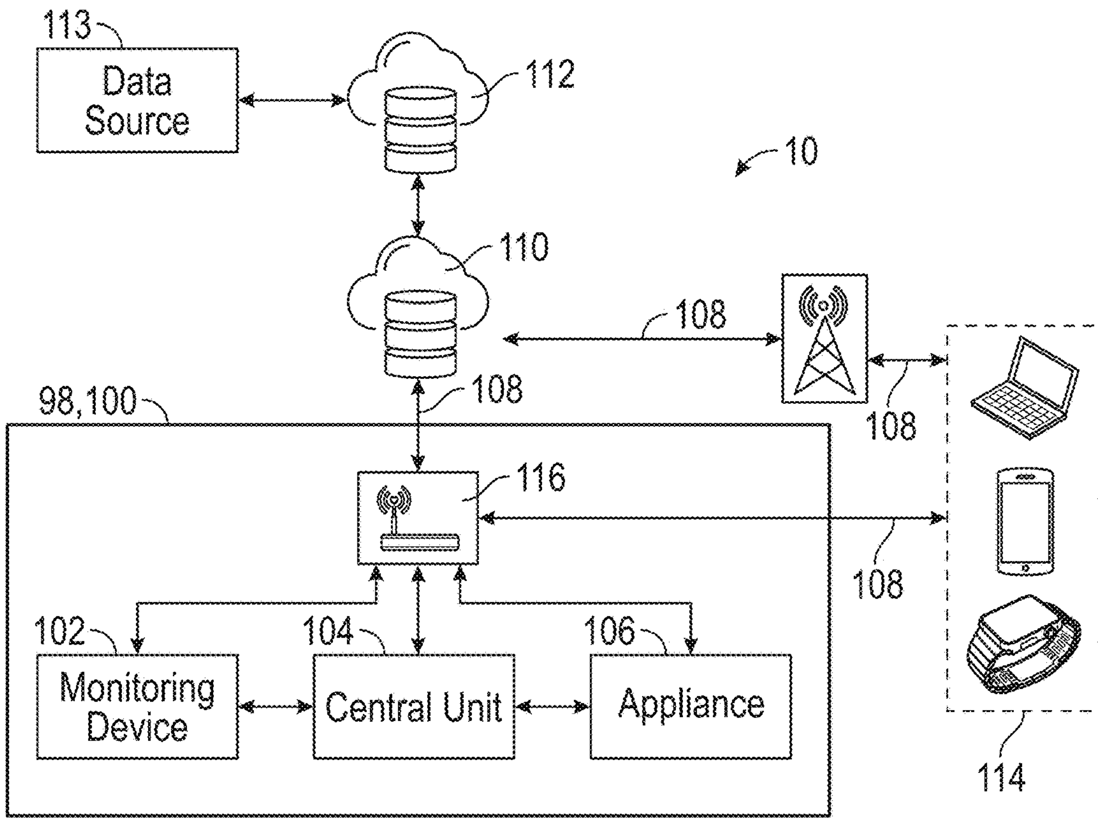

113
Data Source
112
10
110
108
108
98,100
116
108
108
102
Monitoring Device
104
Central Unit
106
Appliance
114

FIG. 1C

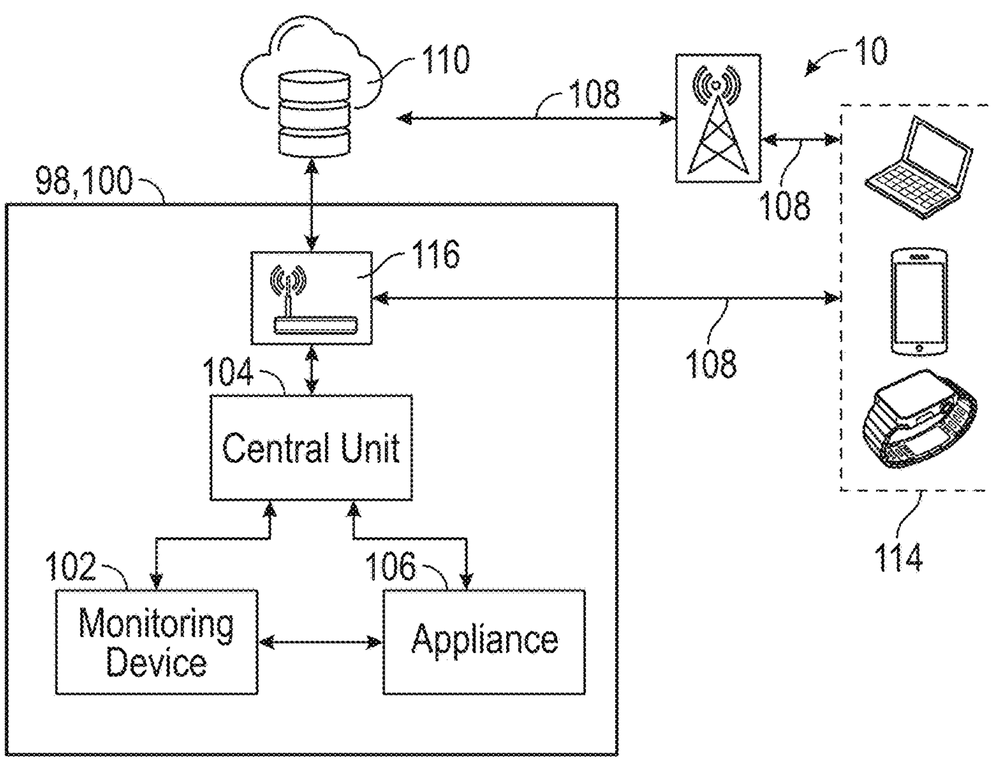

110
108
10
98,100
108
116
108
104
Central Unit
102
Monitoring Device
106
Appliance
114

FIG. 1D

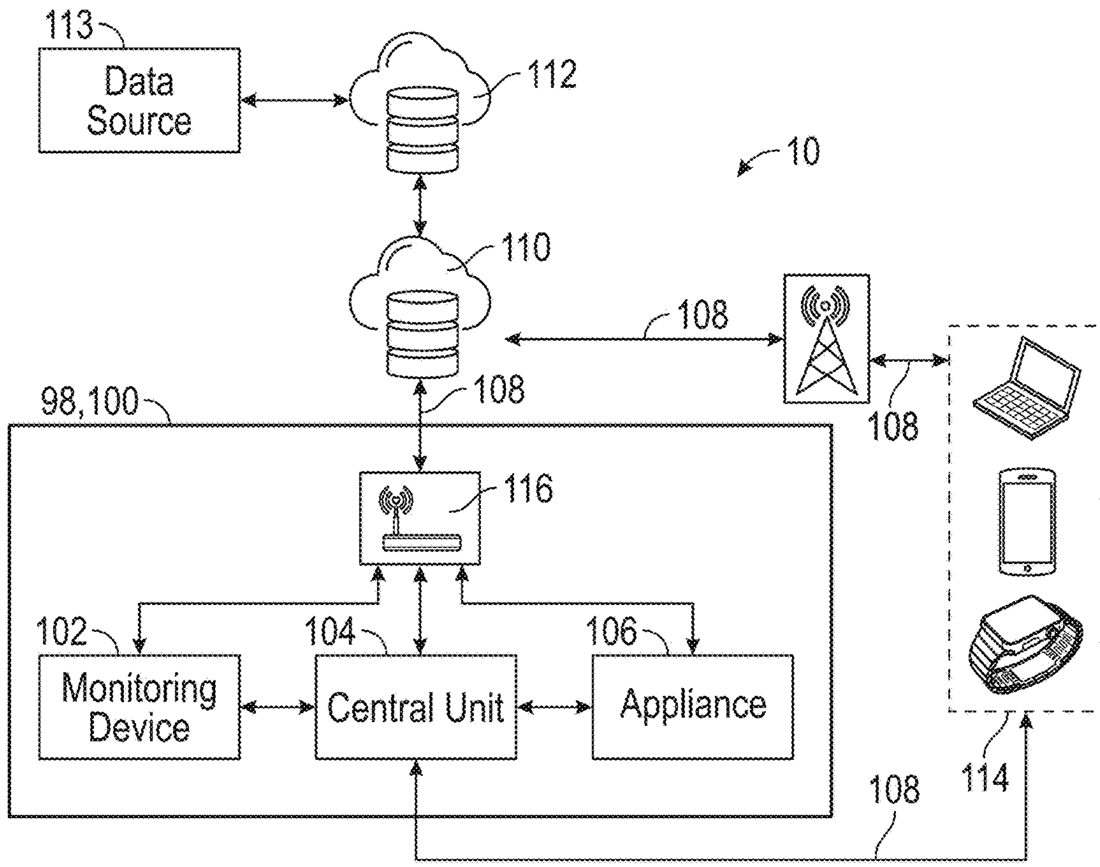

113
Data Source
112
10
110
108
108
98,100
116
108
102
Monitoring Device
104
Central Unit
106
Appliance
108
114

FIG. 1E

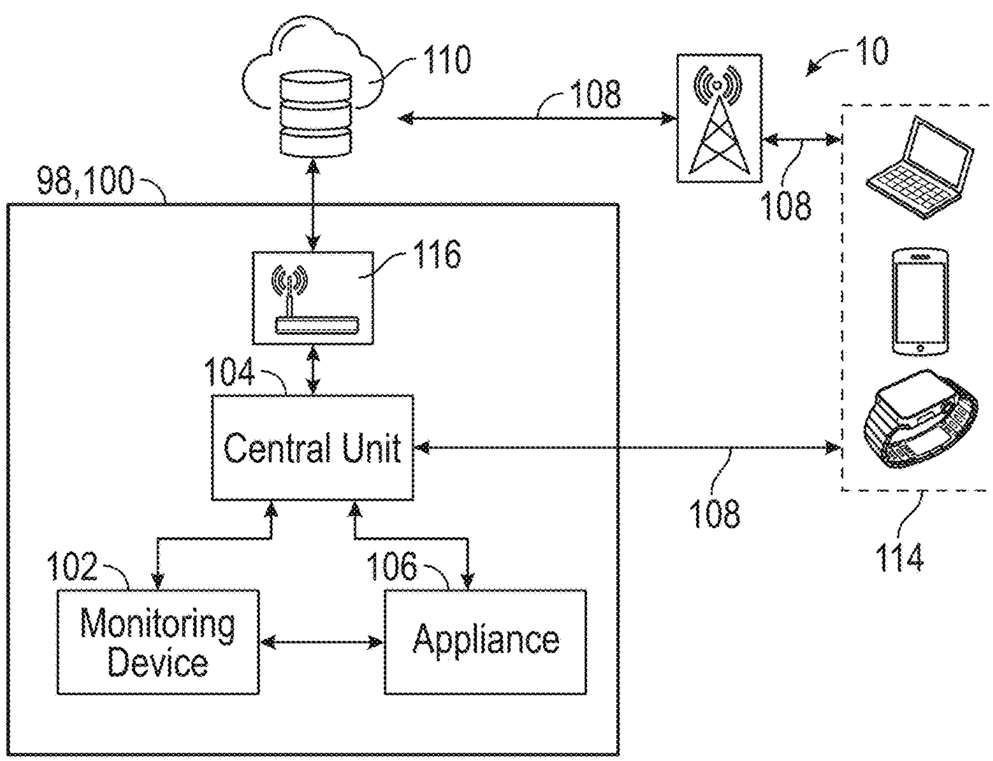

110
108
10
98,100
108
116
104
Central Unit
108
102
Monitoring Device
106
Appliance
114

FIG. 1F

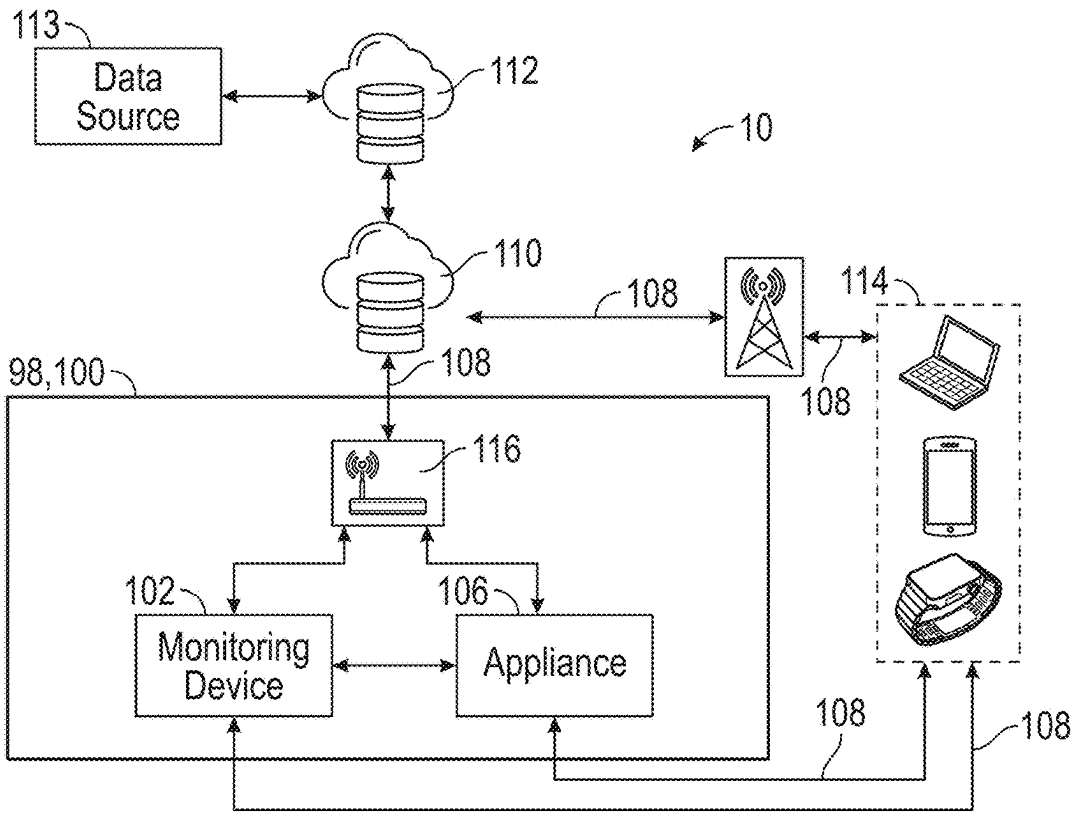

113
Data Source
112
10
110
108
114
98,100
108
108
116
102
106
Monitoring Device
Appliance
108
108

FIG. 1G

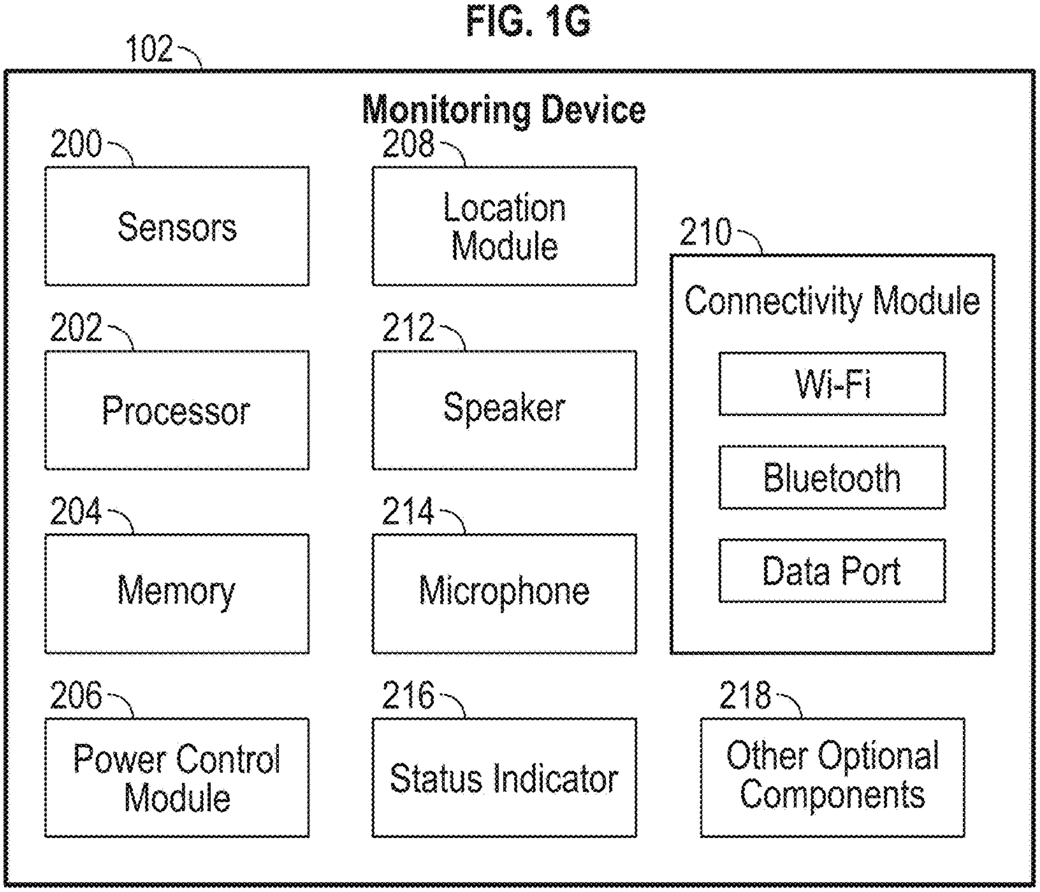

102
Monitoring Device
200
Sensors
208
Location Module
210
Connectivity Module
Wi-Fi
Bluetooth
Data Port
202
Processor
212
Speaker
204
Memory
214
Microphone
206
Power Control Module
216
Status Indicator
218
Other Optional Components

FIG. 2

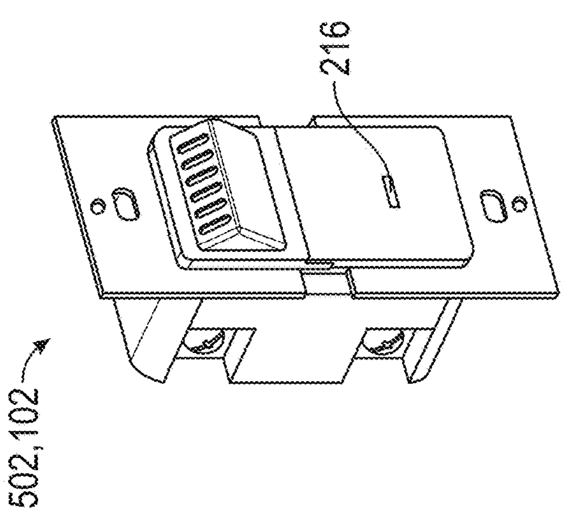


FIG. 3A

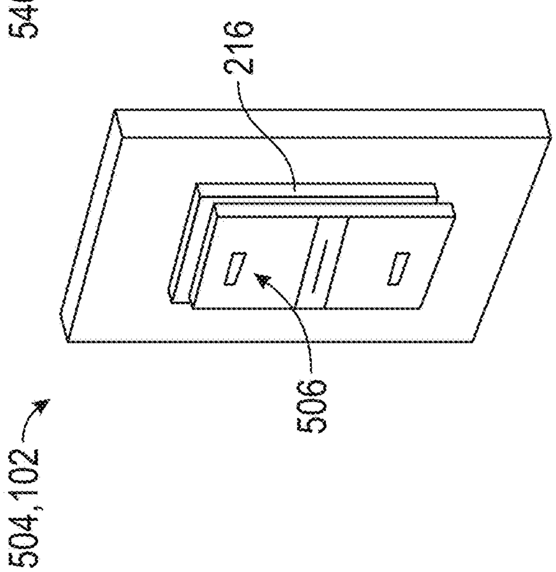


FIG. 3B

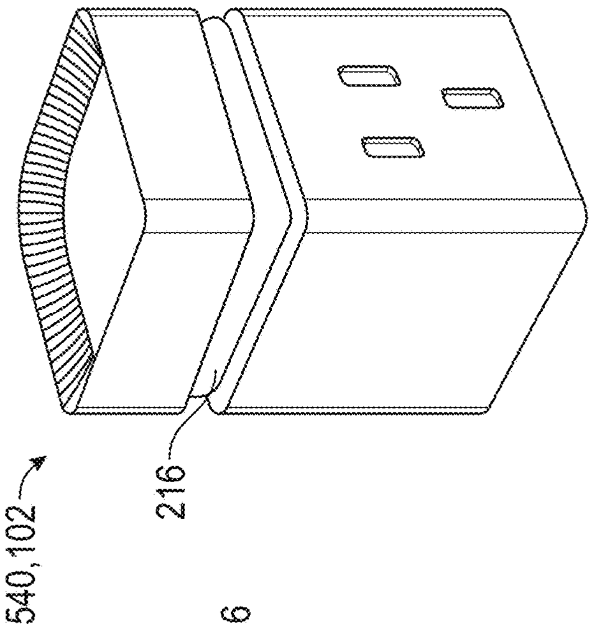


FIG. 4A

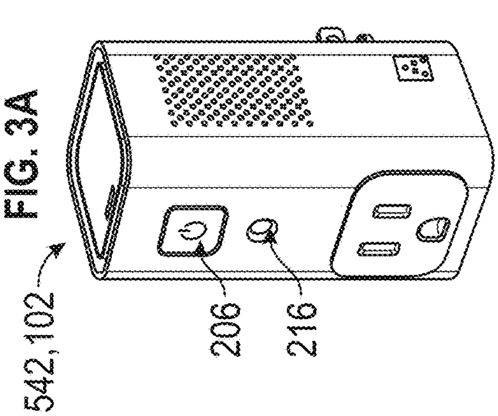


FIG. 4B

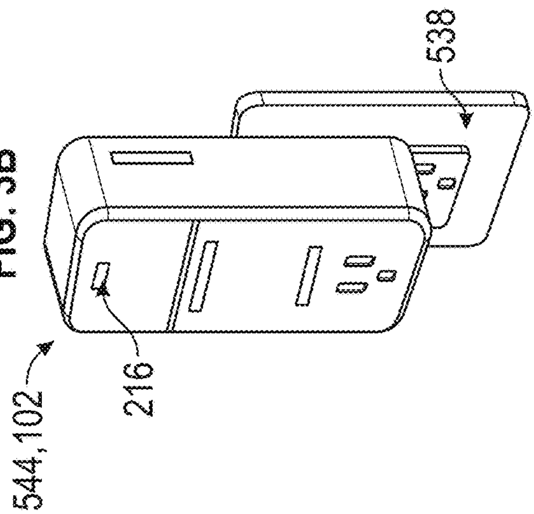


FIG. 4C

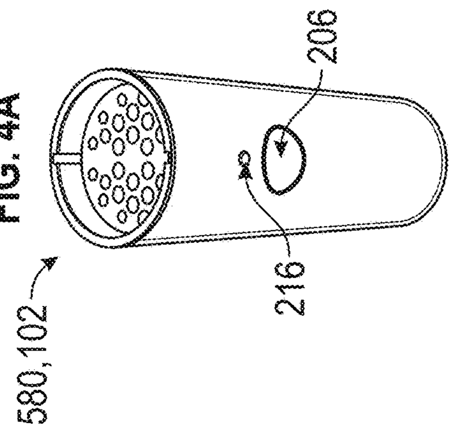


FIG. 5

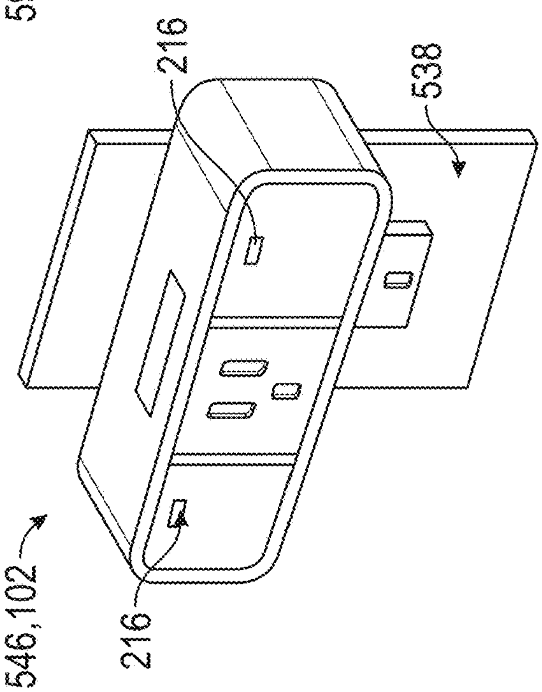


FIG. 4D

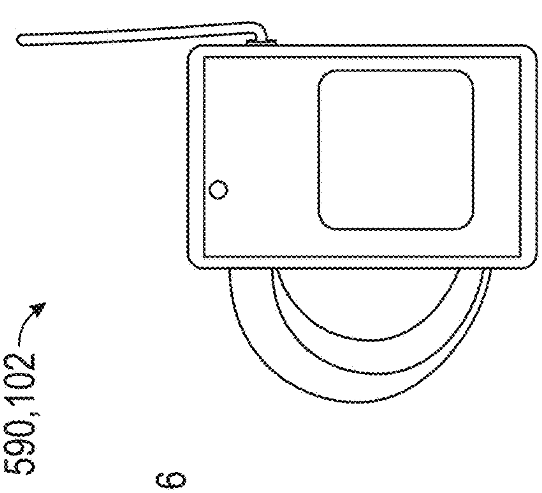


FIG. 6

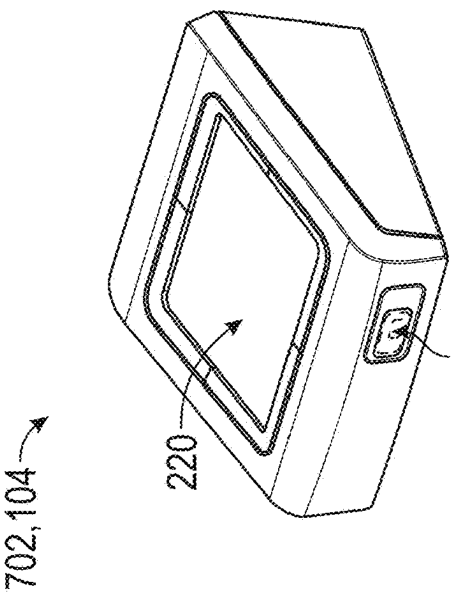


FIG. 7

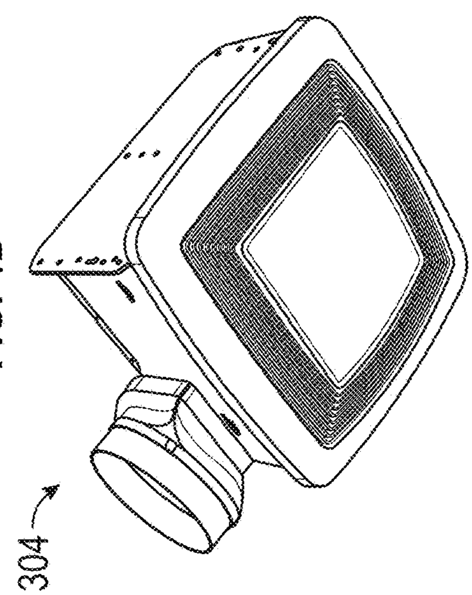


FIG. 8A

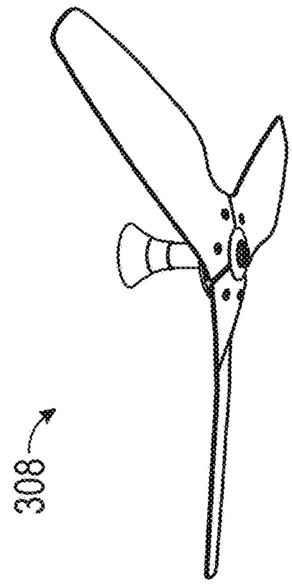


FIG. 8B

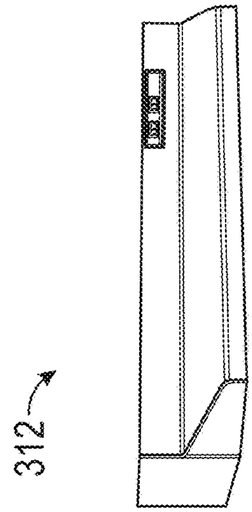


FIG. 8C

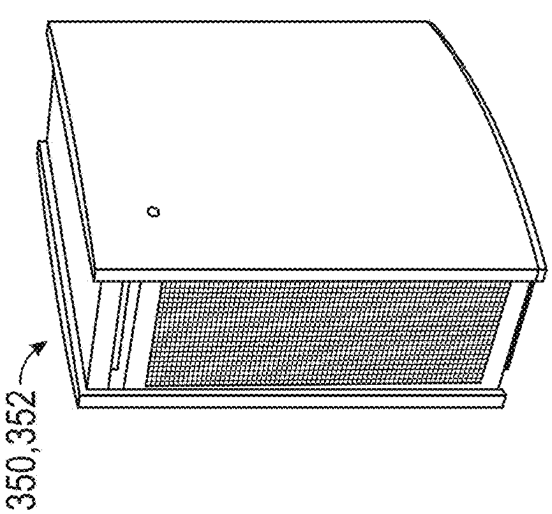


FIG. 9

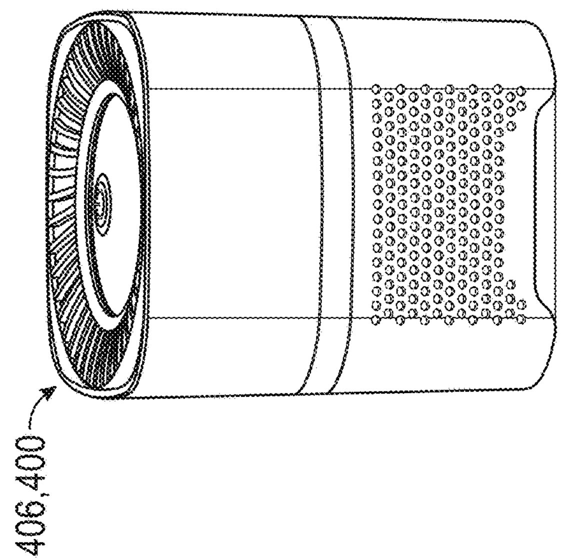


FIG. 10A

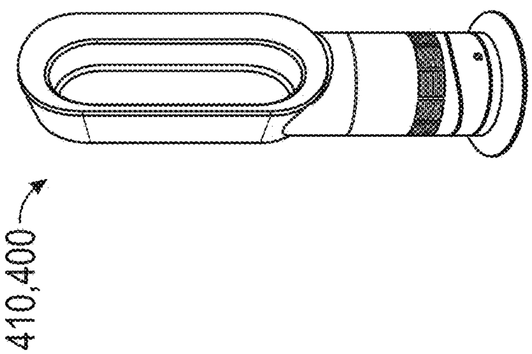


FIG. 10B

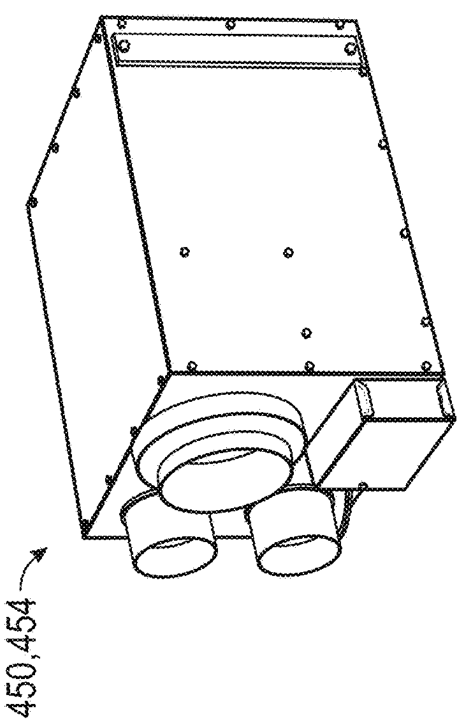


FIG. 11

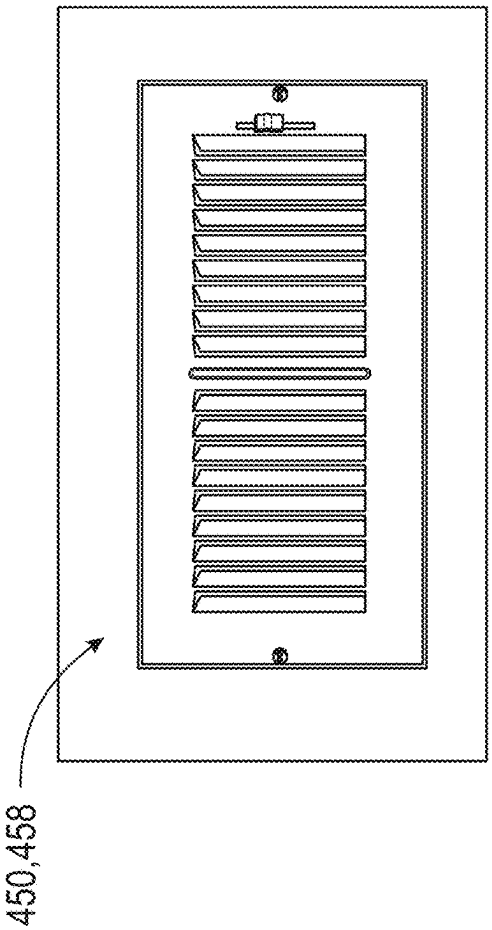


FIG. 12

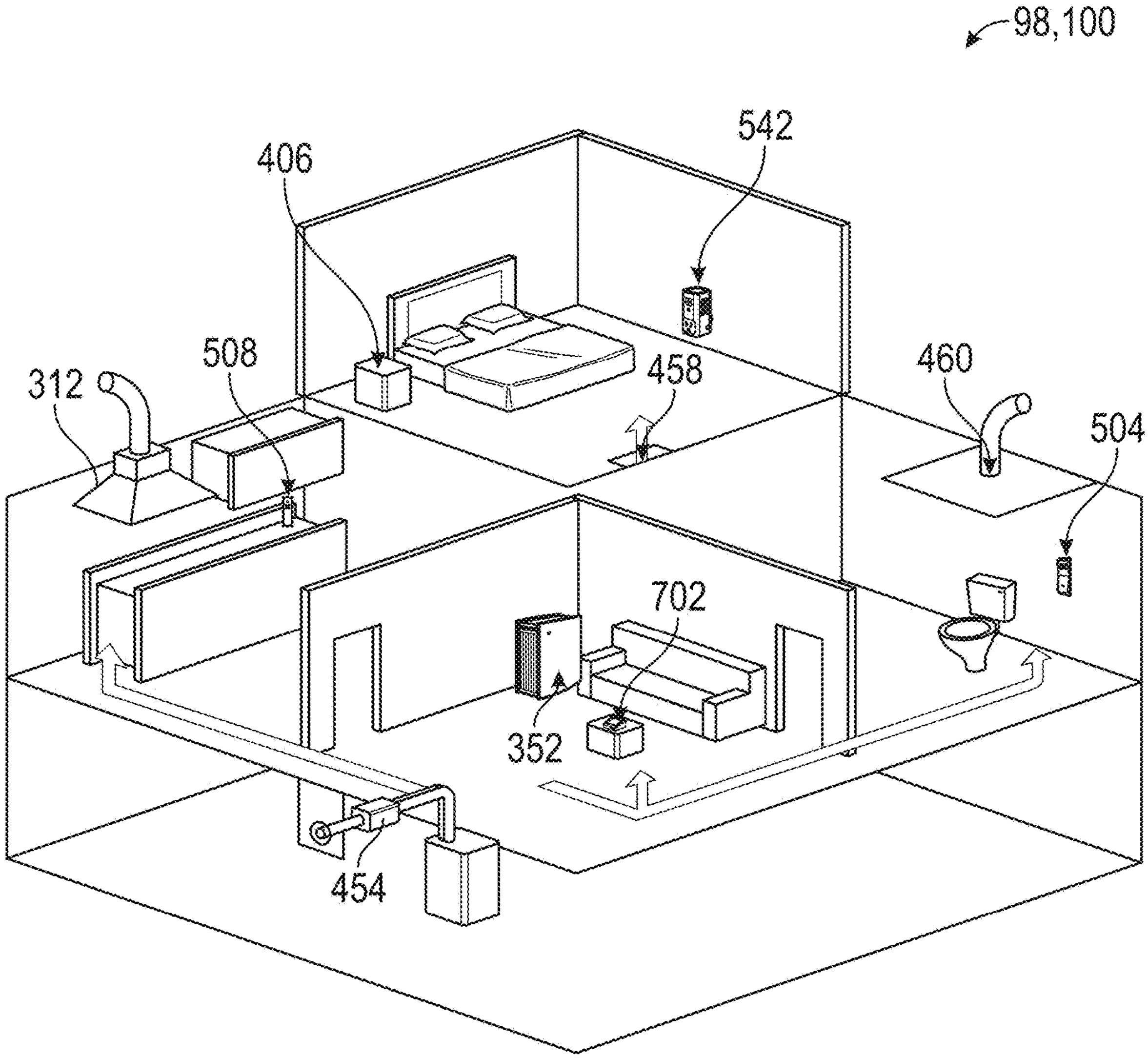


FIG. 13

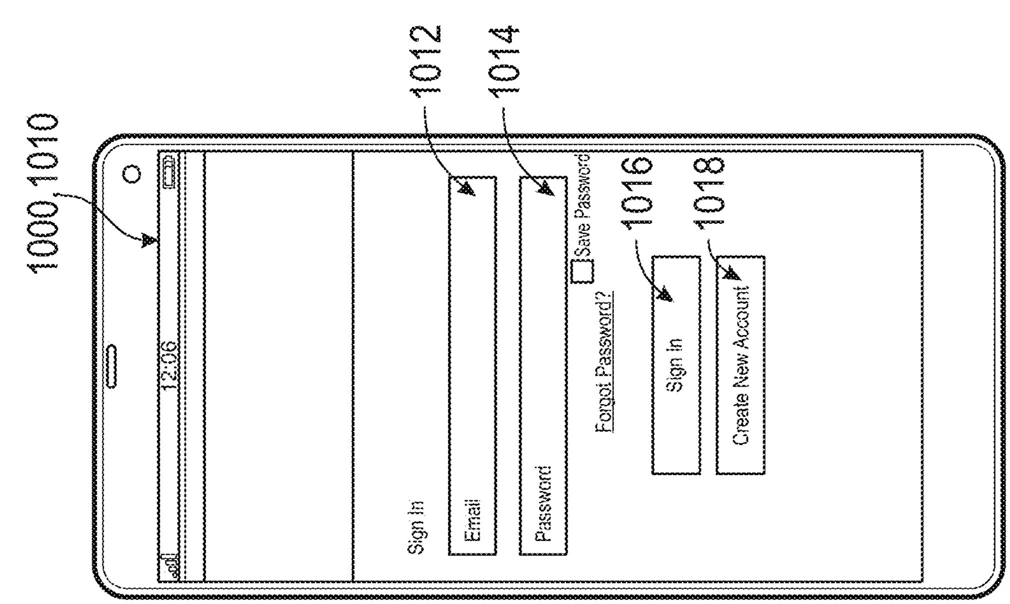


FIG. 14

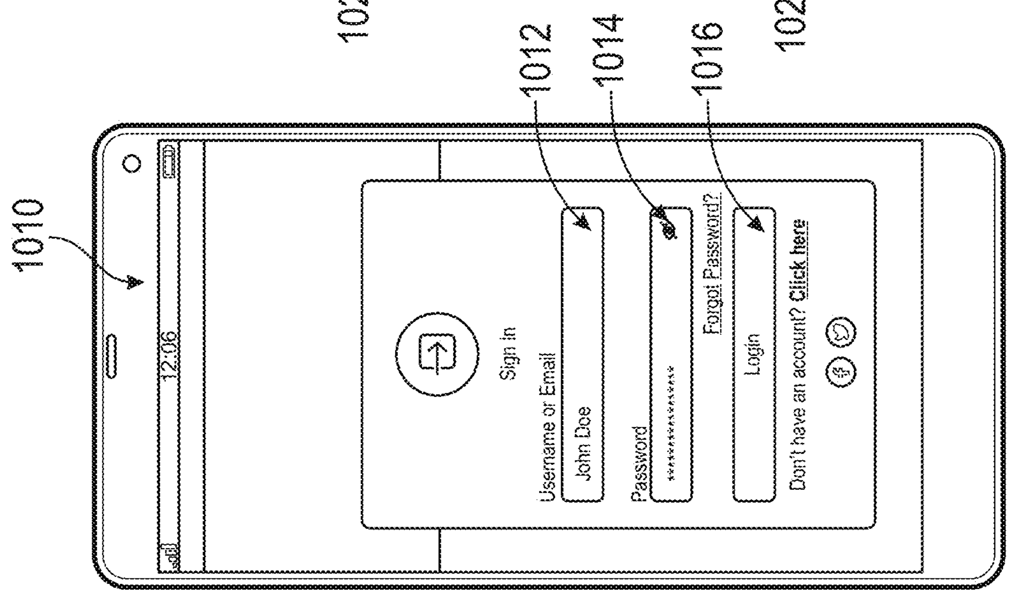


FIG. 15

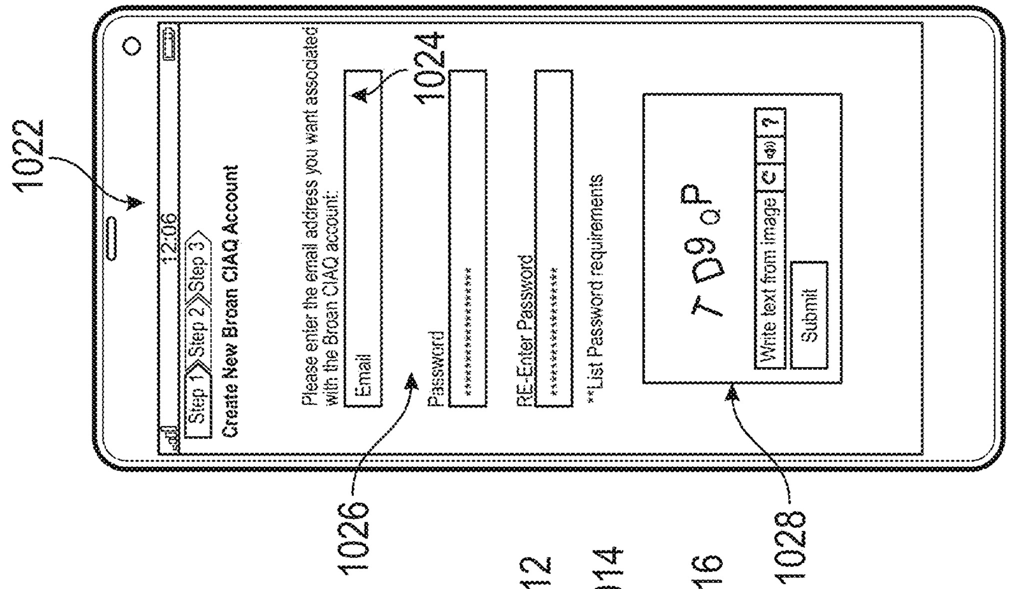


FIG. 16

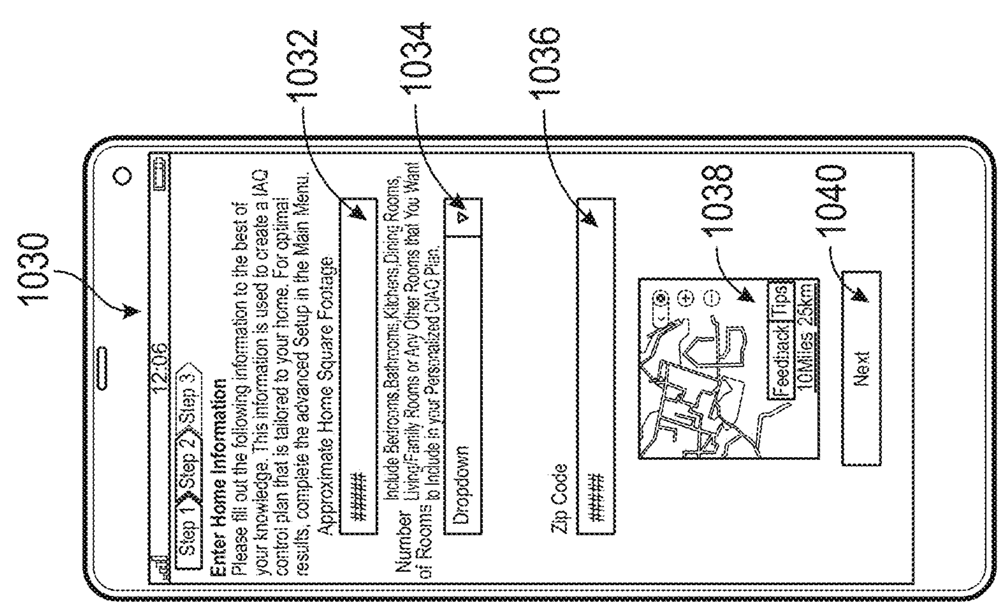


FIG. 17

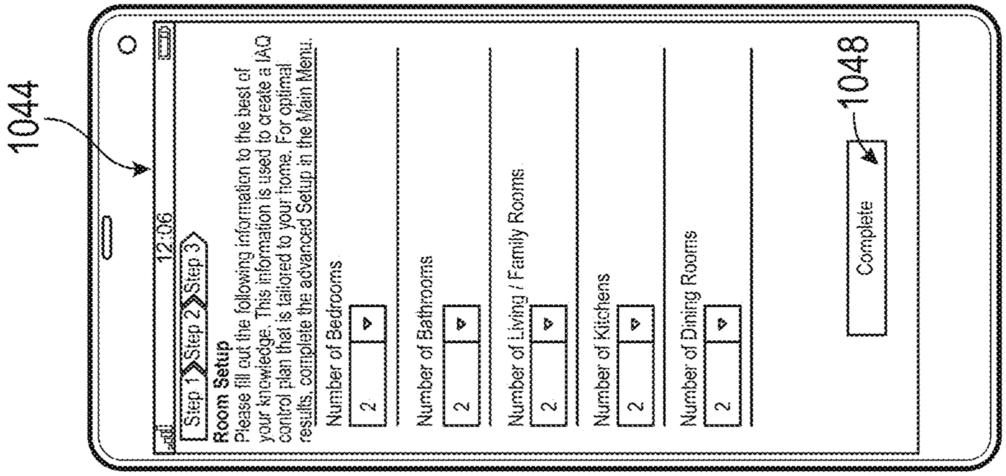


FIG. 18

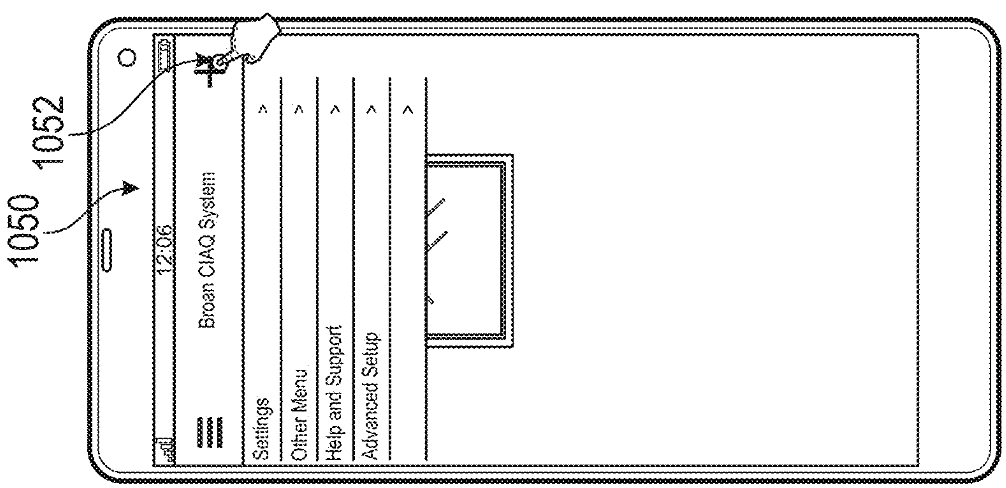


FIG. 19

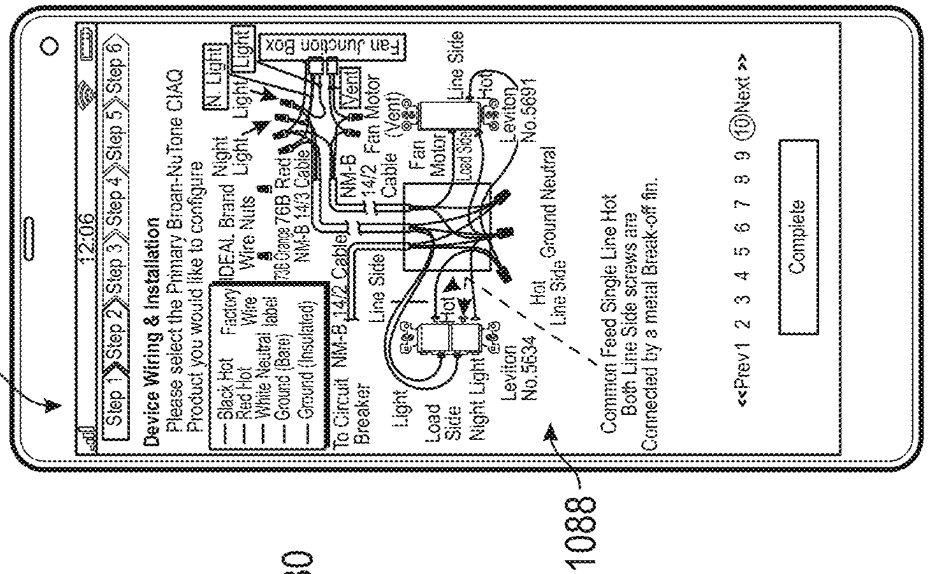


FIG. 22A

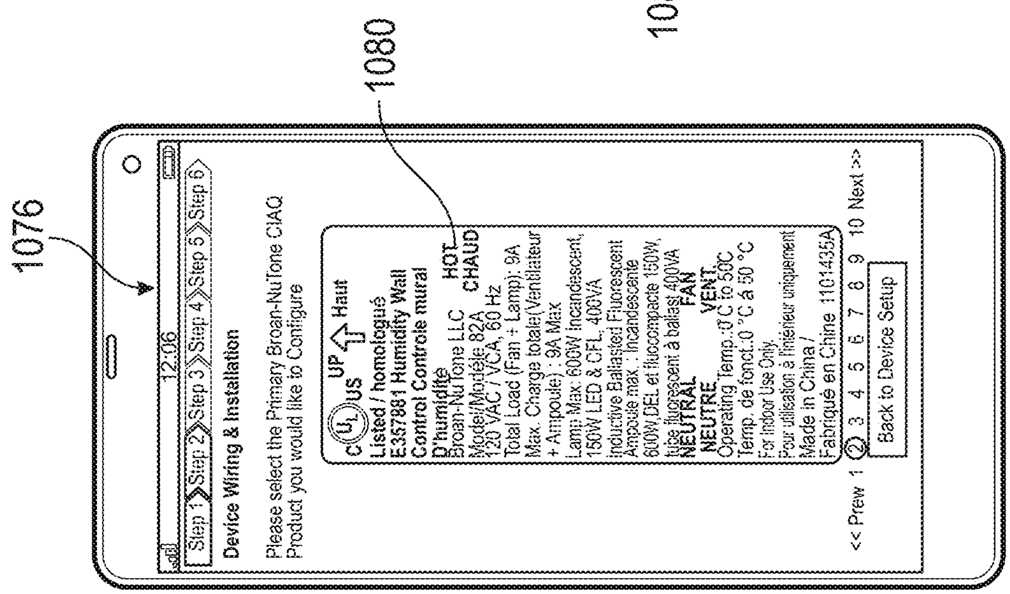


FIG. 21

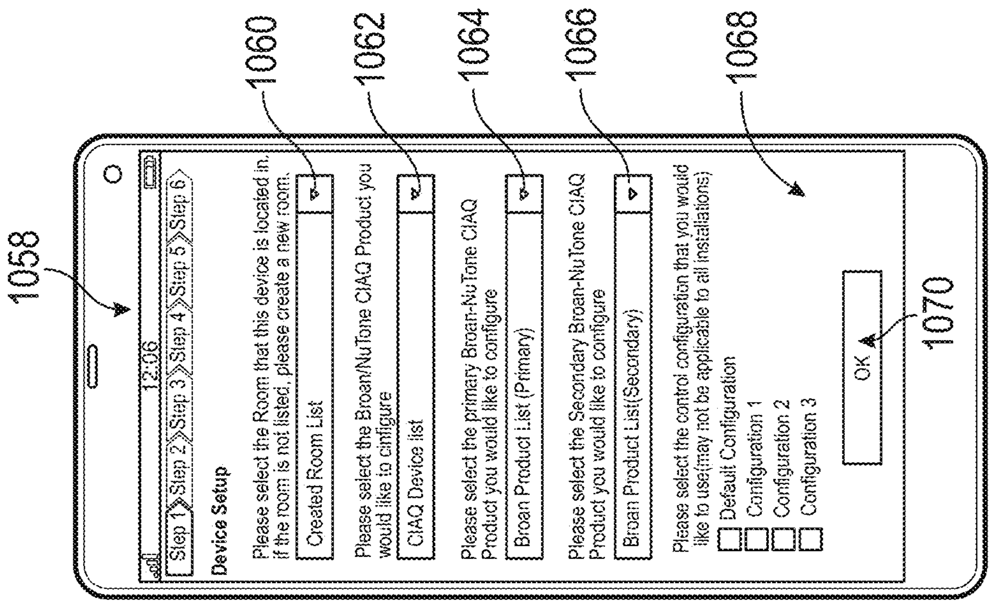


FIG. 20

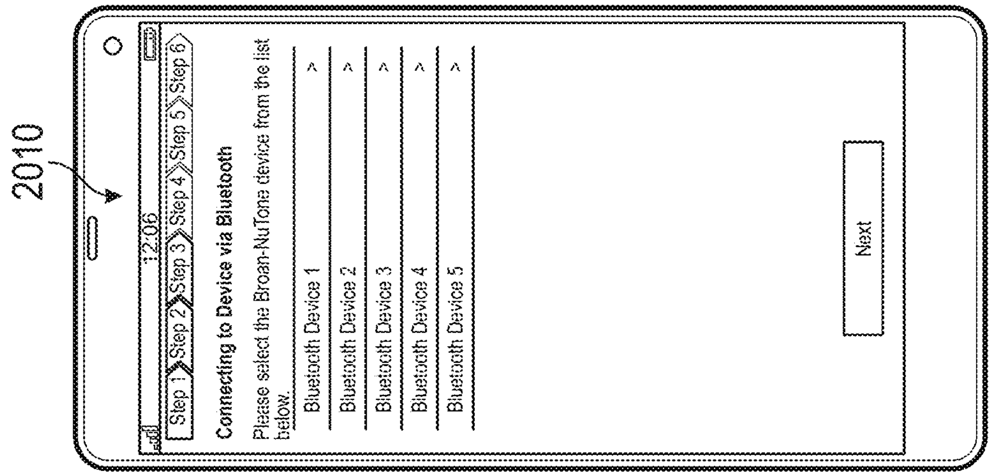


FIG. 24

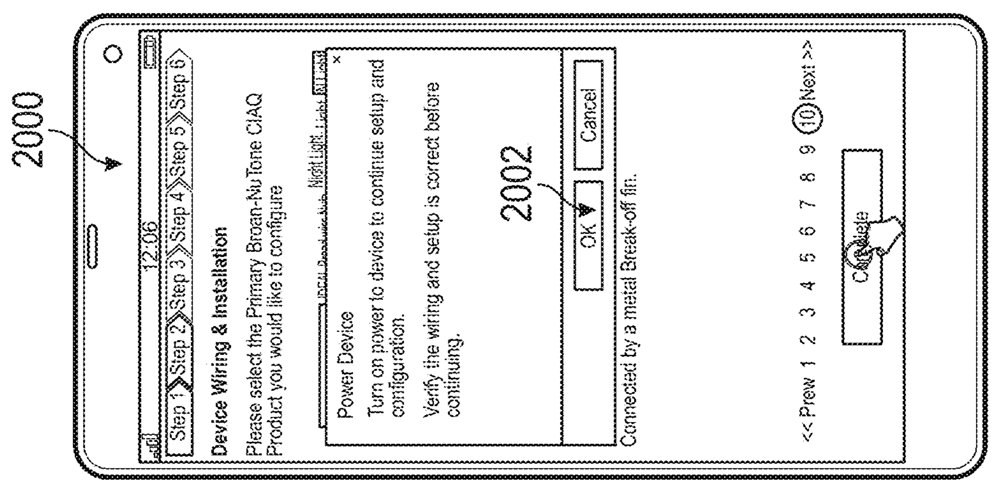


FIG. 23

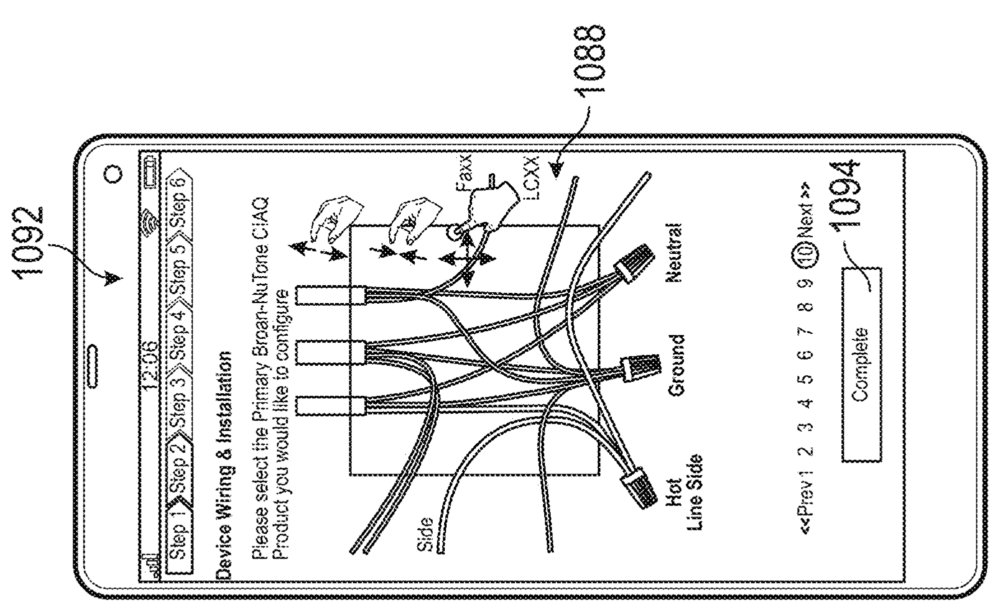


FIG. 22B

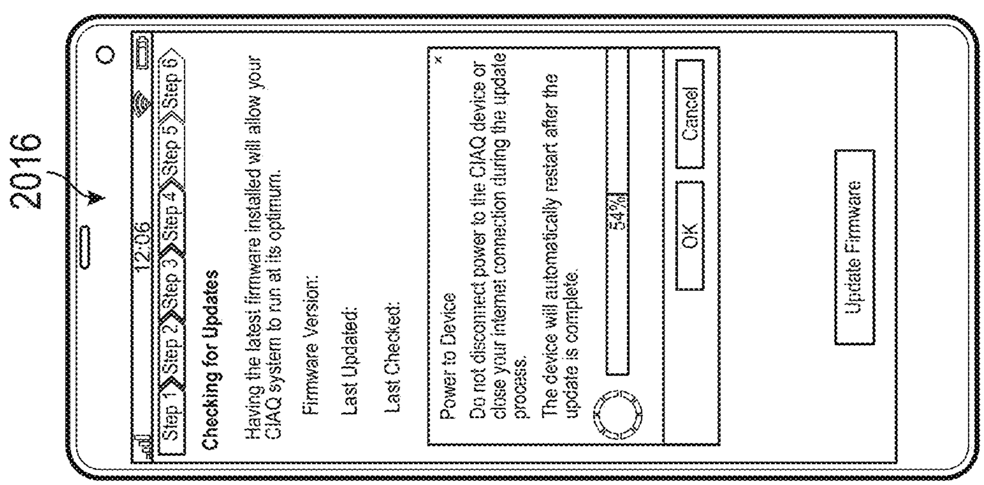


FIG. 25

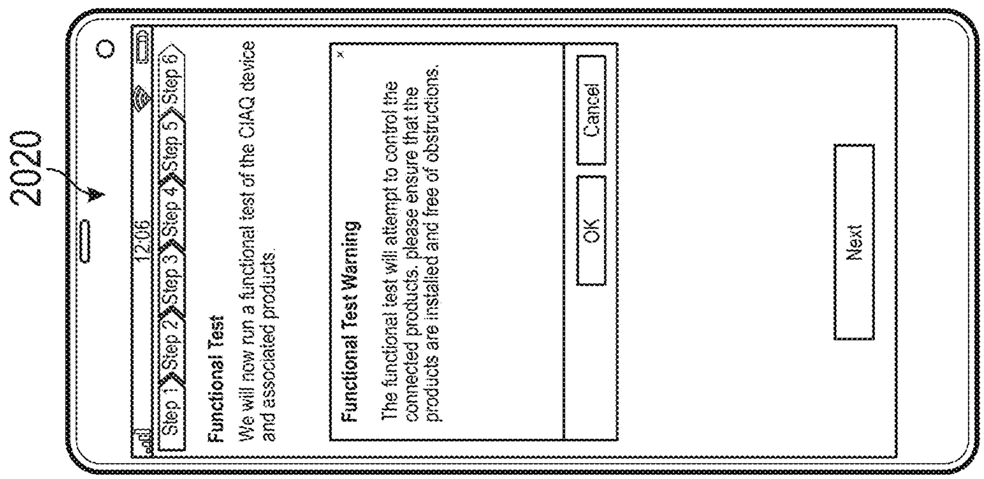


FIG. 26

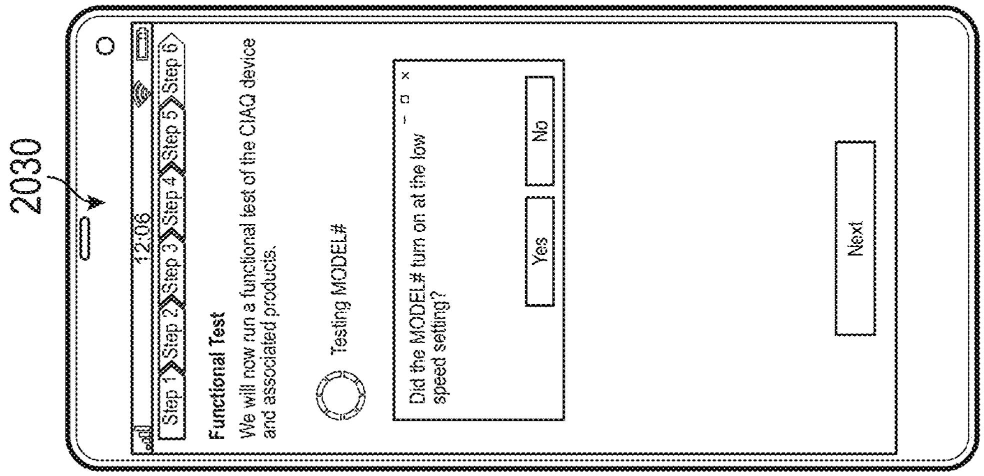


FIG. 27

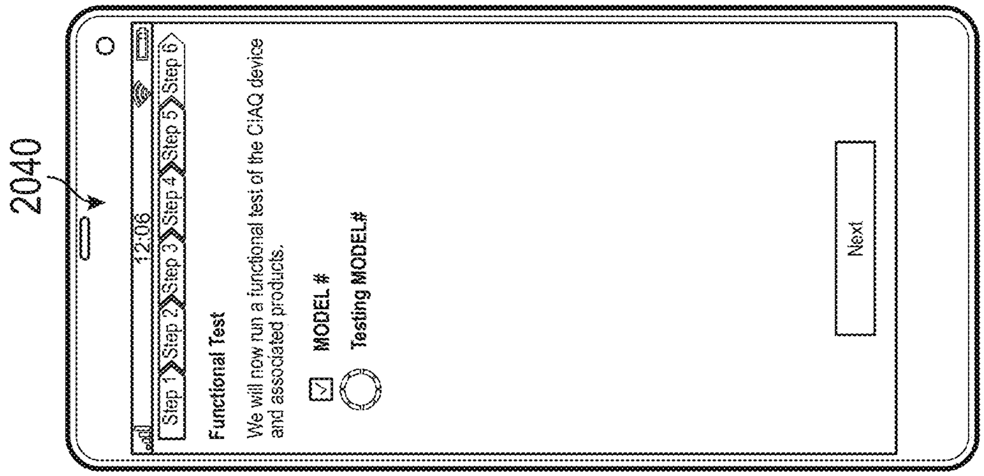


FIG. 30

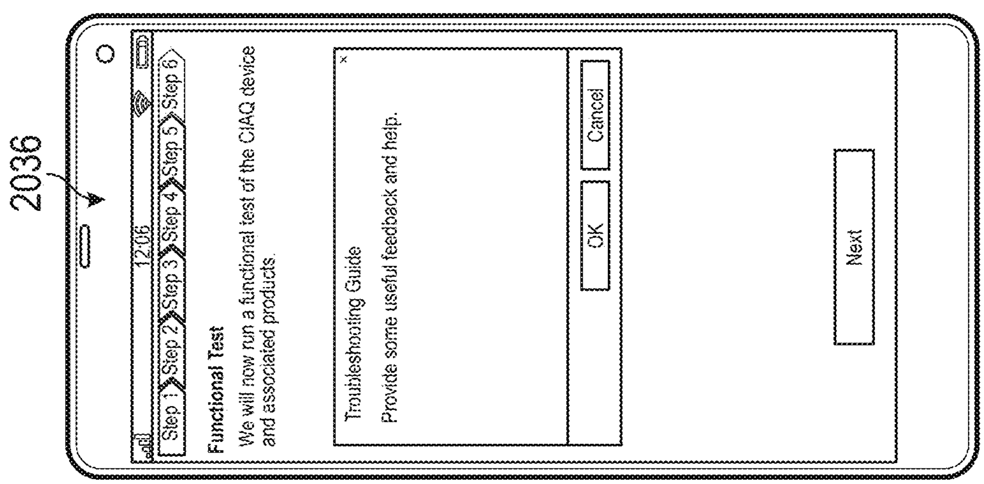


FIG. 29

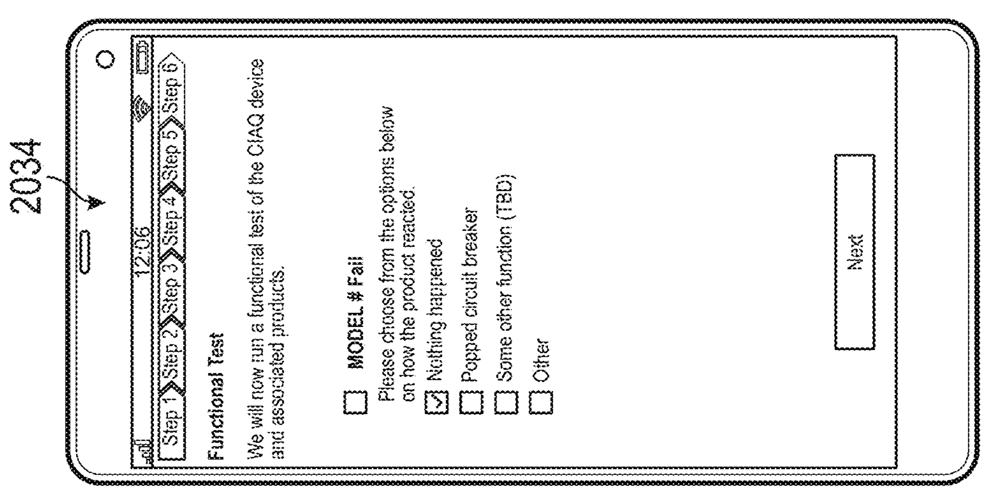


FIG. 28

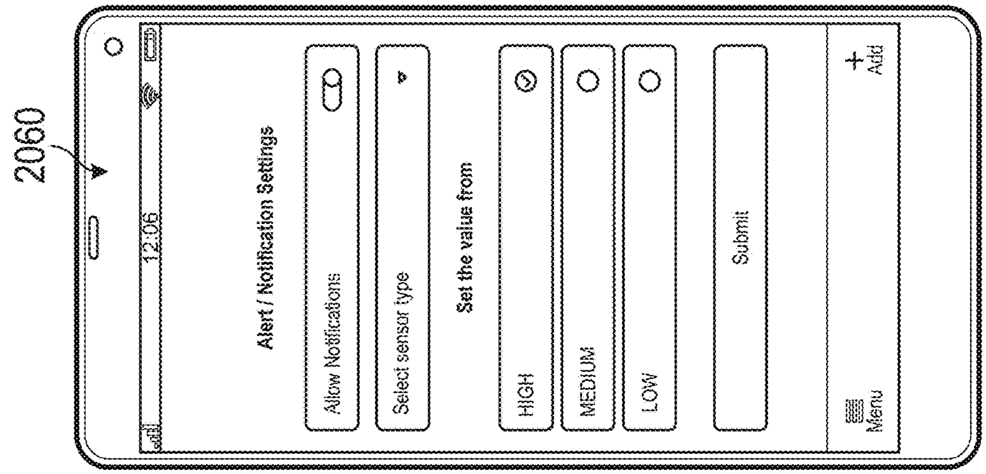


FIG. 33

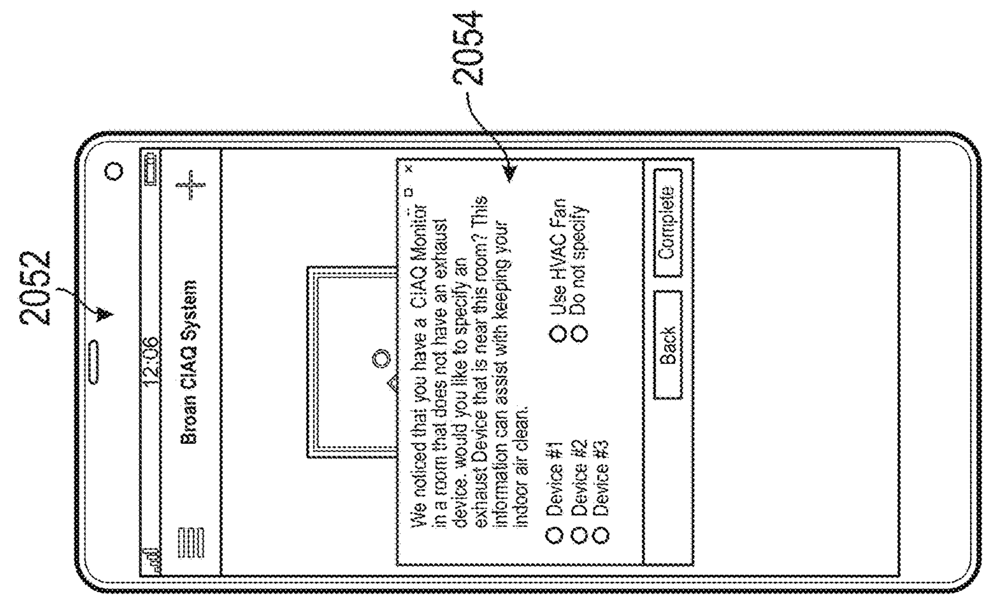


FIG. 32

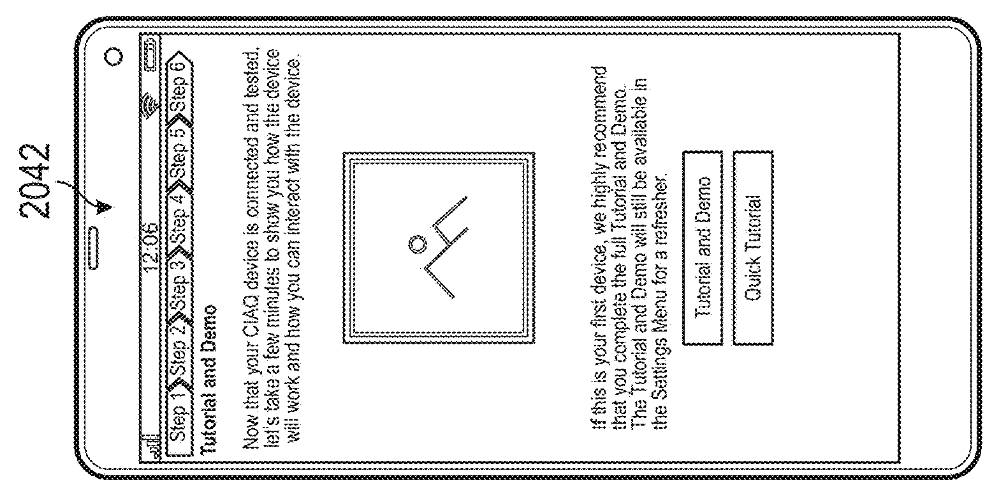


FIG. 31

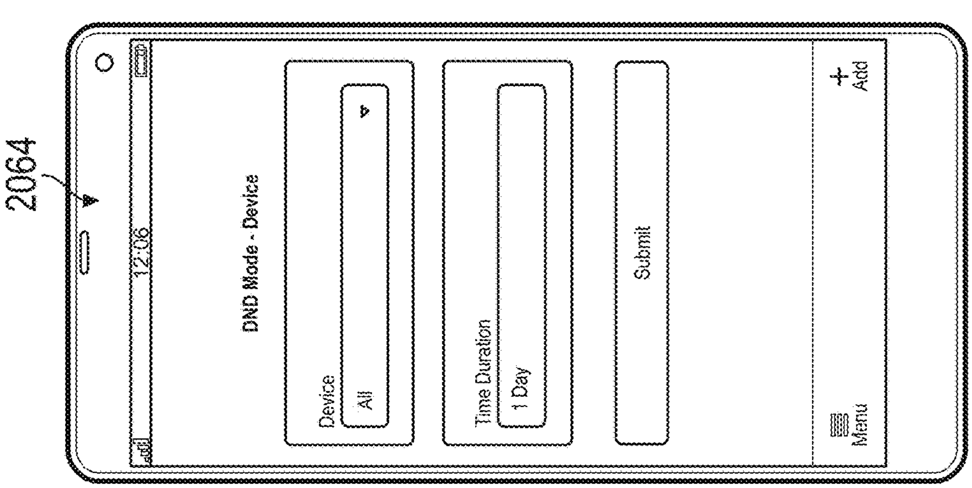

2064
12:06
DND Mode - Device
Device
All
Time Duration
1 Day
Submit
Menu
Add

FIG. 34

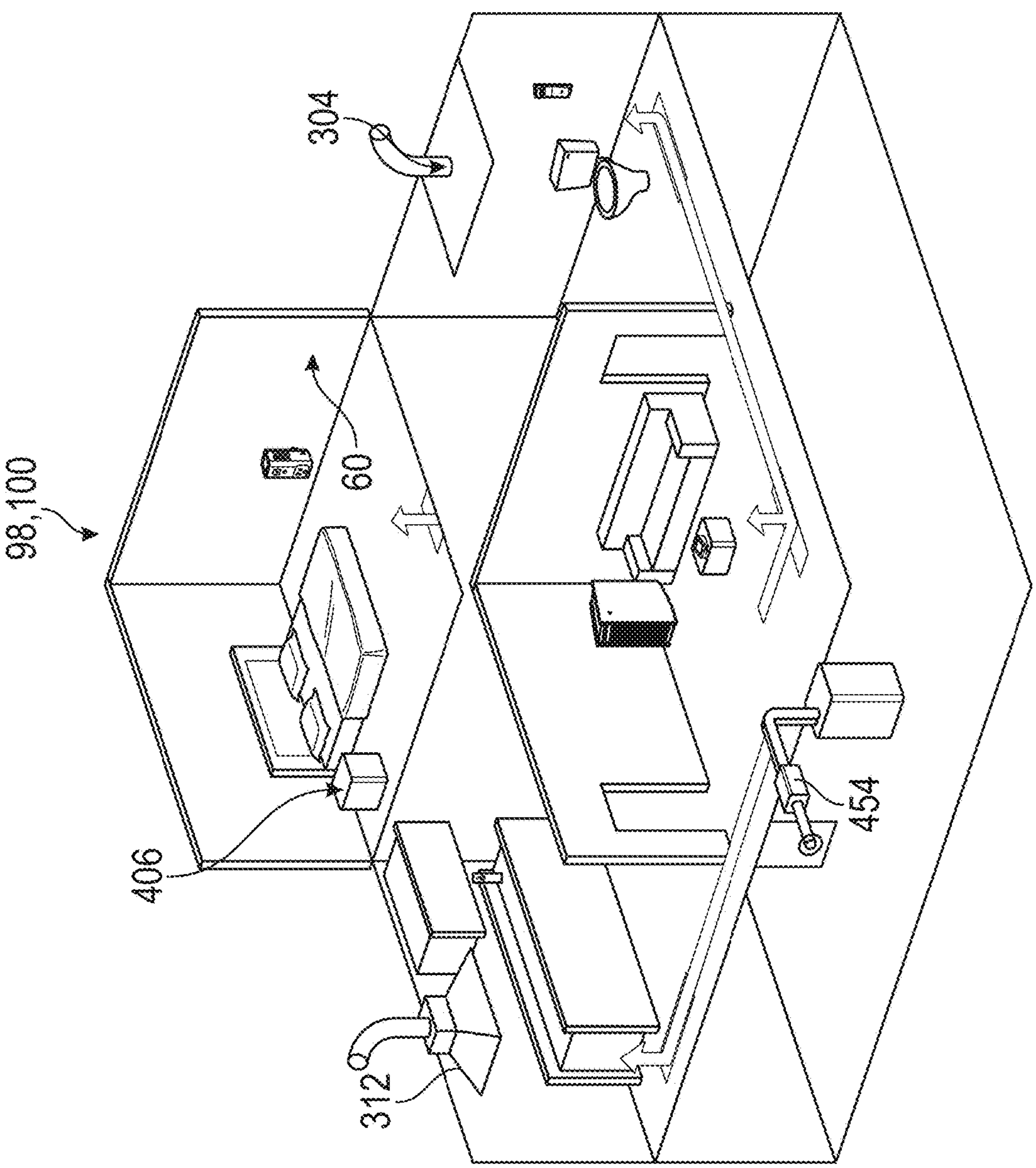

98,100
60
304
406
312
454

FIG. 35

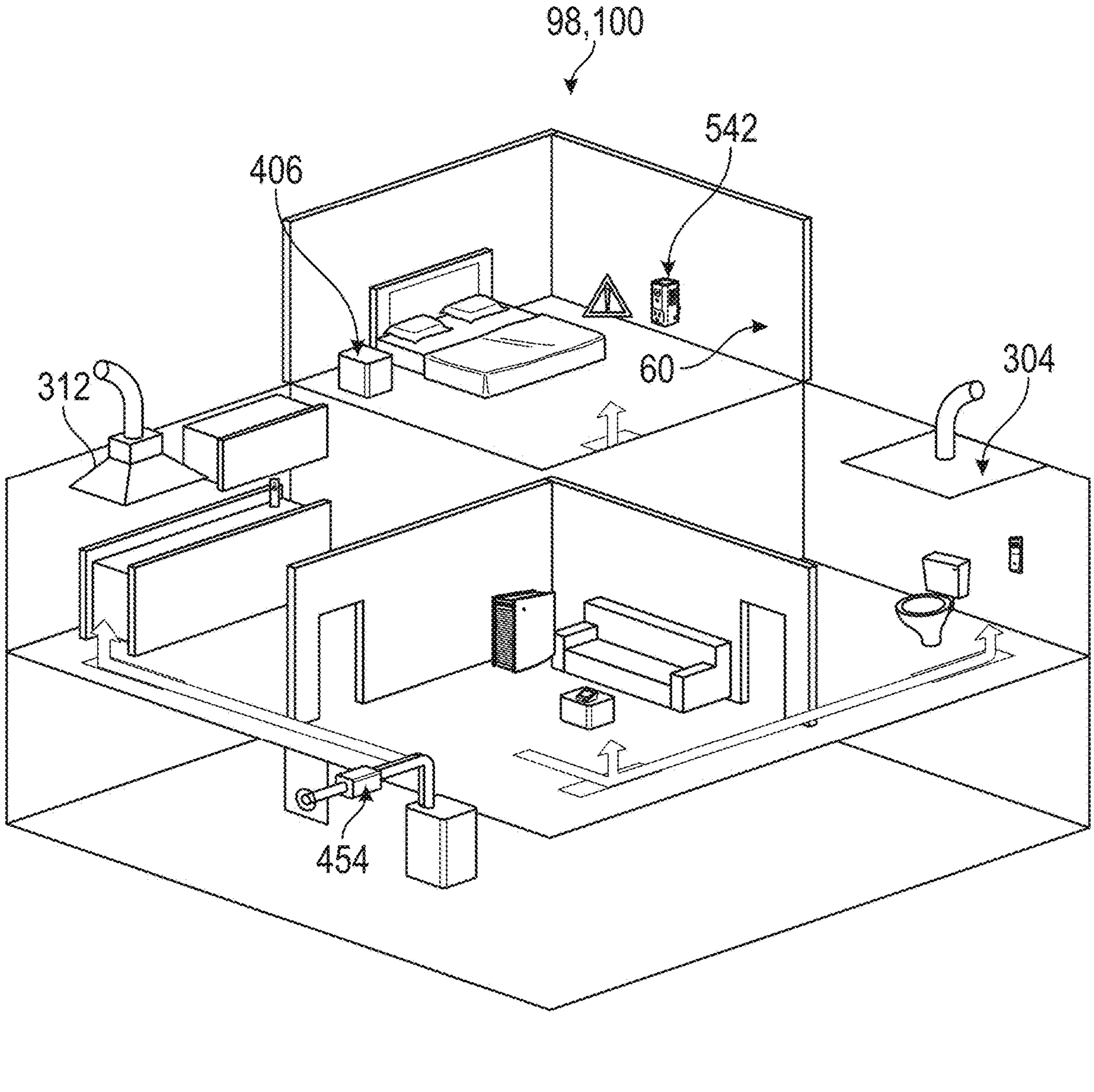


FIG. 36

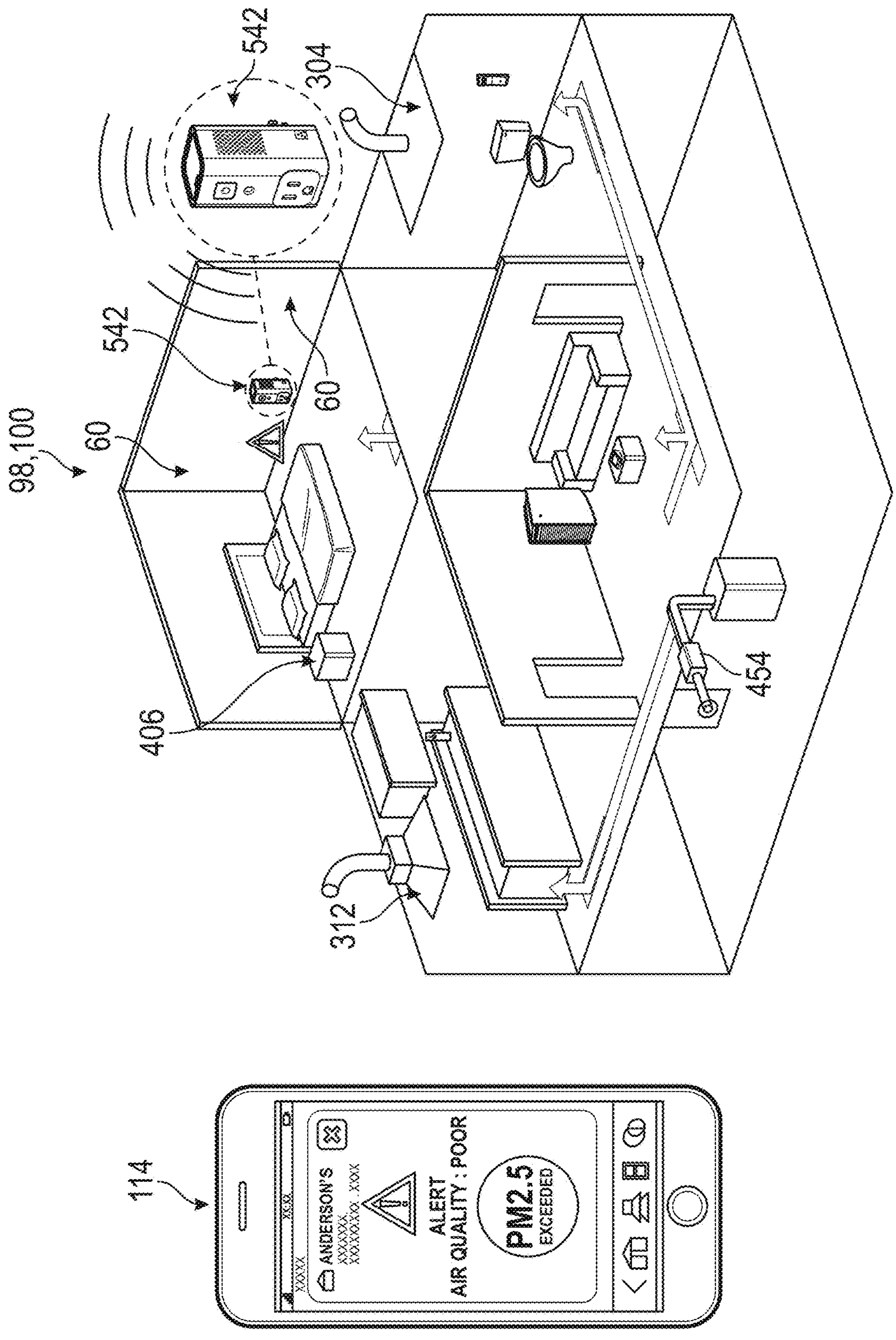

98,100
60
542
60
542
304
406
454
312
114
ANDERSON'S
ALERT
AIR QUALITY : POOR
PM2.5
EXCEEDED

FIG. 37

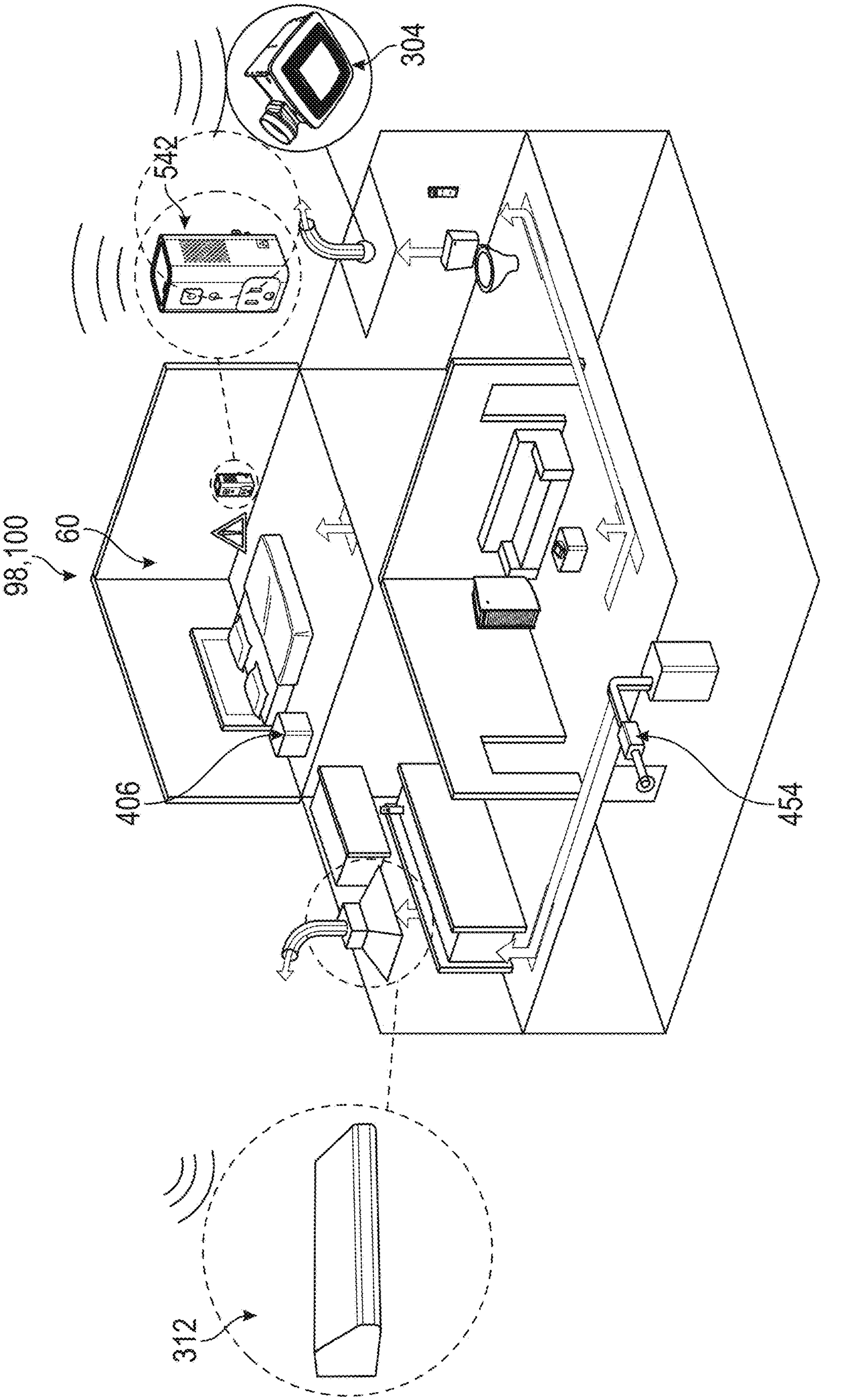


FIG. 38

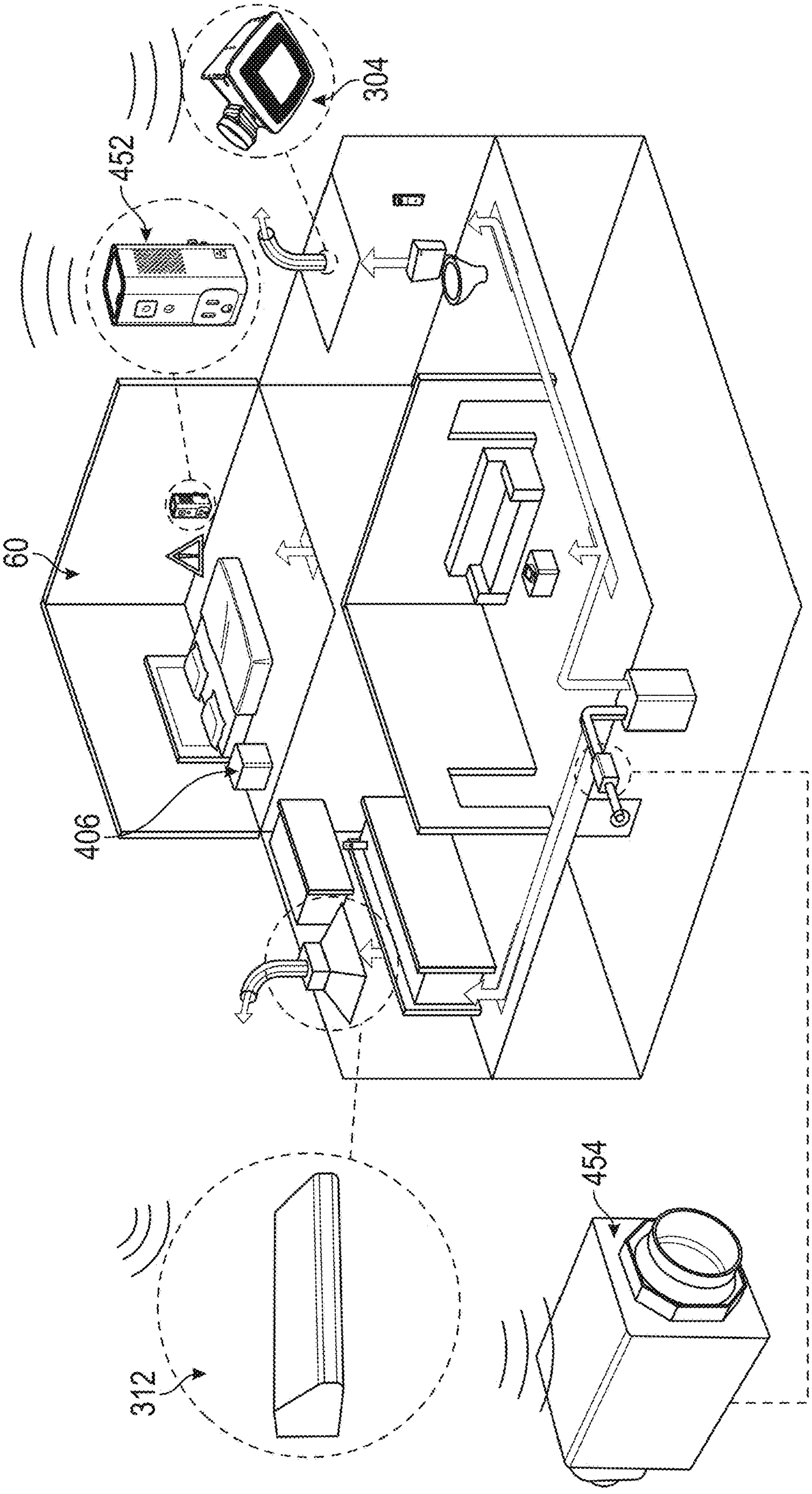


FIG. 39

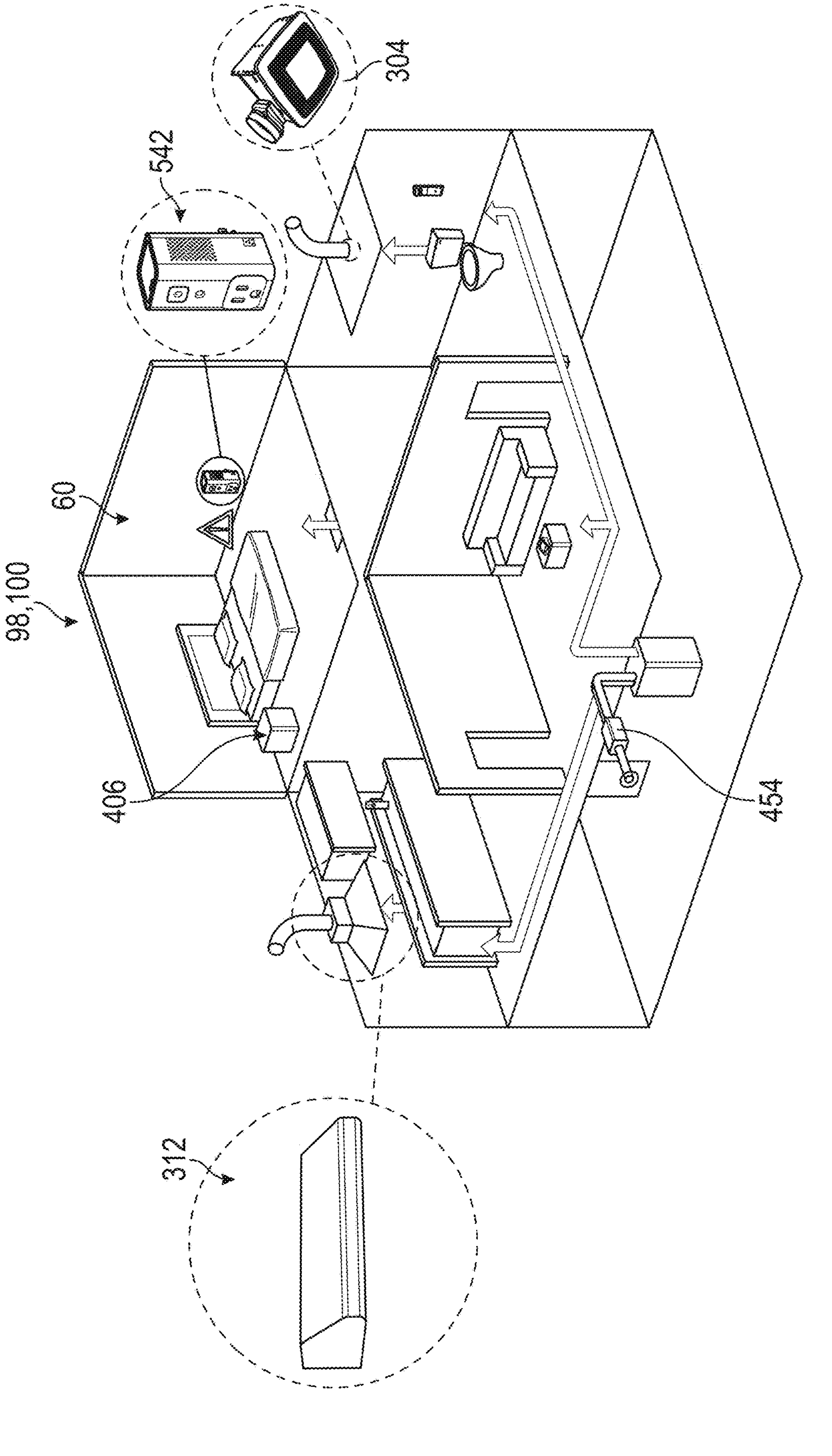


FIG. 40

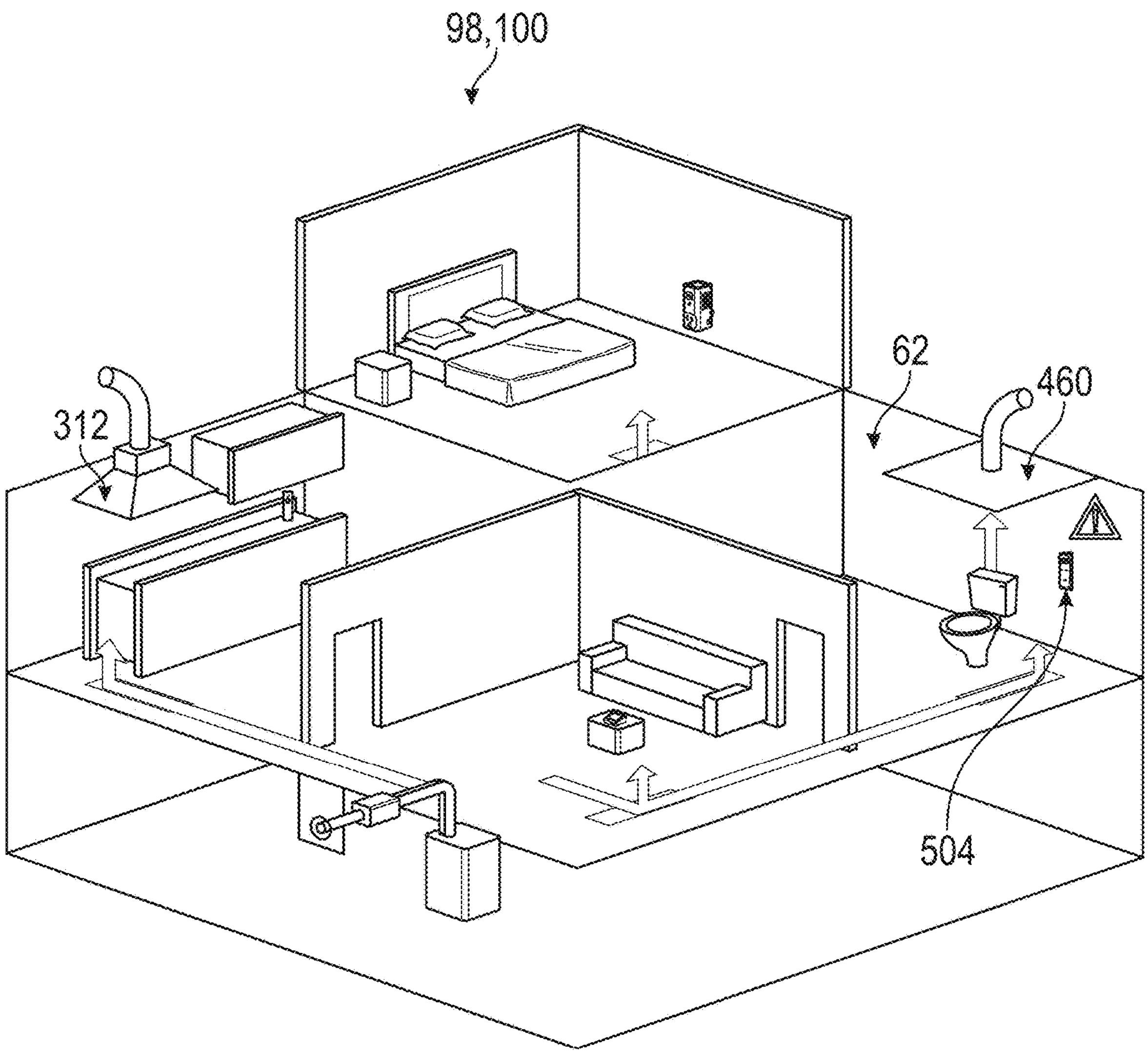


FIG. 41

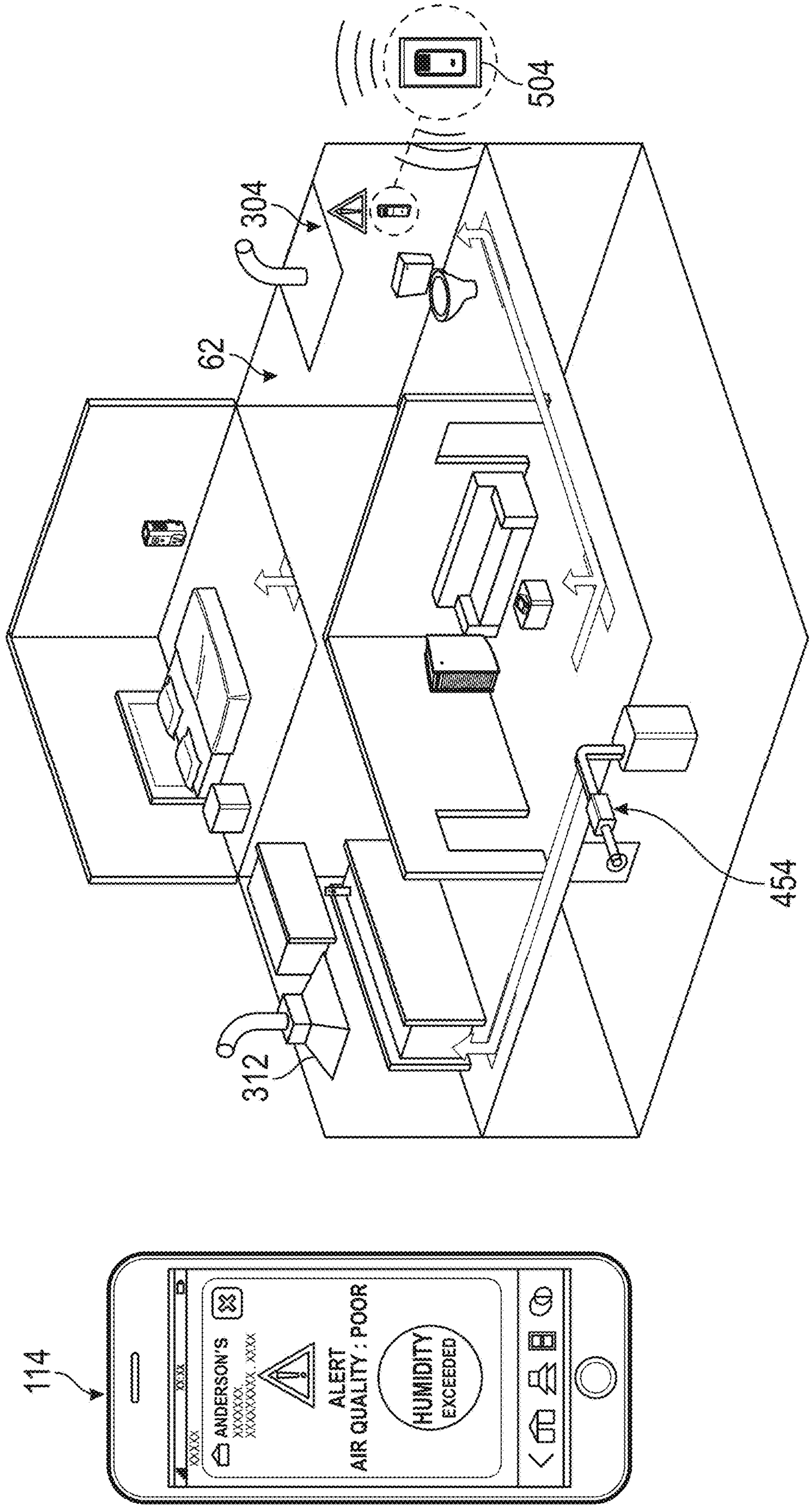

98,100
62
304
504
312
454
114
ANDERSON'S
ALERT
AIR QUALITY : POOR
HUMIDITY
EXCEEDED

FIG. 42

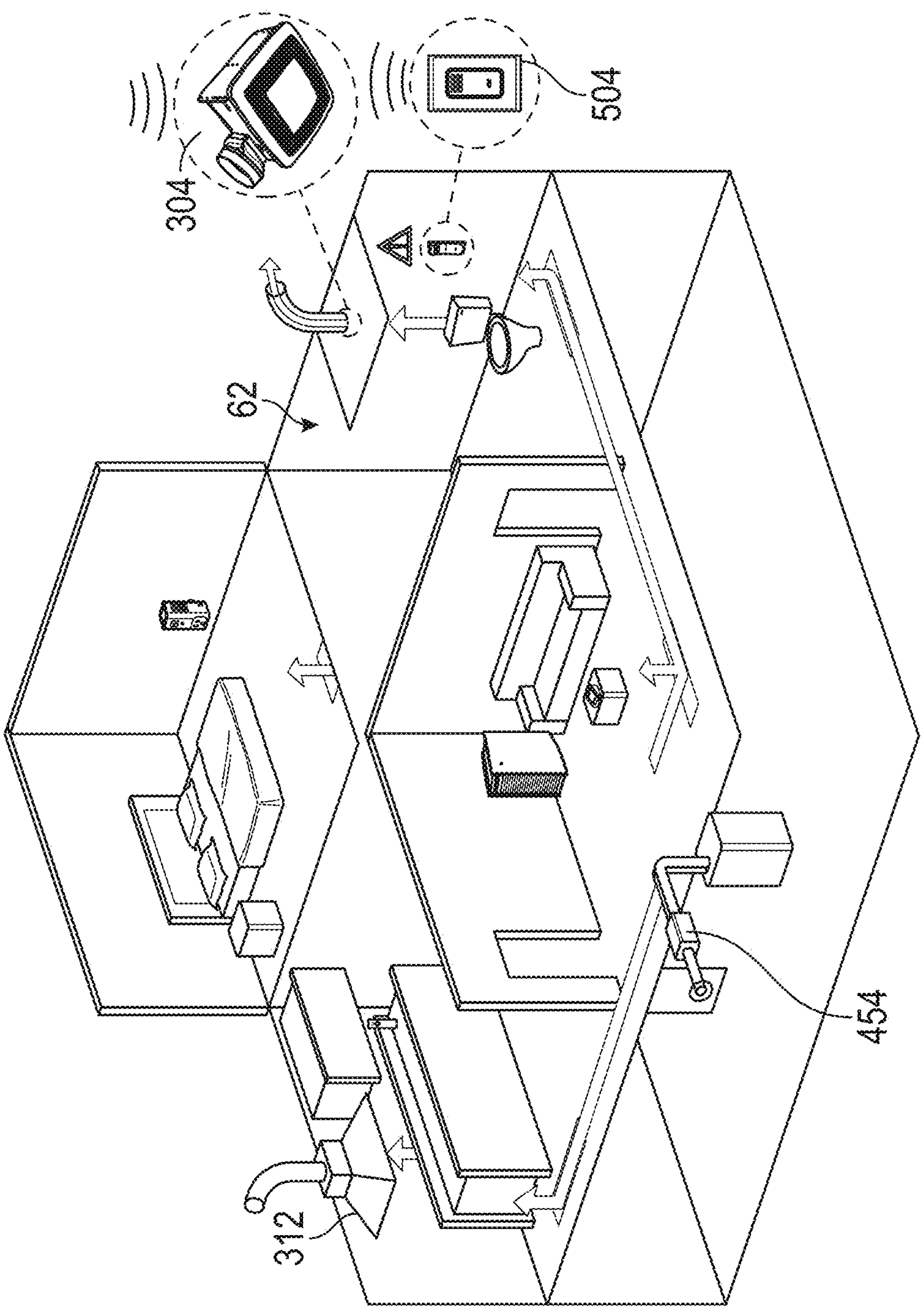


FIG. 43

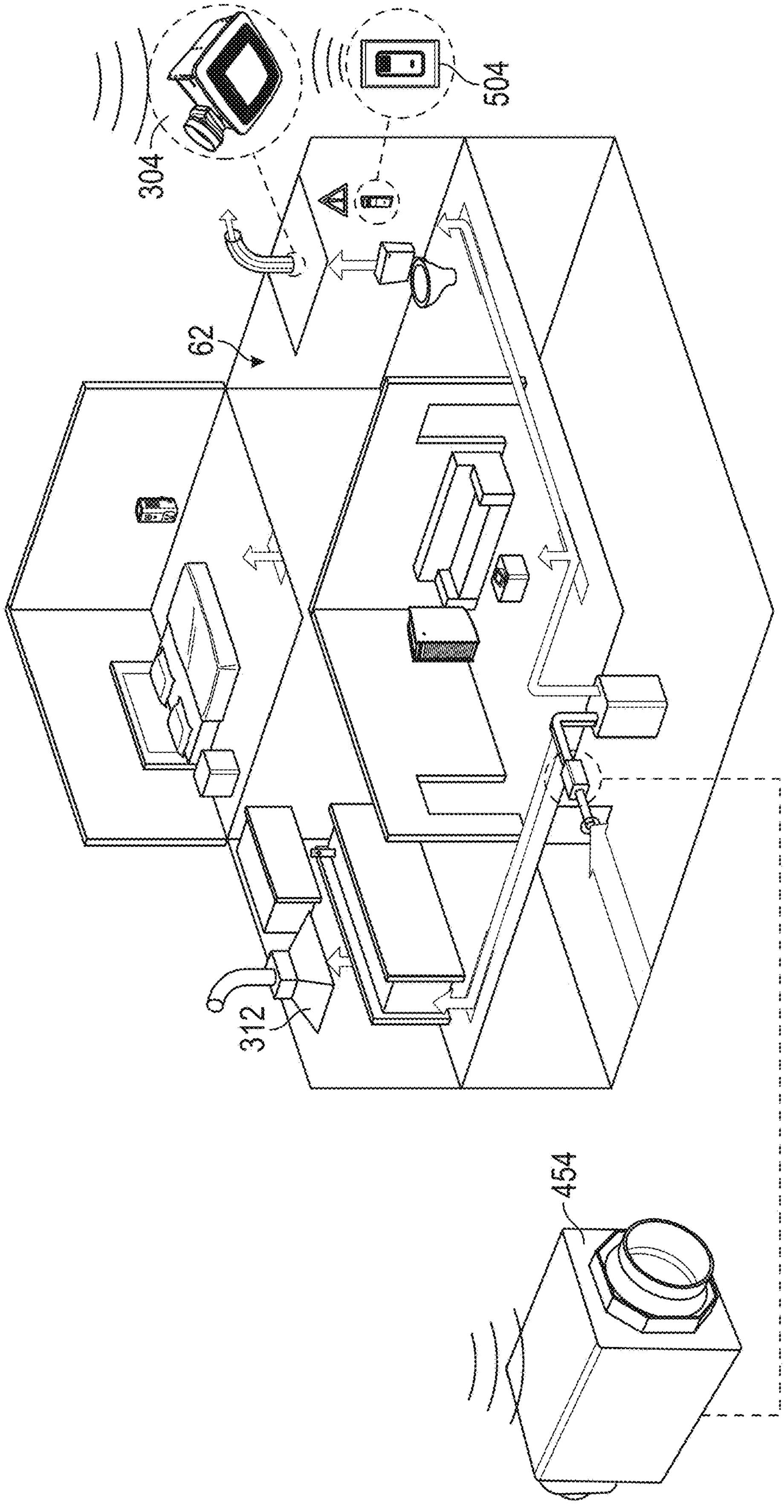


FIG. 44

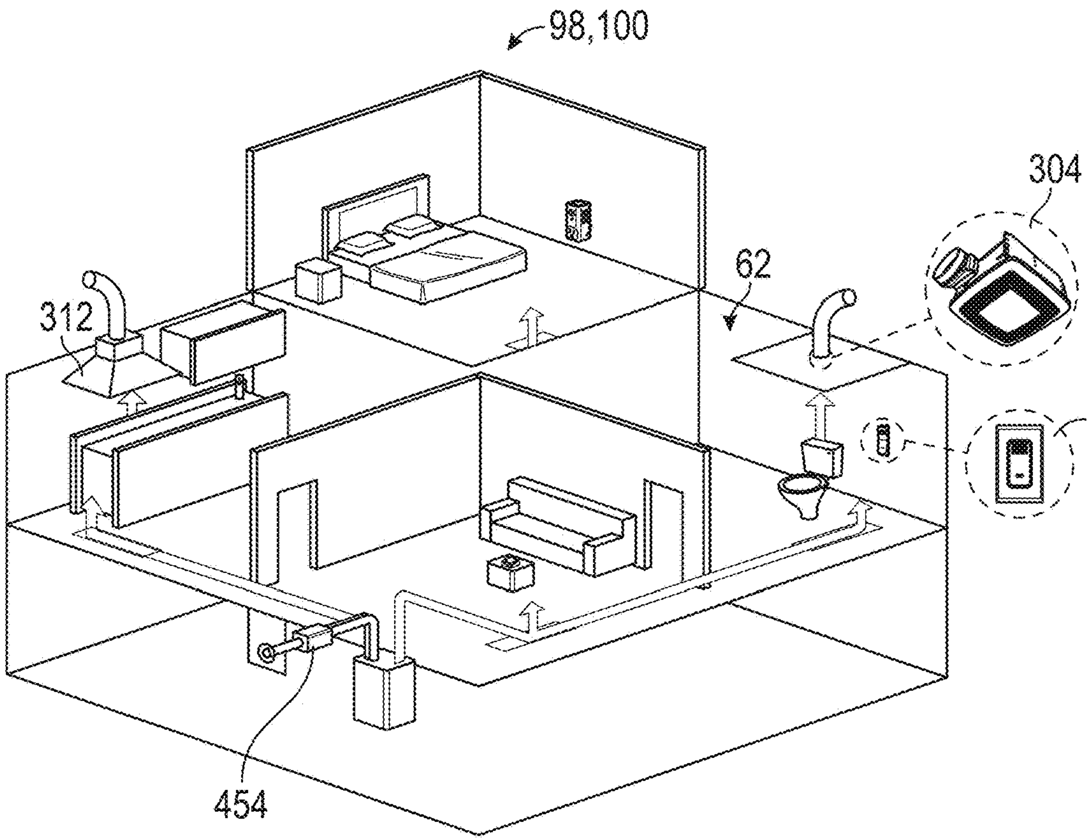


FIG. 45

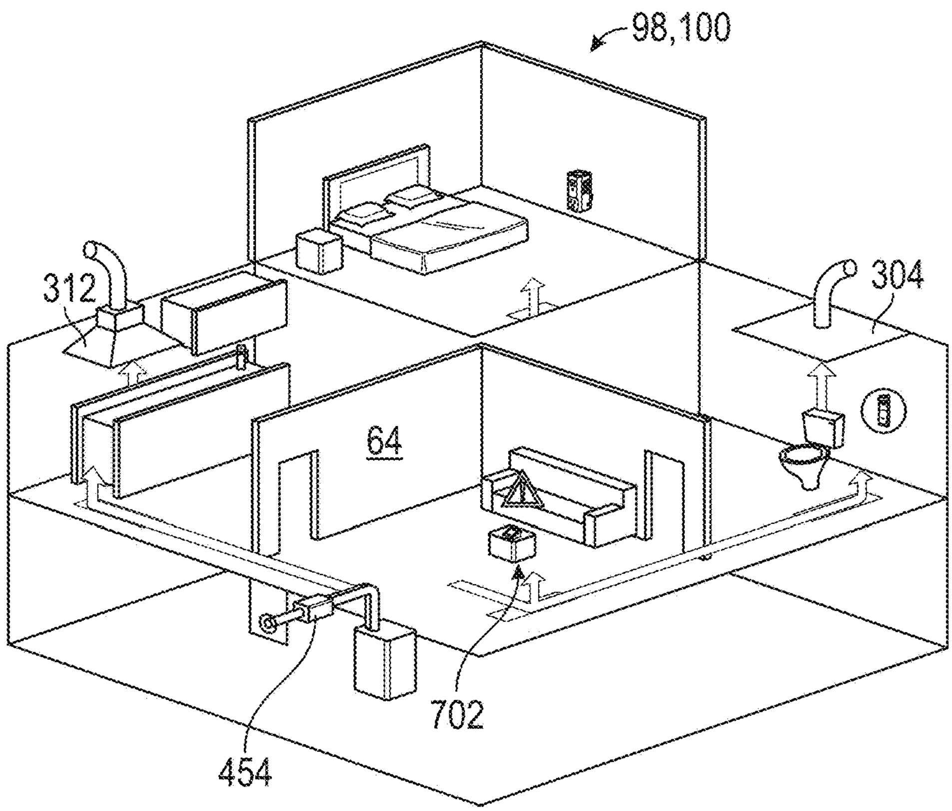


FIG. 46

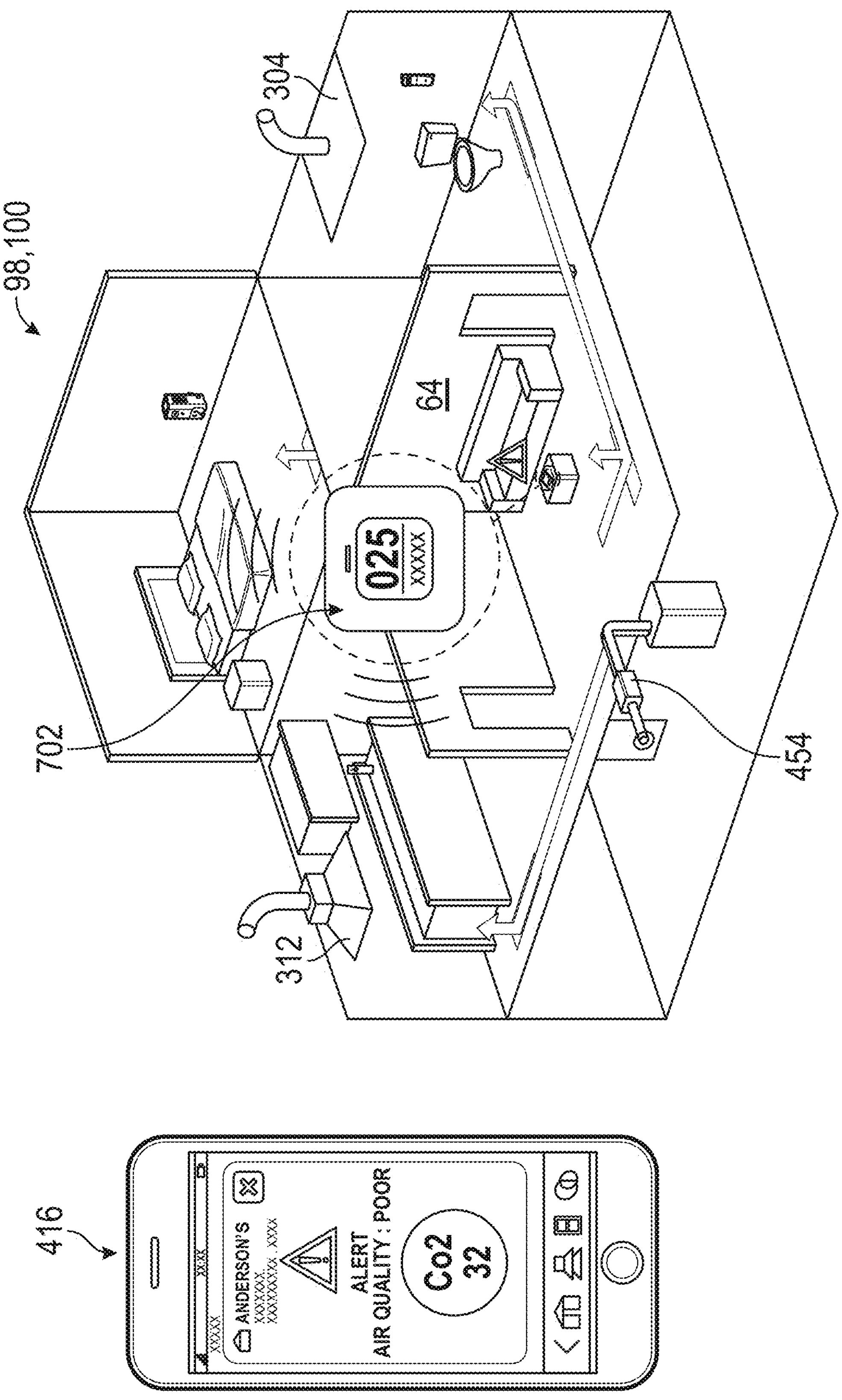

98,100
304
64
025
XXXXX
702
454
312
416
ANDERSON'S
ALERT
AIR QUALITY : POOR
Co2
32

FIG. 47

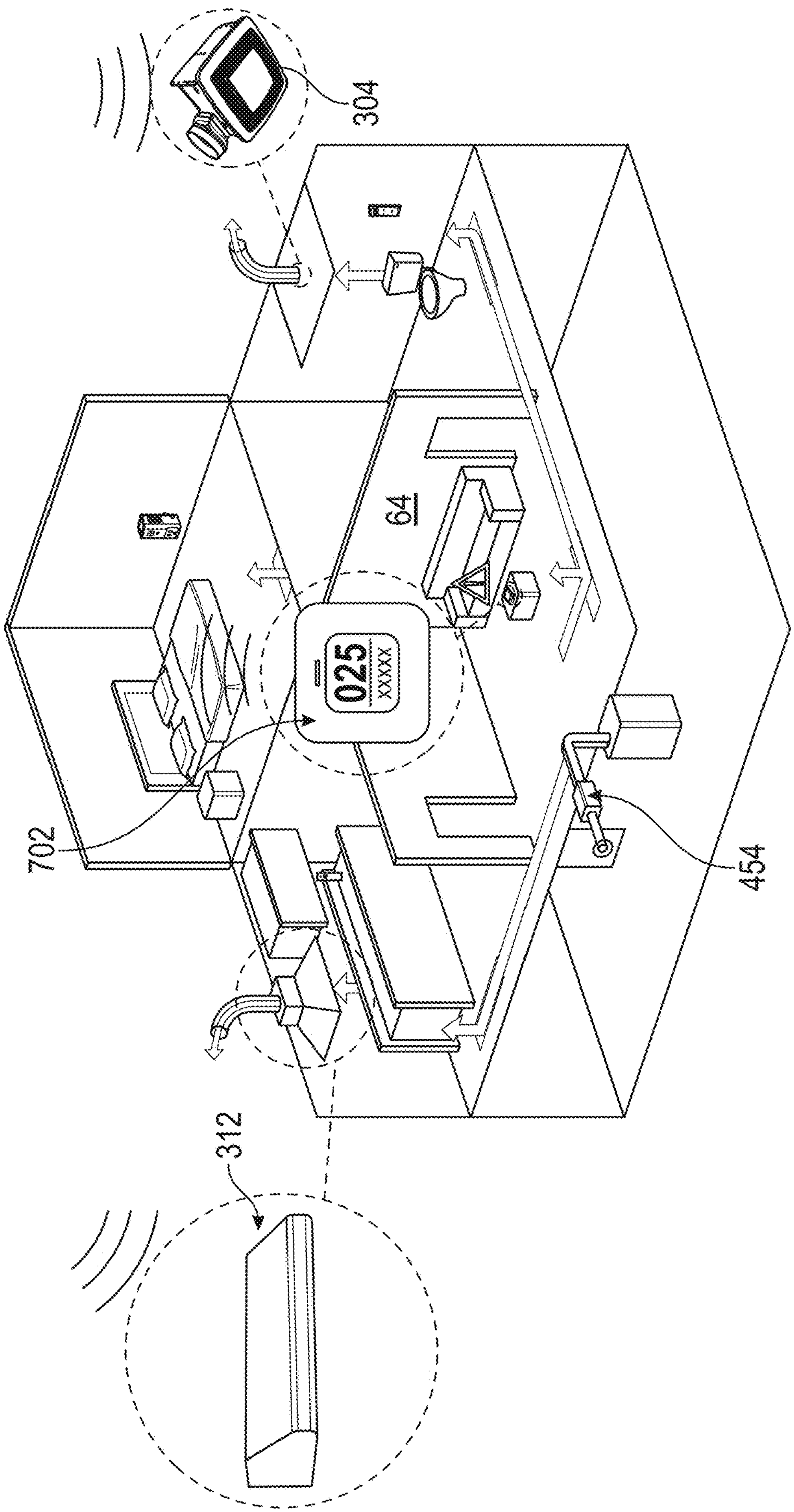


FIG. 48

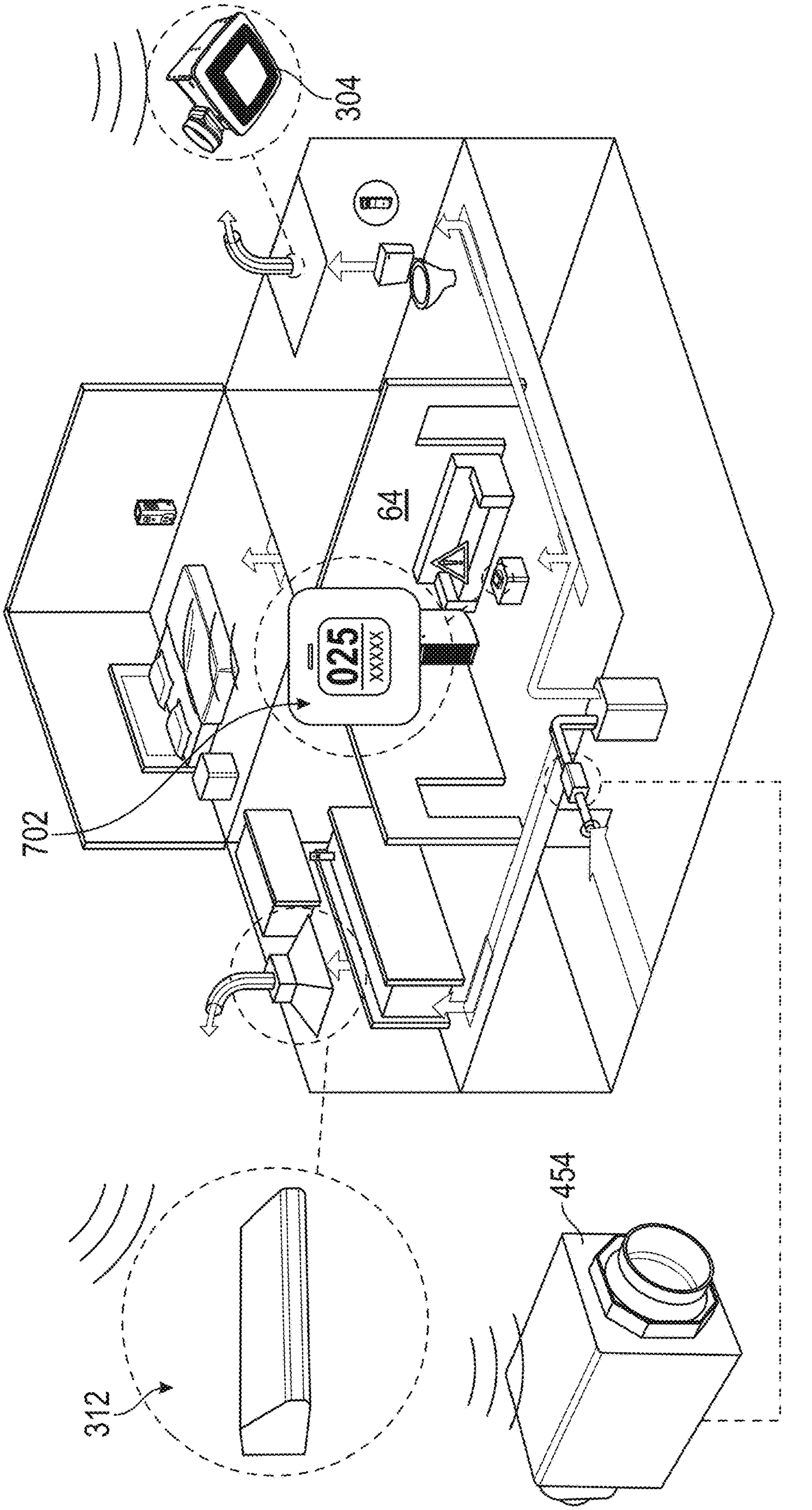


FIG. 49

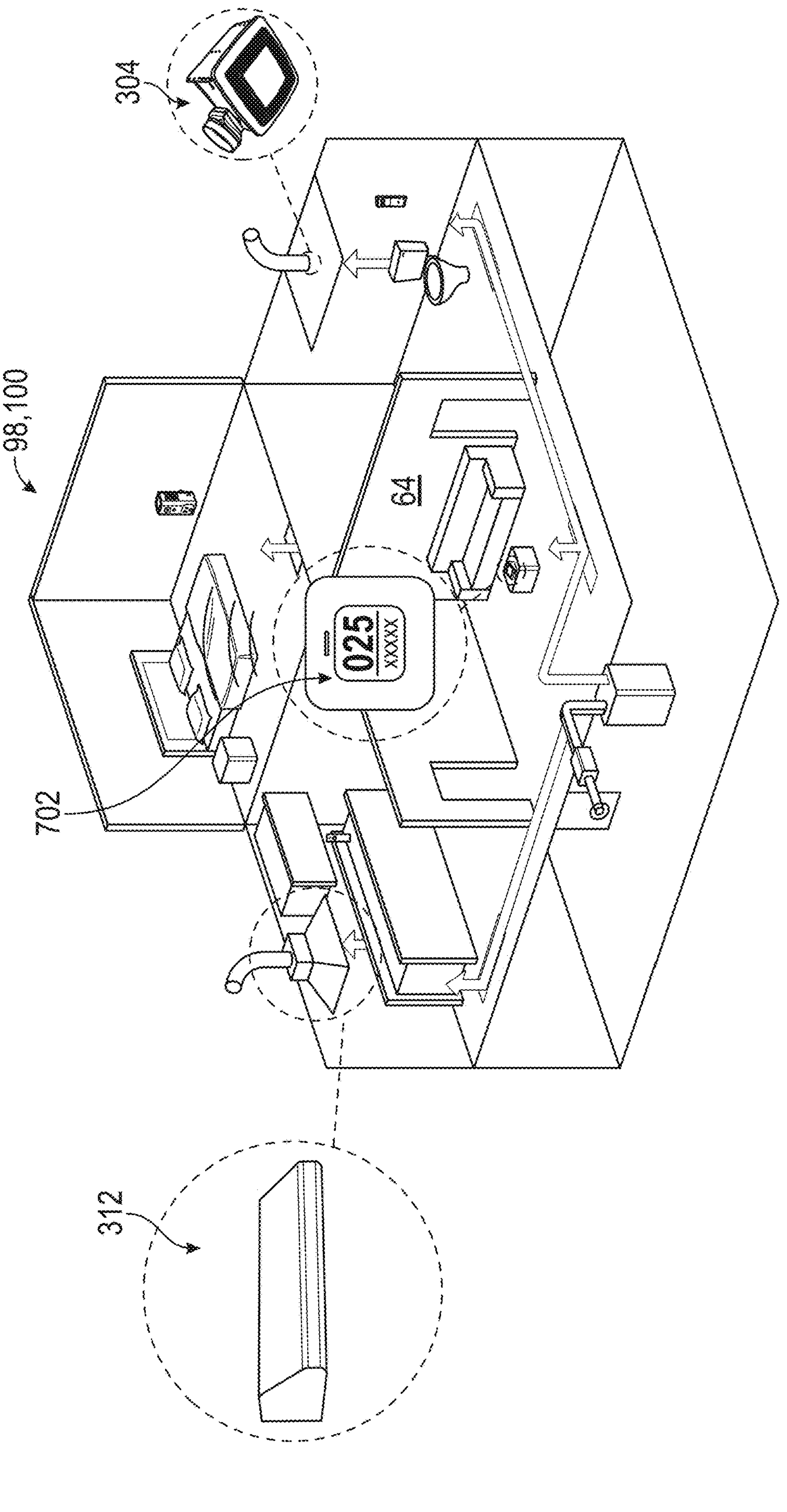


FIG. 50

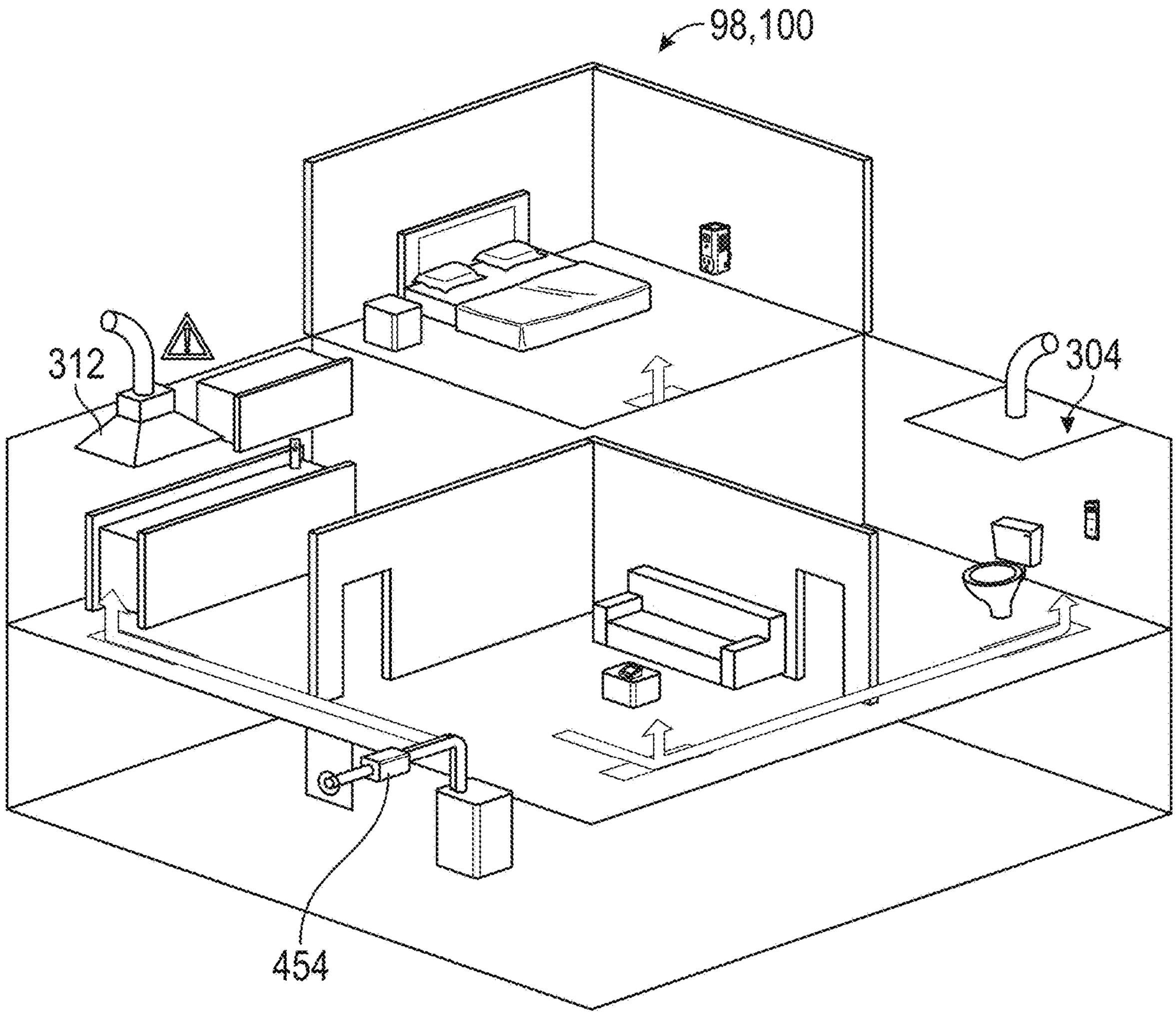


FIG. 51

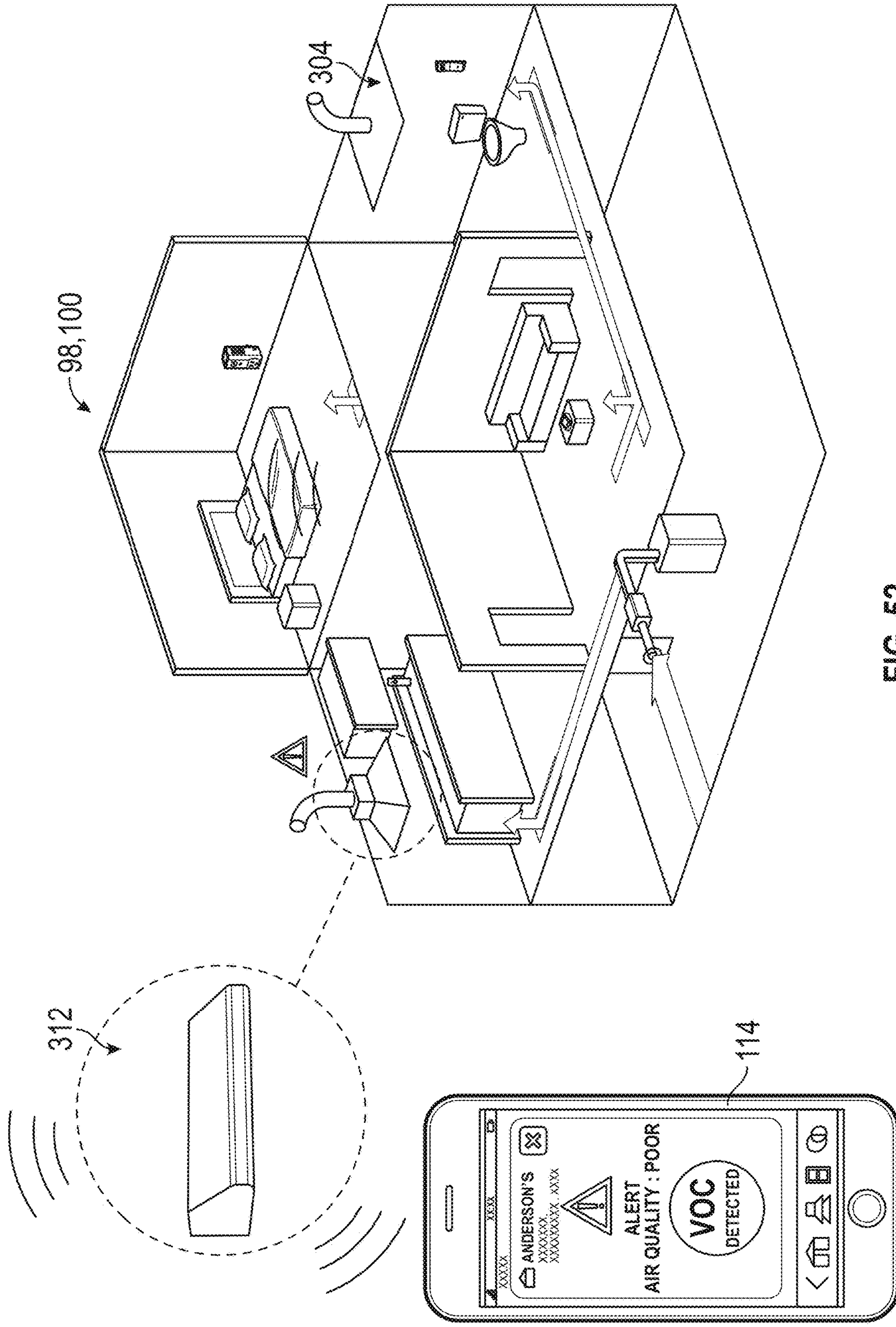

98,100
304
312
114
ANDERSON'S
ALERT
AIR QUALITY : POOR
VOC
DETECTED

FIG. 52

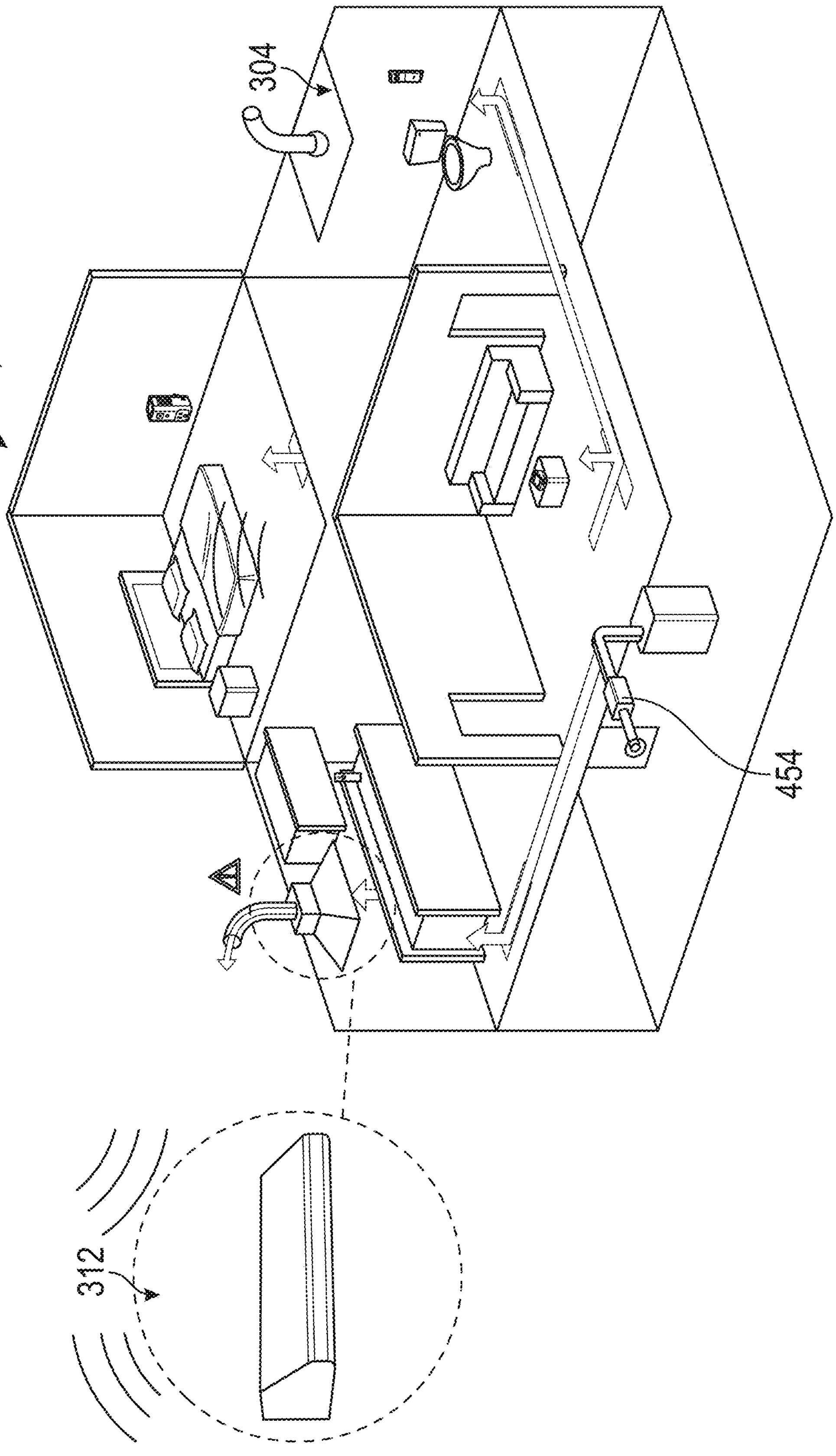


FIG. 53

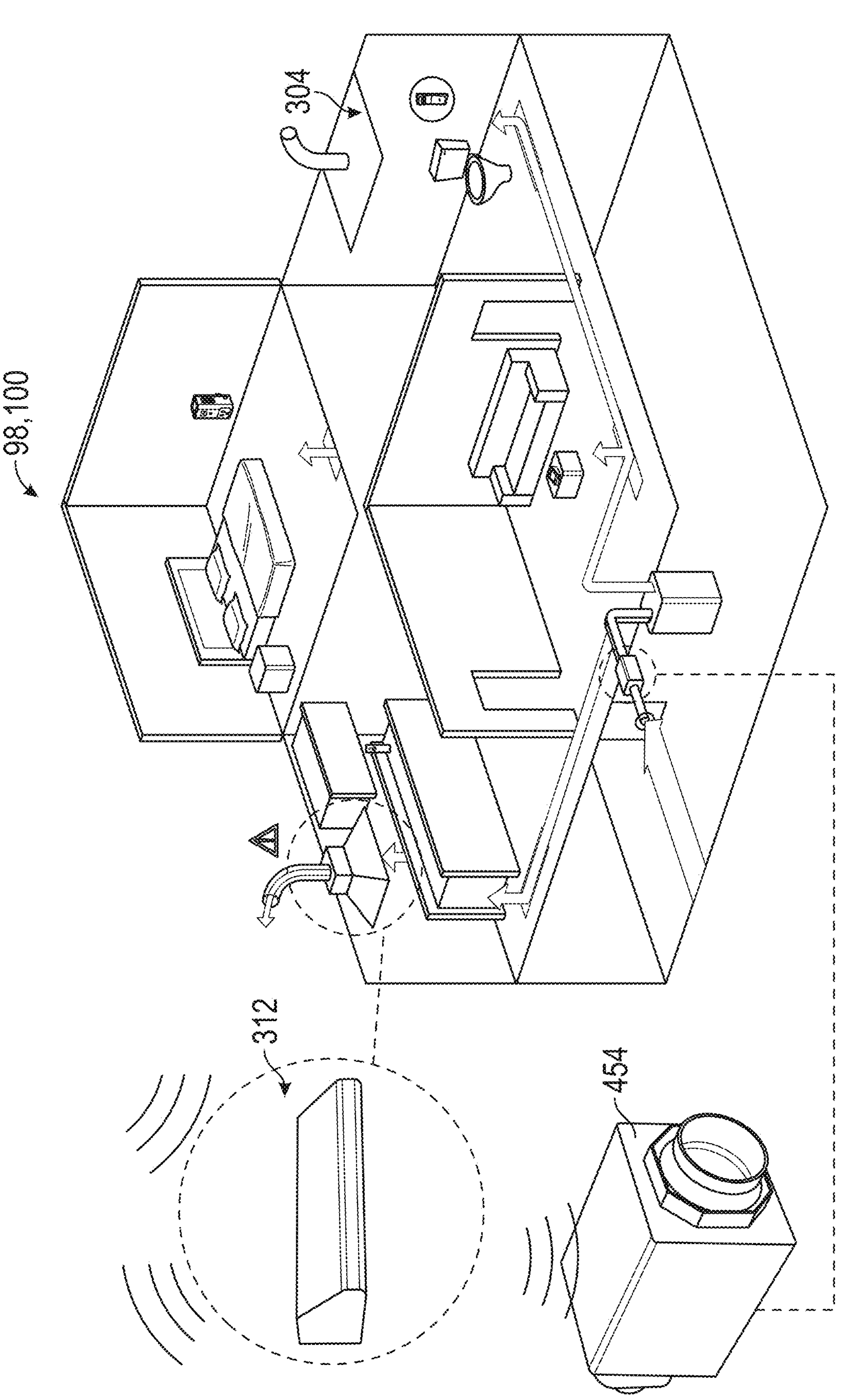


FIG. 54

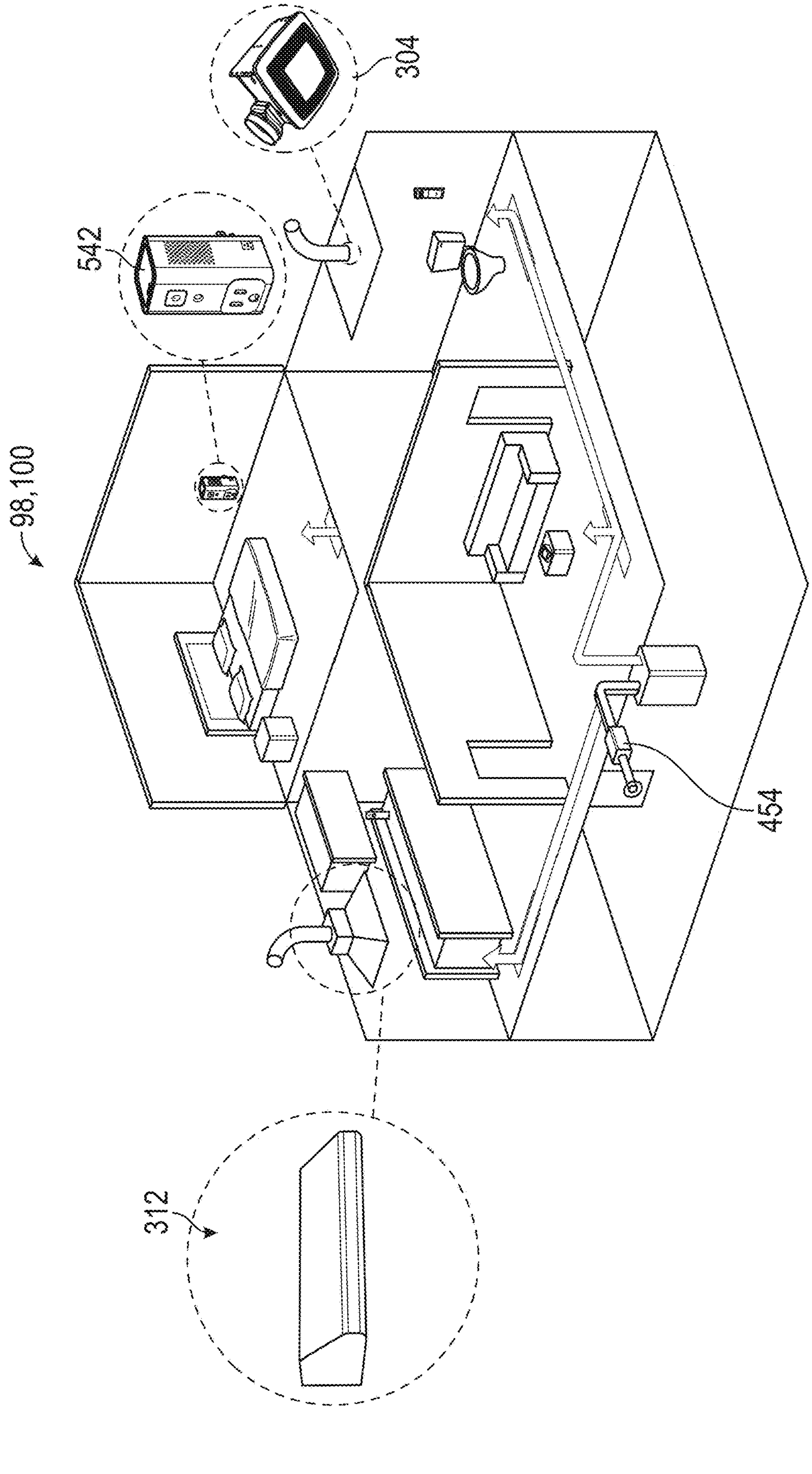


FIG. 55

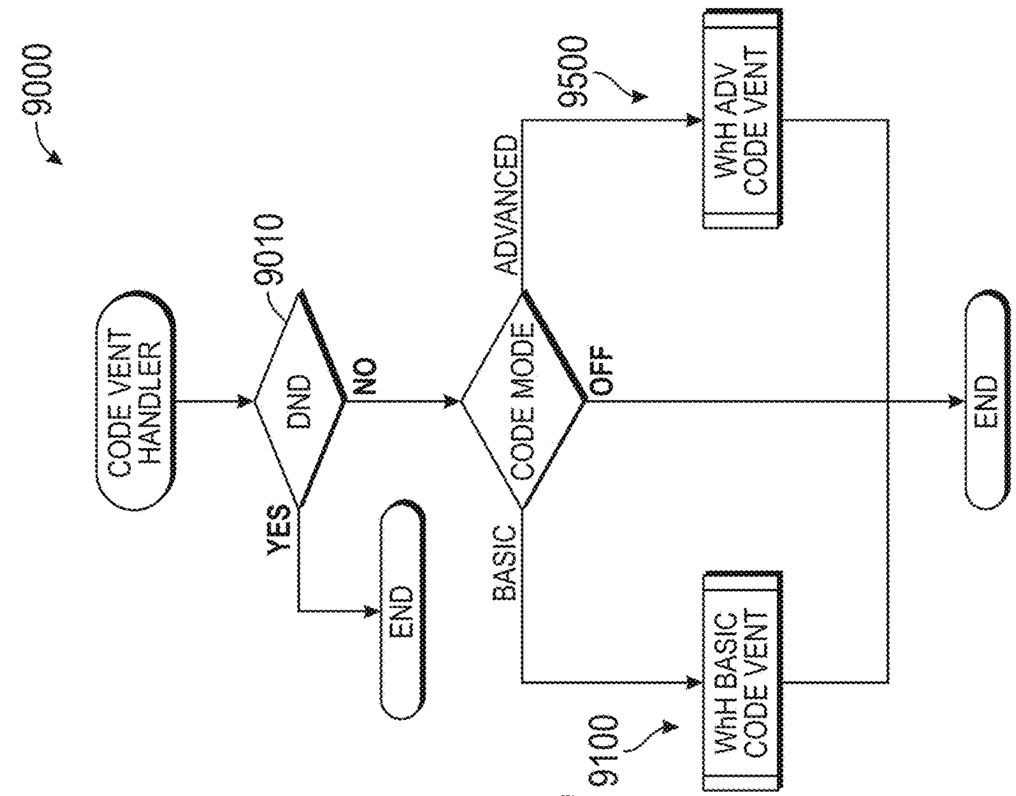

9000
CODE VENT HANDLER
9010
DND
YES
NO
END
CODE MODE
ADVANCED
OFF
BASIC
9500
WhH ADV CODE VENT
9100
WhH BASIC CODE VENT
END

FIG. 57

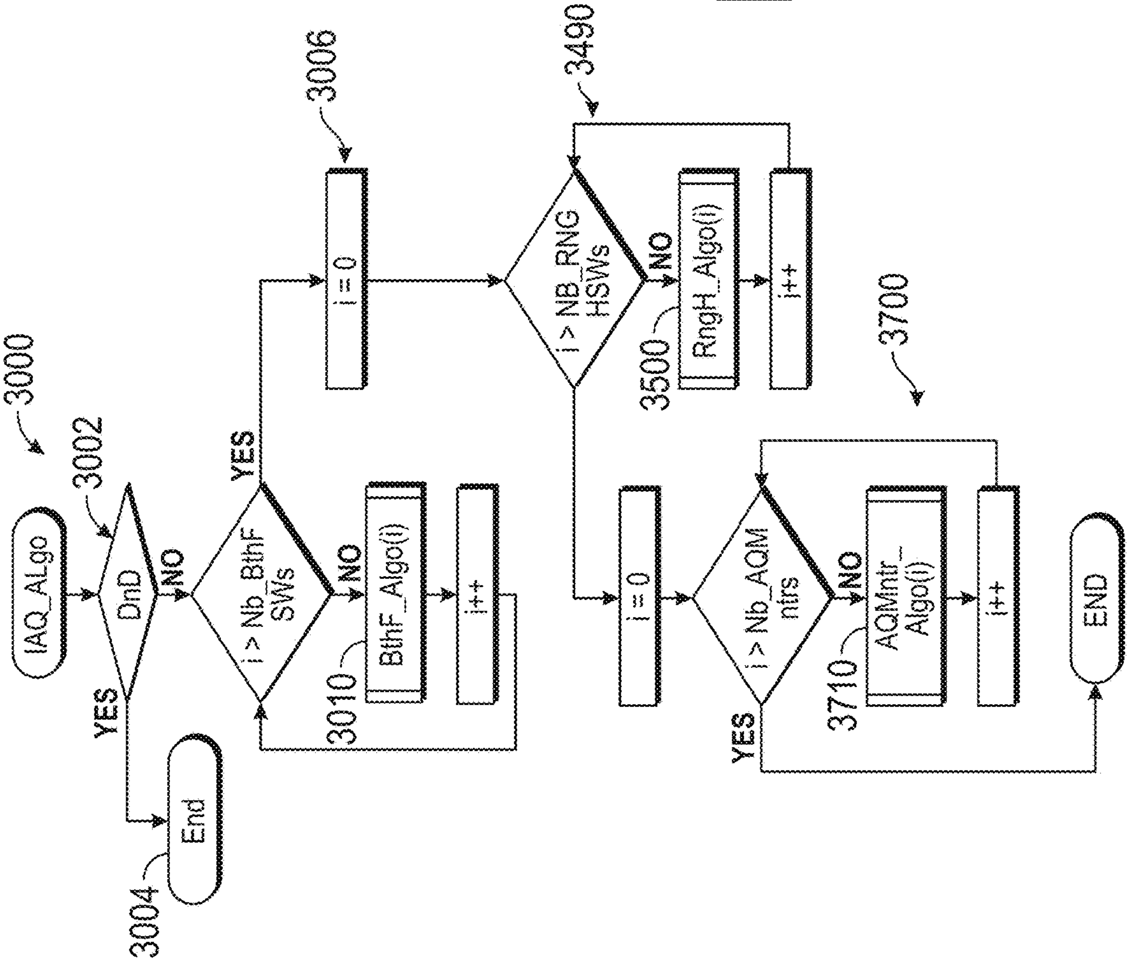
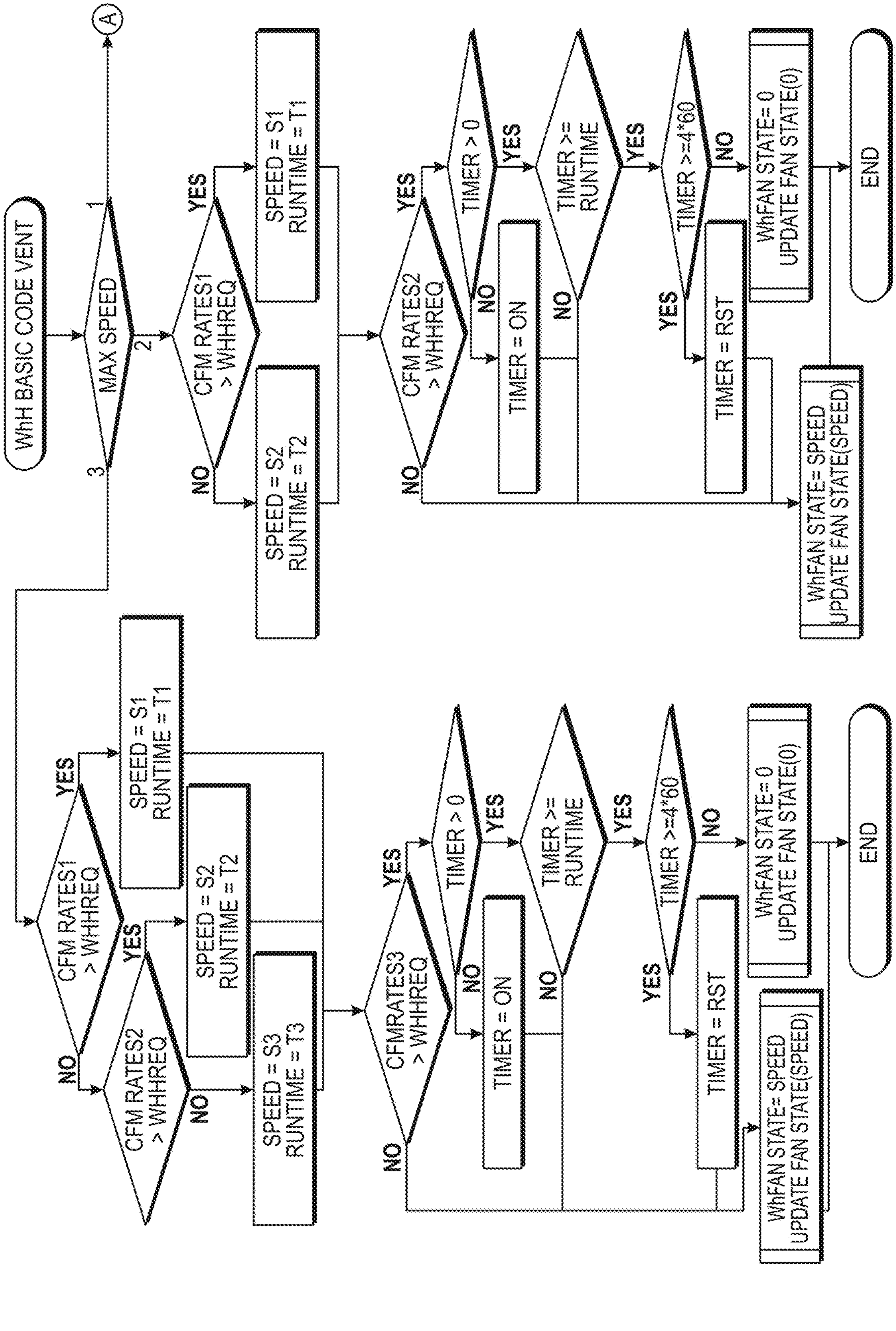

3000
IAQ_ALgo
3002
DnD
YES
NO
End
3004
i > Nb_BthF SWs
YES
NO
3010
BthF_Algo(i)
i++
3006
i = 0
3490
i > Nb_RNG HSWs
NO
3500
RngH_Algo(i)
i++
3700
i = 0
i > Nb_AQM ntrs
NO
YES
3710
AQMntr_ Algo(i)
i++
END

FIG. 56

FIG. 58A



WhH BASIC CODE VENT
MAX SPEED
1
2
3
A
CFM RATES1 > WHHREQ
YES
NO
SPEED = S1
RUNTIME = T1
SPEED = S2
RUNTIME = T2
CFM RATES2 > WHHREQ
TIMER > 0
TIMER = ON
TIMER >= RUNTIME
TIMER >=4*60
TIMER = RST
WhFAN STATE= 0
UPDATE FAN STATE(0)
WhFAN STATE= SPEED
UPDATE FAN STATE(SPEED)
END
SPEED = S3
RUNTIME = T3
CFMRATES3 > WHHREQ

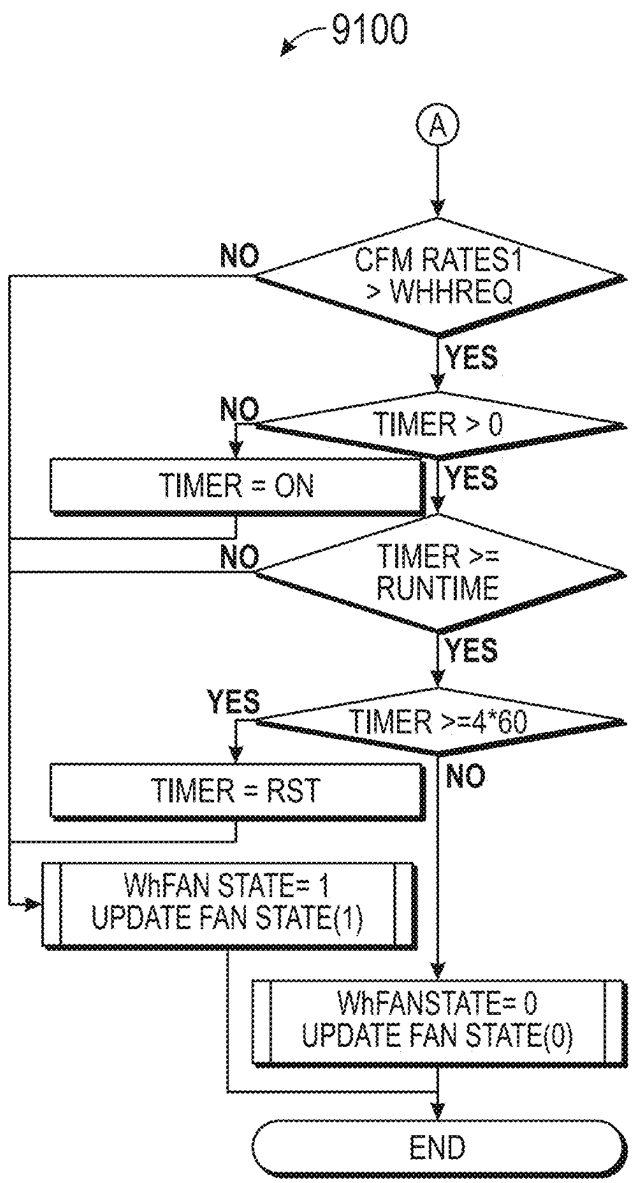
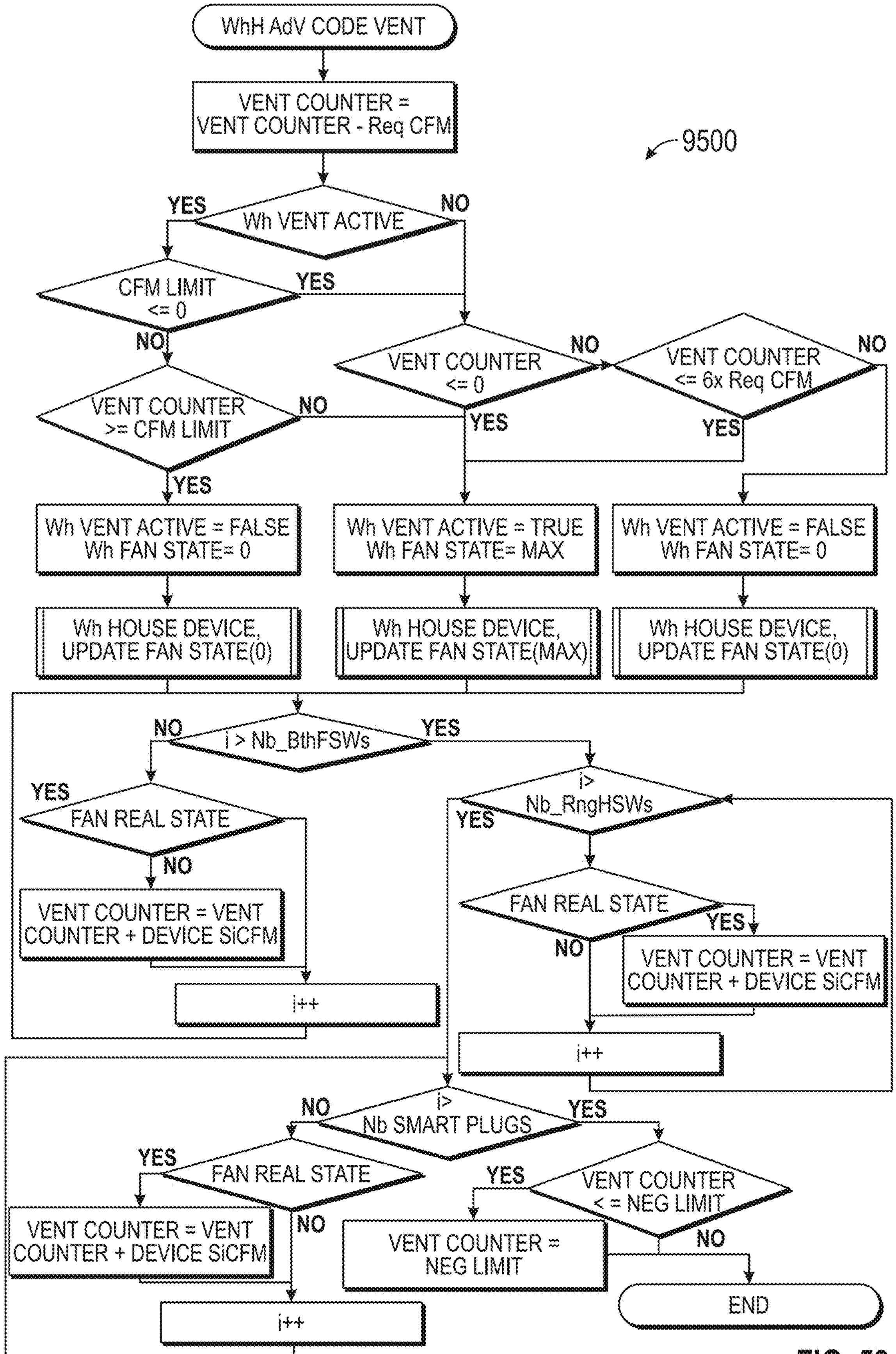

9100
A
CFM RATES1 > WHHREQ
NO
YES
TIMER > 0
NO
YES
TIMER = ON
TIMER >= RUNTIME
NO
YES
TIMER >=4*60
YES
NO
TIMER = RST
WhFAN STATE= 1
UPDATE FAN STATE(1)
WhFANSTATE= 0
UPDATE FAN STATE(0)
END

FIG. 58B



WhH AdV CODE VENT
VENT COUNTER = VENT COUNTER - Req CFM
9500
Wh VENT ACTIVE
YES
NO
CFM LIMIT <= 0
YES
NO
VENT COUNTER <= 0
NO
VENT COUNTER <= 6x Req CFM
NO
VENT COUNTER >= CFM LIMIT
NO
YES
YES
YES
Wh VENT ACTIVE = FALSE Wh FAN STATE= 0
Wh VENT ACTIVE = TRUE Wh FAN STATE= MAX
Wh VENT ACTIVE = FALSE Wh FAN STATE= 0
Wh HOUSE DEVICE, UPDATE FAN STATE(0)
Wh HOUSE DEVICE, UPDATE FAN STATE(MAX)
Wh HOUSE DEVICE, UPDATE FAN STATE(0)
i > Nb_BthFSWs
NO
YES
FAN REAL STATE
YES
NO
VENT COUNTER = VENT COUNTER + DEVICE SiCFM
i++
i> Nb_RngHSWs
YES
FAN REAL STATE
YES
NO
VENT COUNTER = VENT COUNTER + DEVICE SiCFM
i++
i> Nb SMART PLUGS
NO
YES
FAN REAL STATE
YES
NO
VENT COUNTER = VENT COUNTER + DEVICE SiCFM
i++
VENT COUNTER < = NEG LIMIT
YES
NO
VENT COUNTER = NEG LIMIT
END

FIG. 59

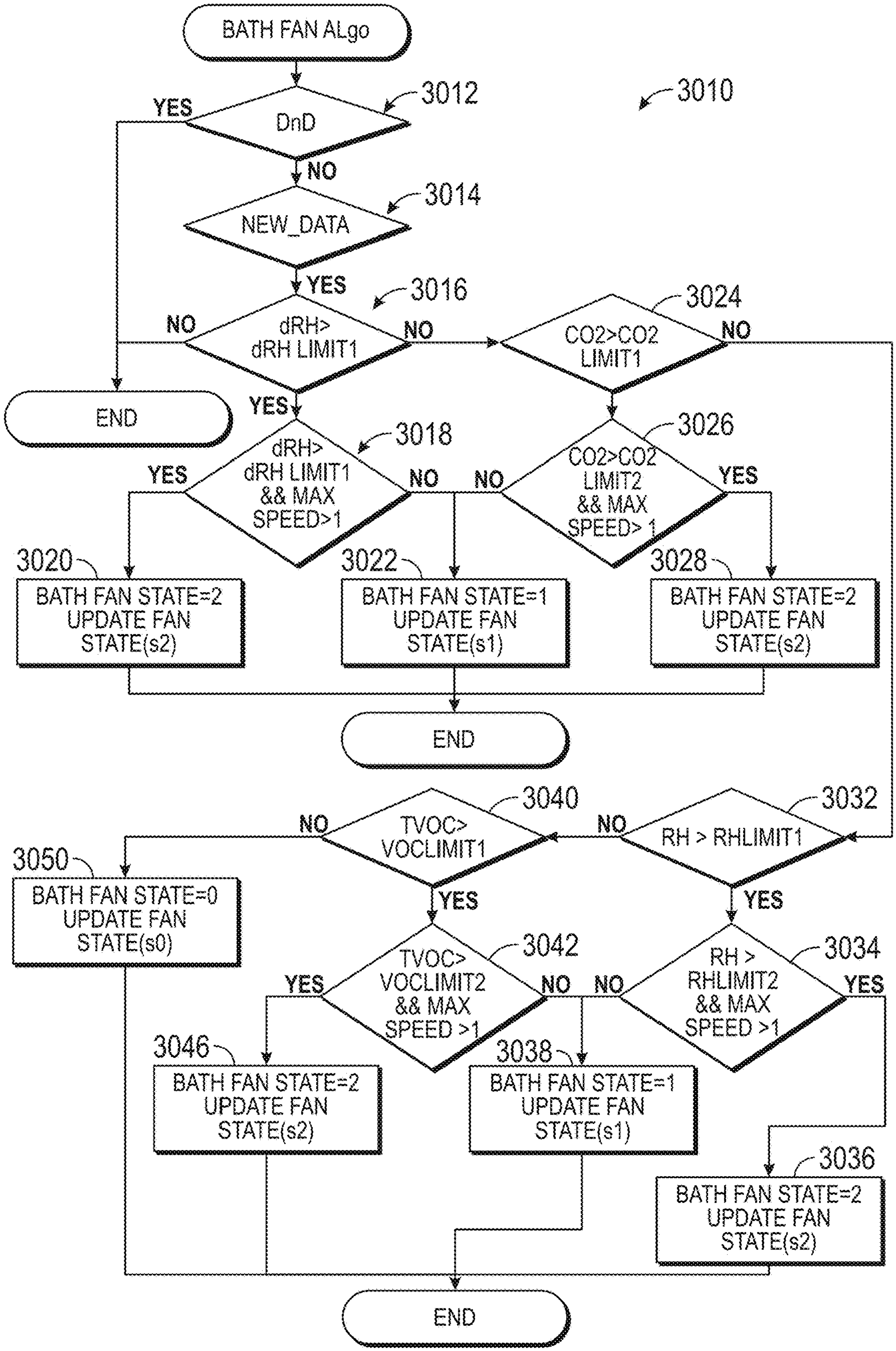

BATH FAN ALgo
3010
3012
DnD
YES
NO
3014
NEW_DATA
YES
3016
dRH> dRH LIMIT1
NO
NO
YES
END
3018
dRH> dRH LIMIT1 && MAX SPEED>1
YES
NO
3024
CO2>CO2 LIMIT1
NO
3026
CO2>CO2 LIMIT2 && MAX SPEED> 1
NO
YES
3020
BATH FAN STATE=2 UPDATE FAN STATE(s2)
3022
BATH FAN STATE=1 UPDATE FAN STATE(s1)
3028
BATH FAN STATE=2 UPDATE FAN STATE(s2)
END
3040
TVOC> VOCLIMIT1
NO
3032
RH > RHLIMIT1
NO
3050
BATH FAN STATE=0 UPDATE FAN STATE(s0)
YES
YES
3042
TVOC> VOCLIMIT2 && MAX SPEED >1
YES
NO
NO
3034
RH > RHLIMIT2 && MAX SPEED >1
YES
3046
BATH FAN STATE=2 UPDATE FAN STATE(s2)
3038
BATH FAN STATE=1 UPDATE FAN STATE(s1)
3036
BATH FAN STATE=2 UPDATE FAN STATE(s2)
END

FIG. 60

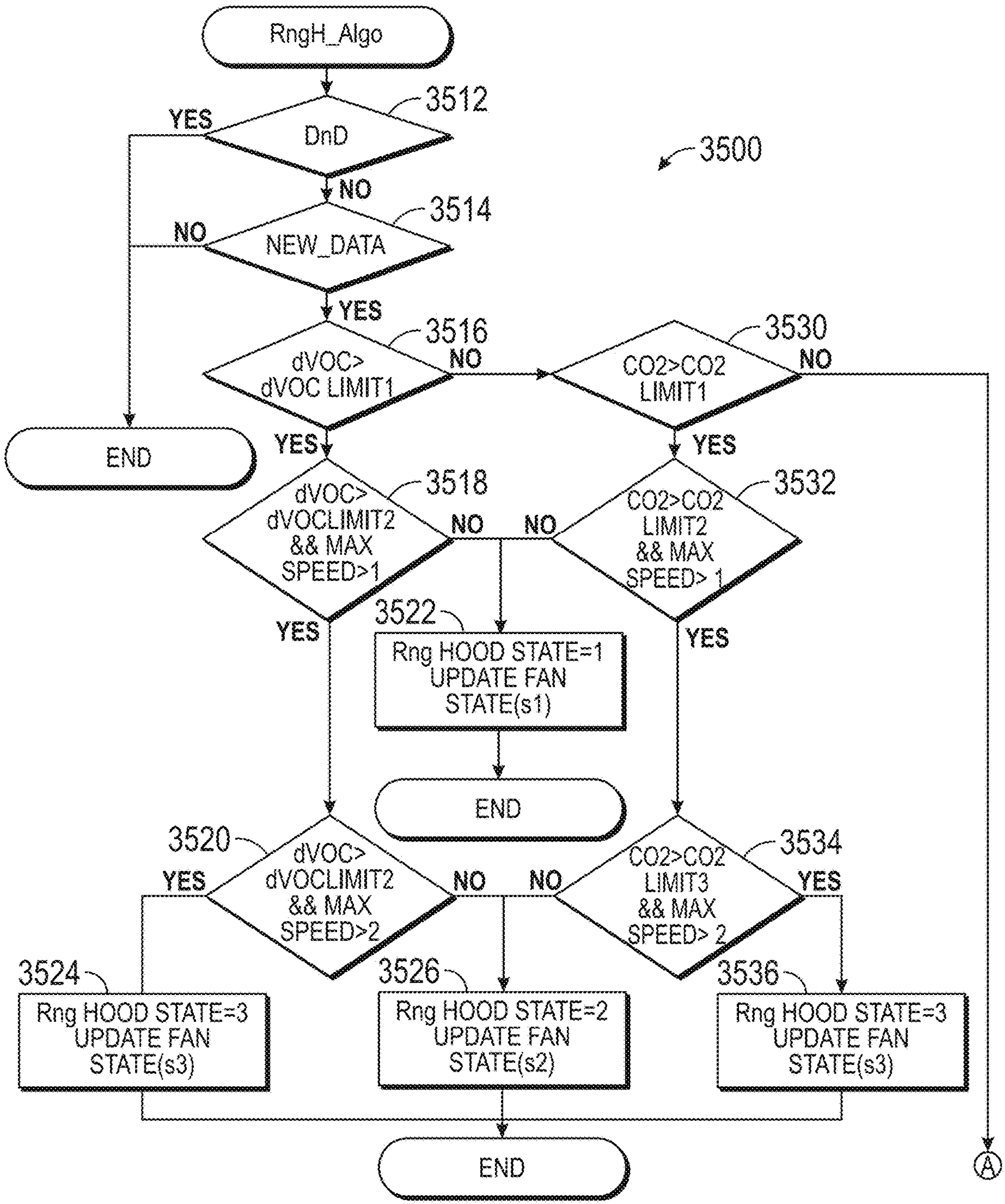

RngH_Algo
3512
DnD
YES
NO
3500
3514
NEW_DATA
NO
YES
3516
dVOC>
dVOC LIMIT1
NO
3530
CO2>CO2
LIMIT1
NO
END
YES
YES
3518
dVOC>
dVOCLIMIT2
&& MAX
SPEED>1
NO
NO
3532
CO2>CO2
LIMIT2
&& MAX
SPEED> 1
YES
3522
Rng HOOD STATE=1
UPDATE FAN
STATE(s1)
YES
END
3520
dVOC>
dVOCLIMIT2
&& MAX
SPEED>2
YES
NO
NO
3534
CO2>CO2
LIMIT3
&& MAX
SPEED> 2
YES
3524
Rng HOOD STATE=3
UPDATE FAN
STATE(s3)
3526
Rng HOOD STATE=2
UPDATE FAN
STATE(s2)
3536
Rng HOOD STATE=3
UPDATE FAN
STATE(s3)
END
A

FIG. 61A

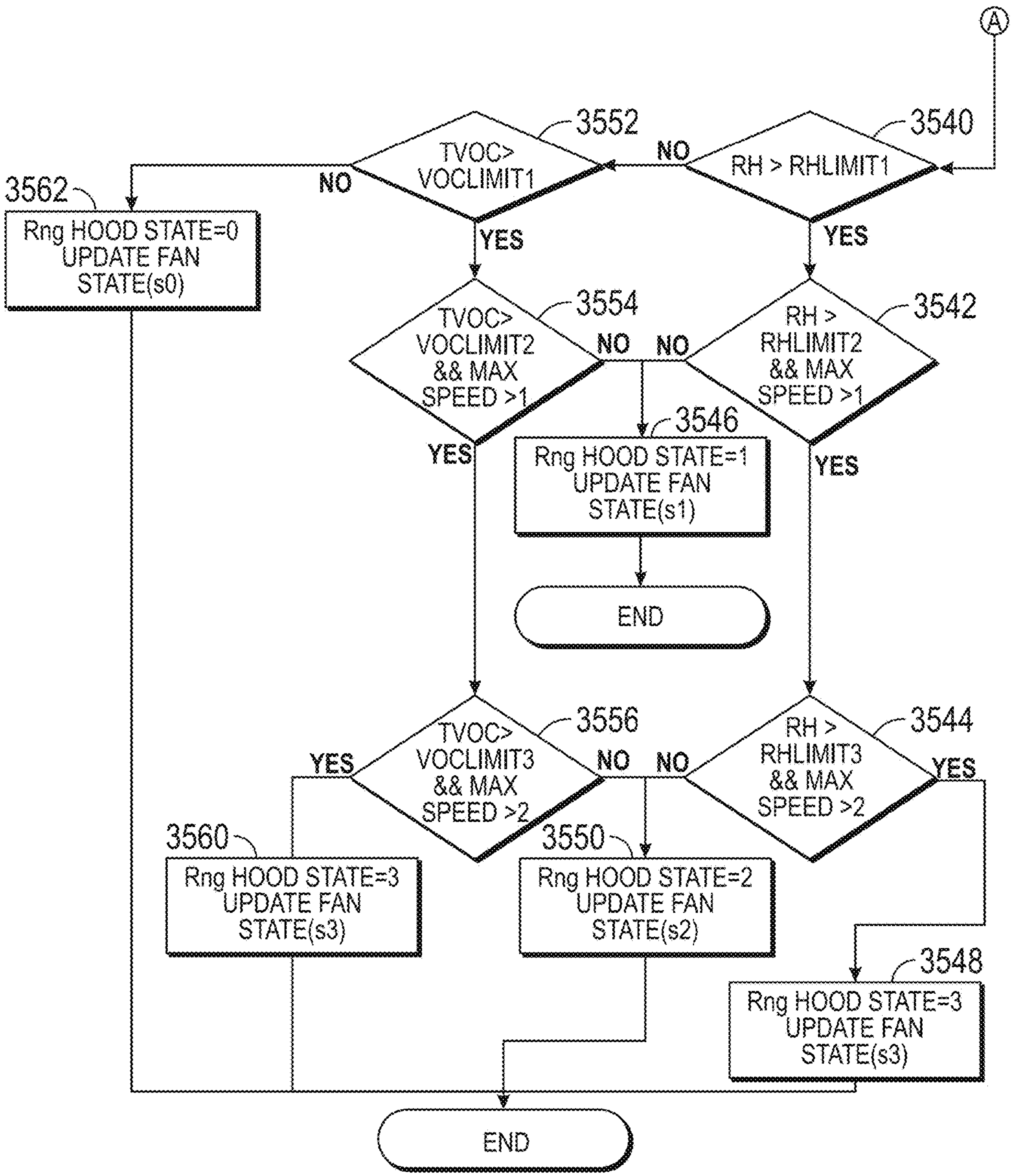

A
3540
RH > RHLIMIT1
NO
YES
3552
TVOC> VOCLIMIT1
NO
YES
3562
Rng HOOD STATE=0 UPDATE FAN STATE(s0)
3554
TVOC> VOCLIMIT2 && MAX SPEED >1
NO
YES
3542
RH > RHLIMIT2 && MAX SPEED >1
NO
YES
3546
Rng HOOD STATE=1 UPDATE FAN STATE(s1)
END
3556
TVOC> VOCLIMIT3 && MAX SPEED >2
YES
NO
3544
RH > RHLIMIT3 && MAX SPEED >2
NO
YES
3560
Rng HOOD STATE=3 UPDATE FAN STATE(s3)
3550
Rng HOOD STATE=2 UPDATE FAN STATE(s2)
3548
Rng HOOD STATE=3 UPDATE FAN STATE(s3)
END

FIG. 61B

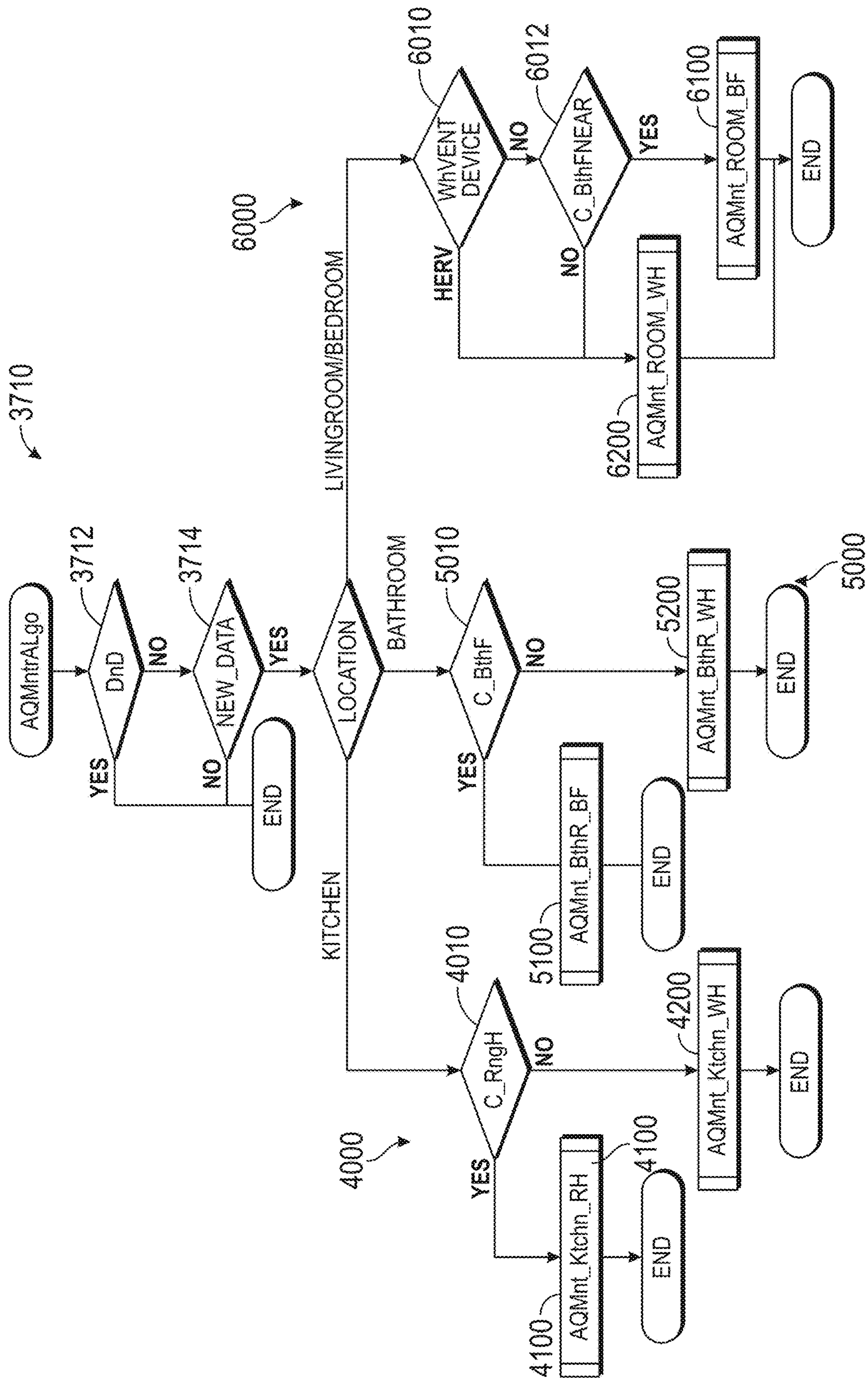
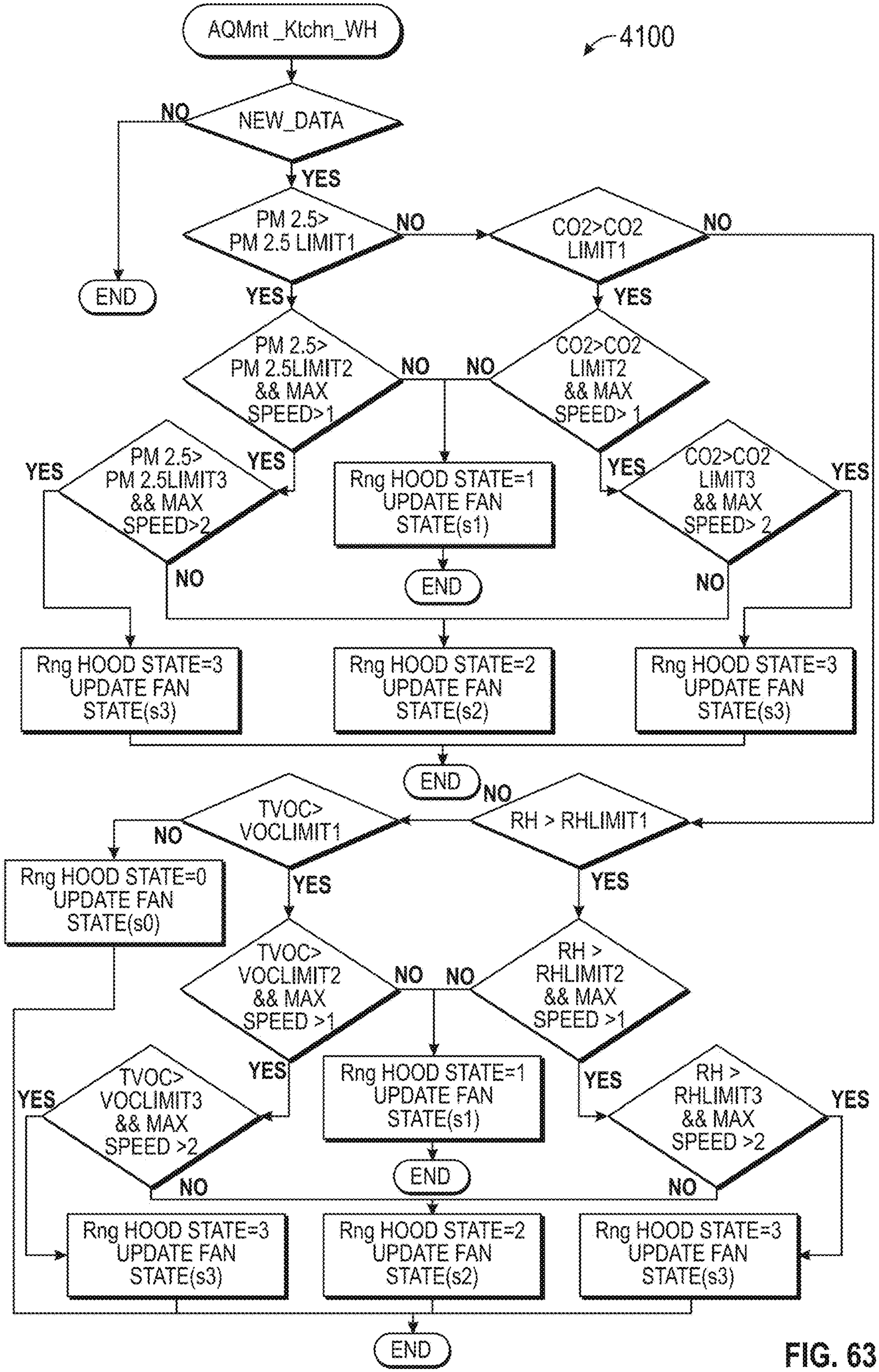

3710
AQMntrALgo
3712
DnD
YES
NO
3714
NEW_DATA
NO
YES
END
LOCATION
KITCHEN
BATHROOM
LIVINGROOM/BEDROOM
4000
4010
C_RngH
YES
NO
4100
AQMnt_Ktchn_RH
4100
END
4200
AQMnt_Ktchn_WH
END
5010
C_BthF
YES
NO
5100
AQMnt_BthR_BF
END
5200
AQMnt_BthR_WH
END
5000
6000
6010
WhVENT DEVICE
HERV
NO
6012
C_BthFNEAR
NO
YES
6100
AQMnt_ROOM_BF
END
6200
AQMnt_ROOM_WH

FIG. 62



4100
AQMnt _Ktchn_WH
NEW_DATA
NO
YES
END
PM 2.5> PM 2.5 LIMIT1
CO2>CO2 LIMIT1
PM 2.5> PM 2.5LIMIT2 && MAX SPEED>1
CO2>CO2 LIMIT2 && MAX SPEED> 1
PM 2.5> PM 2.5LIMIT3 && MAX SPEED>2
CO2>CO2 LIMIT3 && MAX SPEED> 2
Rng HOOD STATE=1 UPDATE FAN STATE(s1)
Rng HOOD STATE=3 UPDATE FAN STATE(s3)
Rng HOOD STATE=2 UPDATE FAN STATE(s2)
TVOC> VOCLIMIT1
RH > RHLIMIT1
Rng HOOD STATE=0 UPDATE FAN STATE(s0)
TVOC> VOCLIMIT2 && MAX SPEED >1
RH > RHLIMIT2 && MAX SPEED >1
TVOC> VOCLIMIT3 && MAX SPEED >2
RH > RHLIMIT3 && MAX SPEED >2

FIG. 63

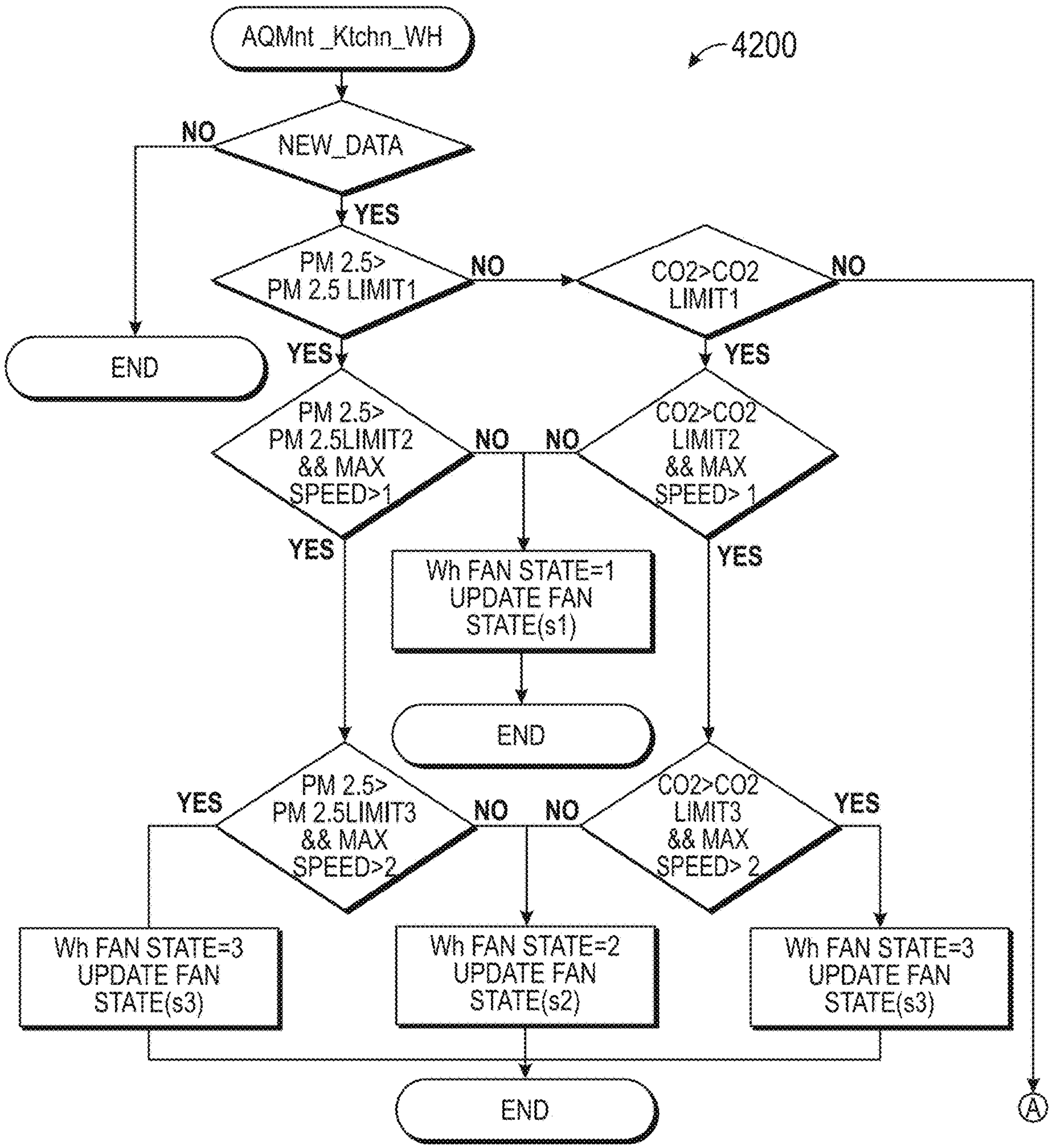

4200
AQMnt _Ktchn_WH
NEW_DATA
NO
YES
END
PM 2.5>
PM 2.5 LIMIT1
NO
CO2>CO2
LIMIT1
NO
YES
YES
PM 2.5>
PM 2.5LIMIT2
&& MAX
SPEED>1
NO
NO
CO2>CO2
LIMIT2
&& MAX
SPEED> 1
YES
YES
Wh FAN STATE=1
UPDATE FAN
STATE(s1)
END
PM 2.5>
PM 2.5LIMIT3
&& MAX
SPEED>2
YES
NO
NO
CO2>CO2
LIMIT3
&& MAX
SPEED> 2
YES
Wh FAN STATE=3
UPDATE FAN
STATE(s3)
Wh FAN STATE=2
UPDATE FAN
STATE(s2)
Wh FAN STATE=3
UPDATE FAN
STATE(s3)
END
A

FIG. 64A

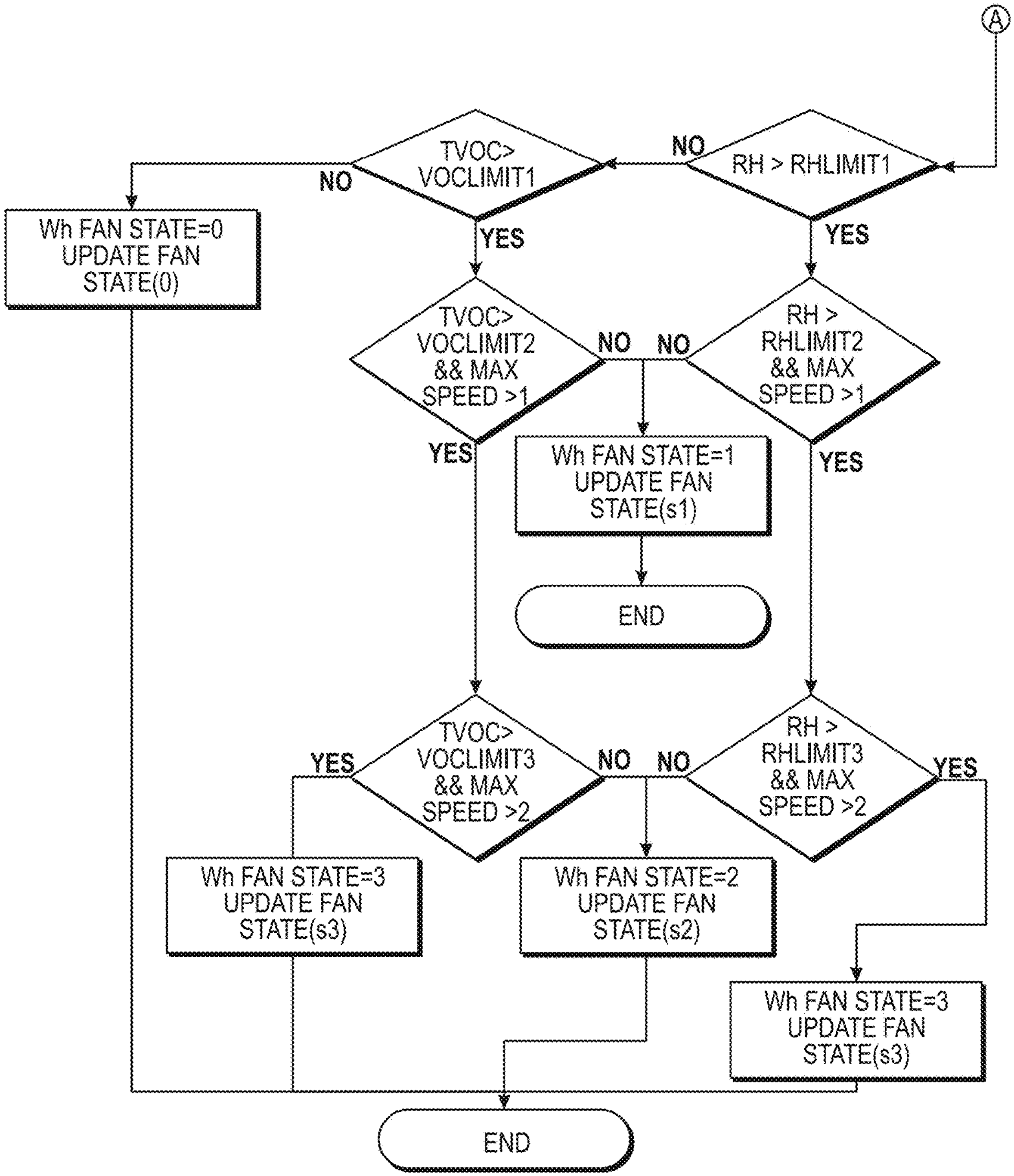

A
RH > RHLIMIT1
NO
YES
TVOC>
VOCLIMIT1
NO
YES
Wh FAN STATE=0
UPDATE FAN
STATE(0)
TVOC>
VOCLIMIT2
&& MAX
SPEED >1
NO
YES
RH >
RHLIMIT2
&& MAX
SPEED >1
NO
YES
Wh FAN STATE=1
UPDATE FAN
STATE(s1)
END
TVOC>
VOCLIMIT3
&& MAX
SPEED >2
YES
NO
RH >
RHLIMIT3
&& MAX
SPEED >2
NO
YES
Wh FAN STATE=3
UPDATE FAN
STATE(s3)
Wh FAN STATE=2
UPDATE FAN
STATE(s2)
Wh FAN STATE=3
UPDATE FAN
STATE(s3)
END

FIG. 64B

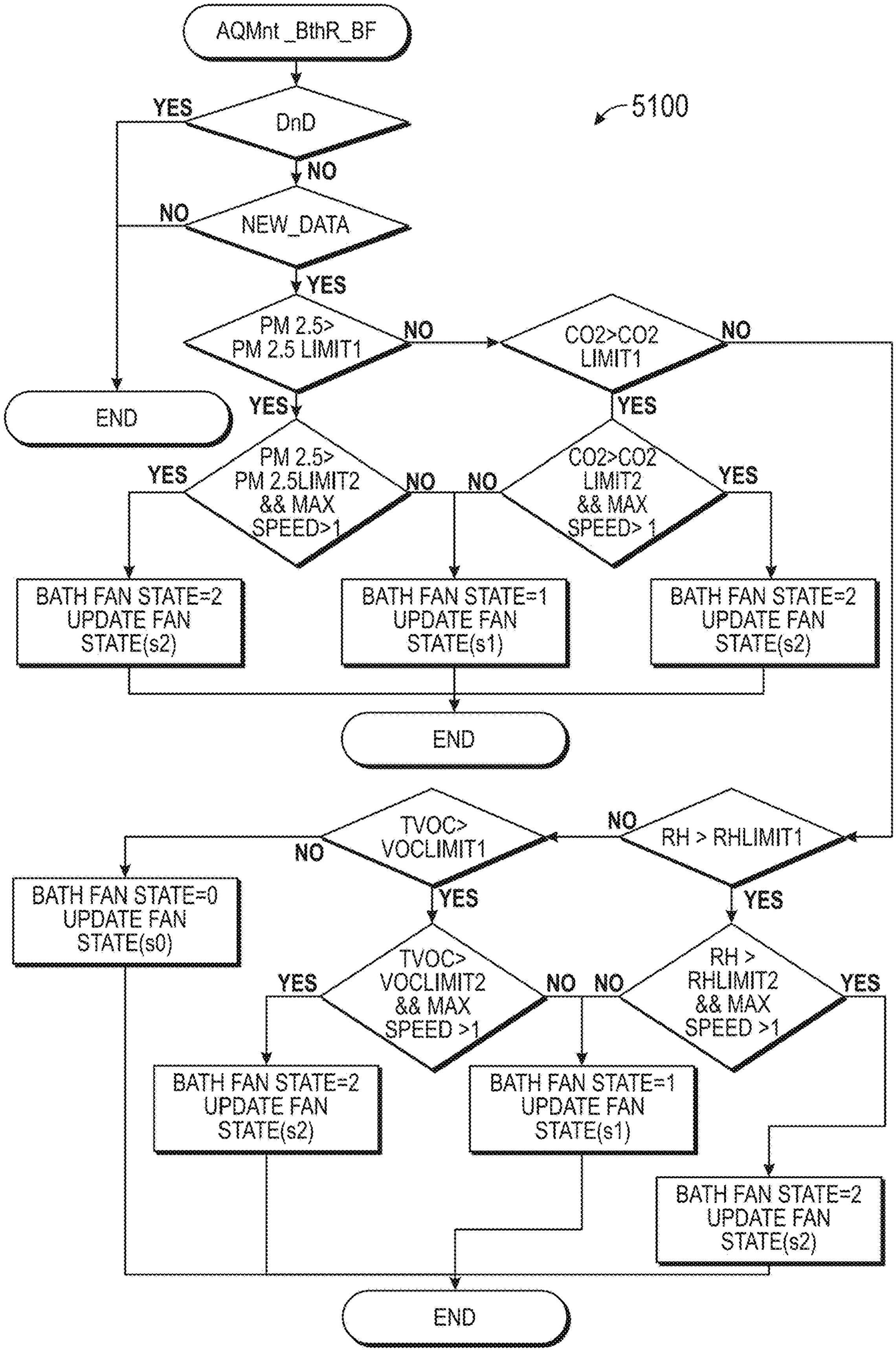

AQMnt _BthR_BF
5100
DnD
YES
NO
NEW_DATA
NO
YES
PM 2.5> PM 2.5 LIMIT1
NO
CO2>CO2 LIMIT1
NO
END
YES
YES
PM 2.5> PM 2.5LIMIT2 && MAX SPEED>1
YES
NO
NO
CO2>CO2 LIMIT2 && MAX SPEED> 1
YES
BATH FAN STATE=2 UPDATE FAN STATE(s2)
BATH FAN STATE=1 UPDATE FAN STATE(s1)
BATH FAN STATE=2 UPDATE FAN STATE(s2)
END
TVOC> VOCLIMIT1
NO
NO
RH > RHLIMIT1
BATH FAN STATE=0 UPDATE FAN STATE(s0)
YES
YES
TVOC> VOCLIMIT2 && MAX SPEED >1
YES
NO
NO
RH > RHLIMIT2 && MAX SPEED >1
YES
BATH FAN STATE=2 UPDATE FAN STATE(s2)
BATH FAN STATE=1 UPDATE FAN STATE(s1)
BATH FAN STATE=2 UPDATE FAN STATE(s2)
END

FIG. 65

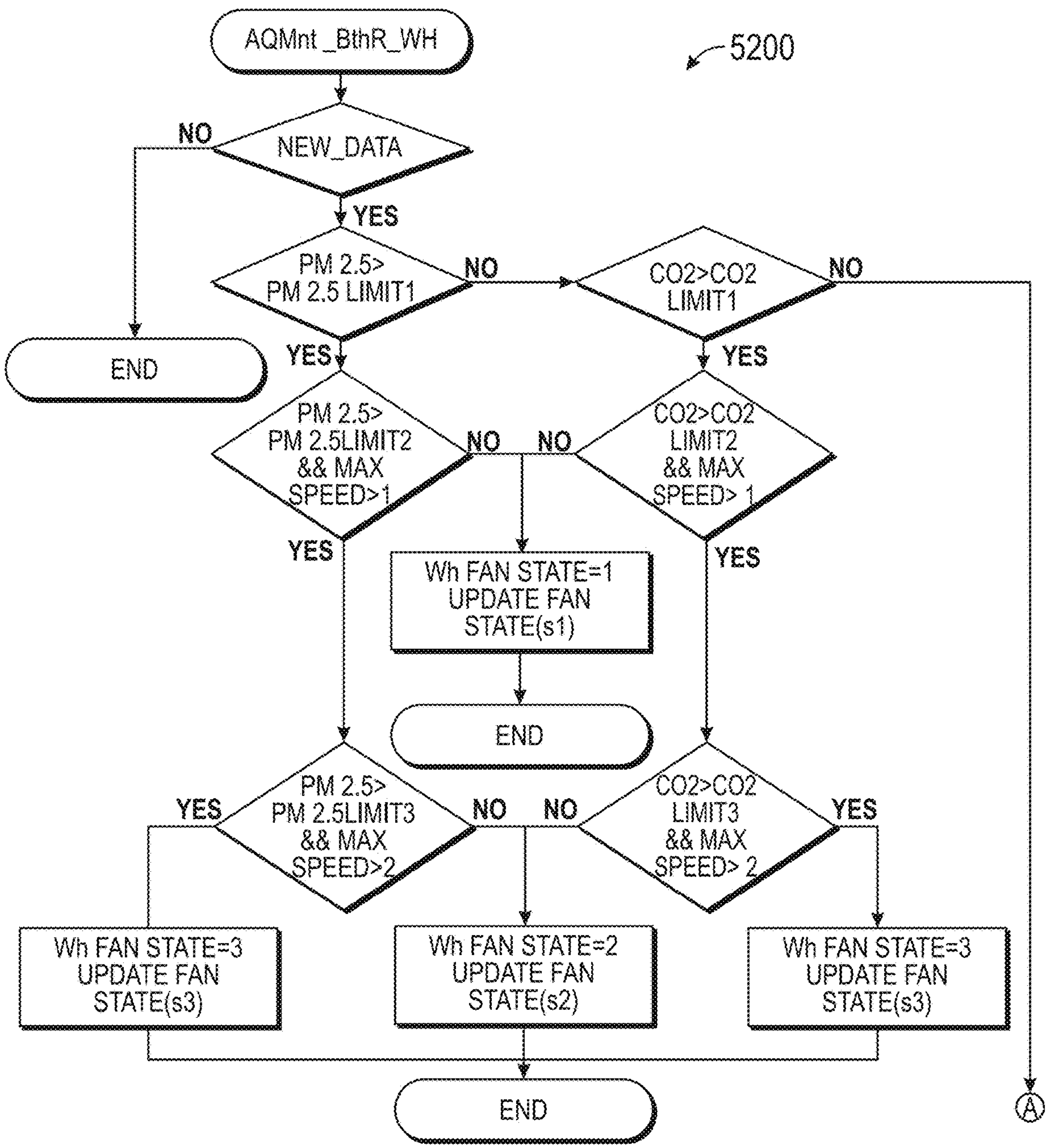

5200
AQMnt _BthR_WH
NEW_DATA
NO
YES
END
PM 2.5>
PM 2.5 LIMIT1
NO
CO2>CO2
LIMIT1
NO
YES
YES
PM 2.5>
PM 2.5LIMIT2
&& MAX
SPEED>1
NO
NO
CO2>CO2
LIMIT2
&& MAX
SPEED> 1
YES
YES
Wh FAN STATE=1
UPDATE FAN
STATE(s1)
END
PM 2.5>
PM 2.5LIMIT3
&& MAX
SPEED>2
YES
NO
NO
CO2>CO2
LIMIT3
&& MAX
SPEED> 2
YES
Wh FAN STATE=3
UPDATE FAN
STATE(s3)
Wh FAN STATE=2
UPDATE FAN
STATE(s2)
Wh FAN STATE=3
UPDATE FAN
STATE(s3)
END
A

FIG. 66A

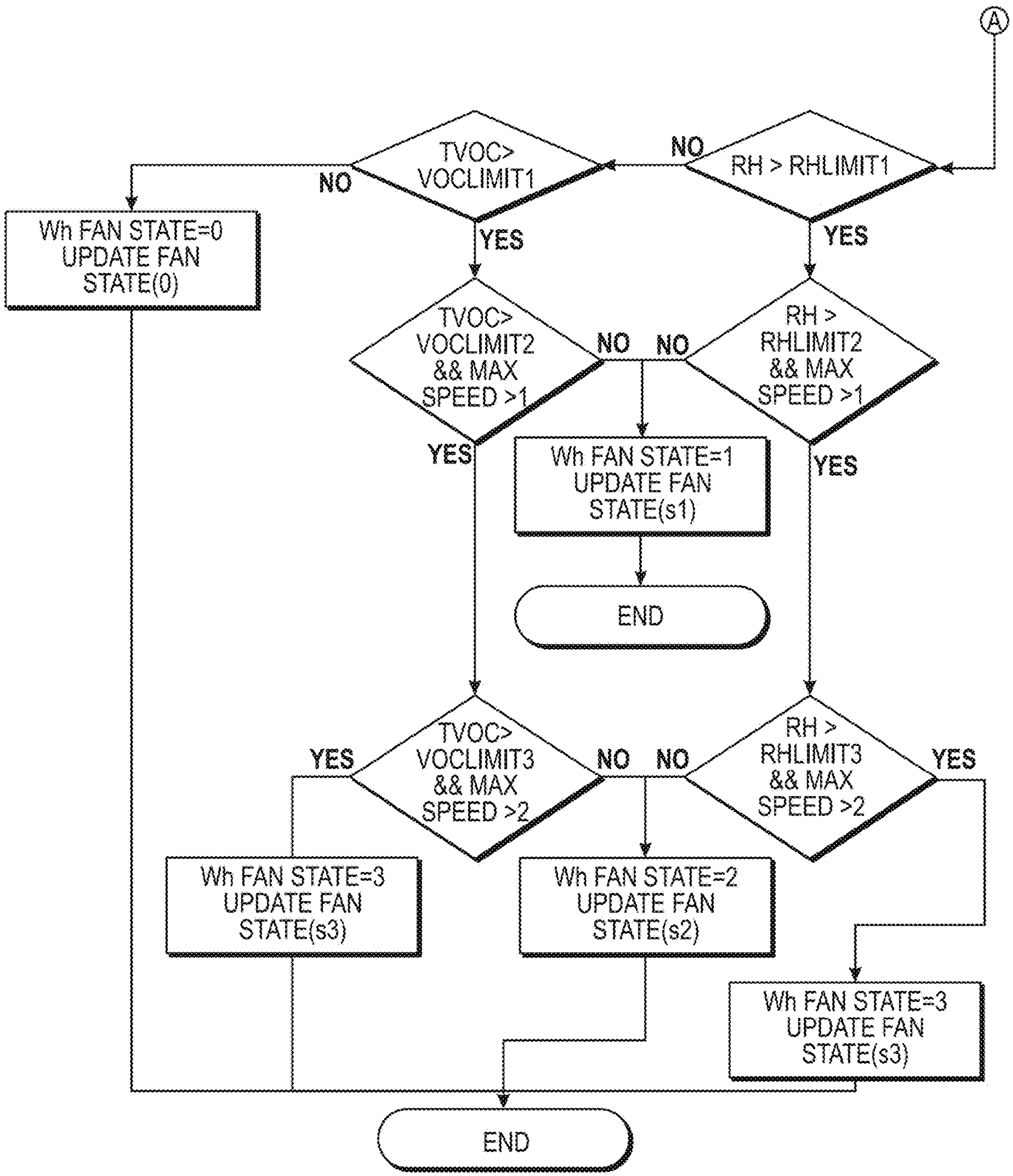

A
RH > RHLIMIT1
NO
YES
TVOC>
VOCLIMIT1
NO
YES
Wh FAN STATE=0
UPDATE FAN
STATE(0)
TVOC>
VOCLIMIT2
&& MAX
SPEED >1
NO
YES
RH >
RHLIMIT2
&& MAX
SPEED >1
NO
YES
Wh FAN STATE=1
UPDATE FAN
STATE(s1)
END
TVOC>
VOCLIMIT3
&& MAX
SPEED >2
YES
NO
RH >
RHLIMIT3
&& MAX
SPEED >2
NO
YES
Wh FAN STATE=3
UPDATE FAN
STATE(s3)
Wh FAN STATE=2
UPDATE FAN
STATE(s2)
Wh FAN STATE=3
UPDATE FAN
STATE(s3)
END

FIG. 66B

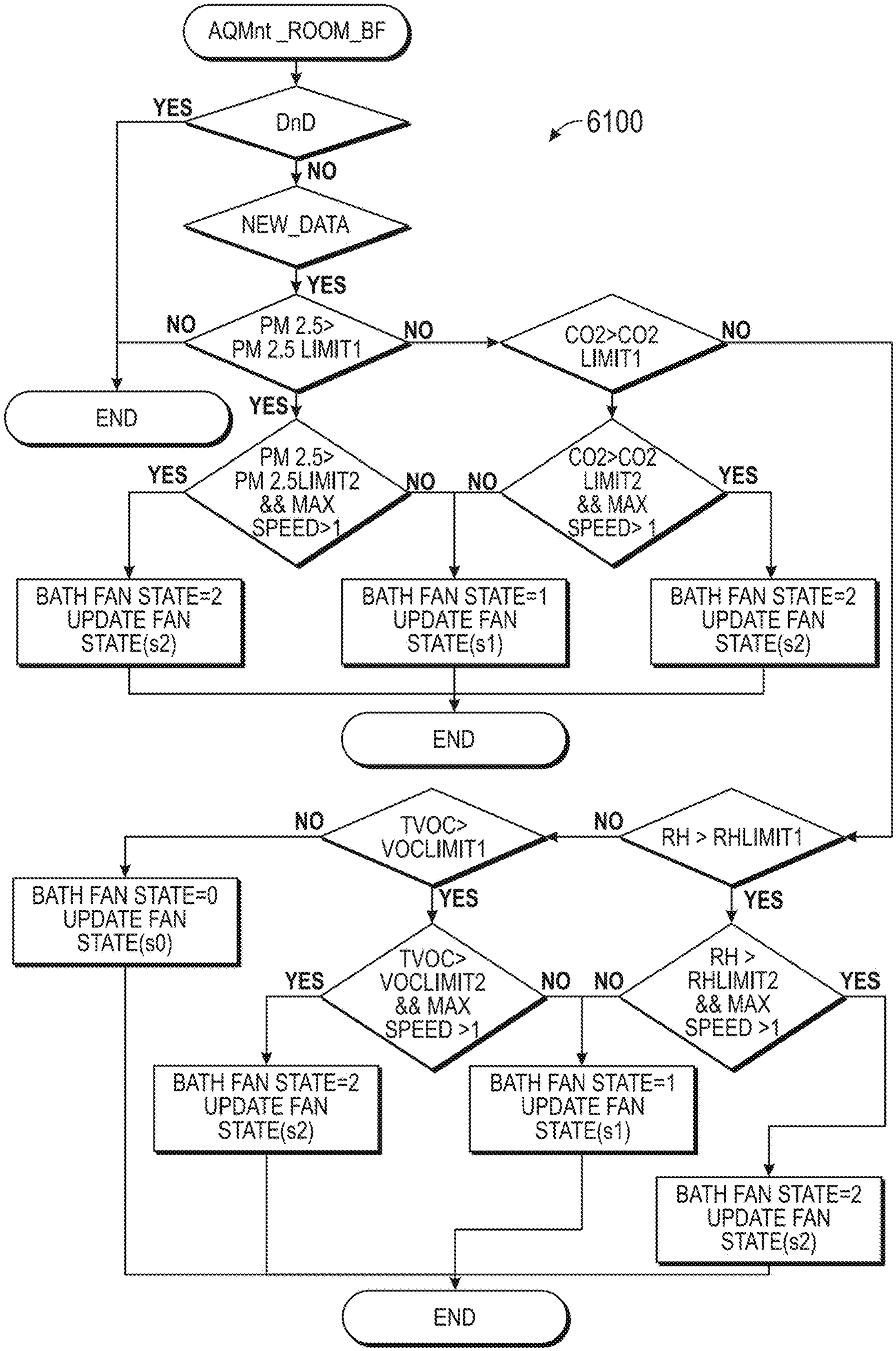

AQMnt _ROOM_BF
DnD
YES
NO
6100
NEW_DATA
YES
PM 2.5> PM 2.5 LIMIT1
NO
NO
CO2>CO2 LIMIT1
NO
END
YES
PM 2.5> PM 2.5LIMIT2 && MAX SPEED>1
YES
NO
NO
CO2>CO2 LIMIT2 && MAX SPEED> 1
YES
BATH FAN STATE=2 UPDATE FAN STATE(s2)
BATH FAN STATE=1 UPDATE FAN STATE(s1)
BATH FAN STATE=2 UPDATE FAN STATE(s2)
END
NO
TVOC> VOCLIMIT1
NO
RH > RHLIMIT1
BATH FAN STATE=0 UPDATE FAN STATE(s0)
YES
YES
TVOC> VOCLIMIT2 && MAX SPEED >1
YES
NO
NO
RH > RHLIMIT2 && MAX SPEED >1
YES
BATH FAN STATE=2 UPDATE FAN STATE(s2)
BATH FAN STATE=1 UPDATE FAN STATE(s1)
BATH FAN STATE=2 UPDATE FAN STATE(s2)
END

FIG. 67

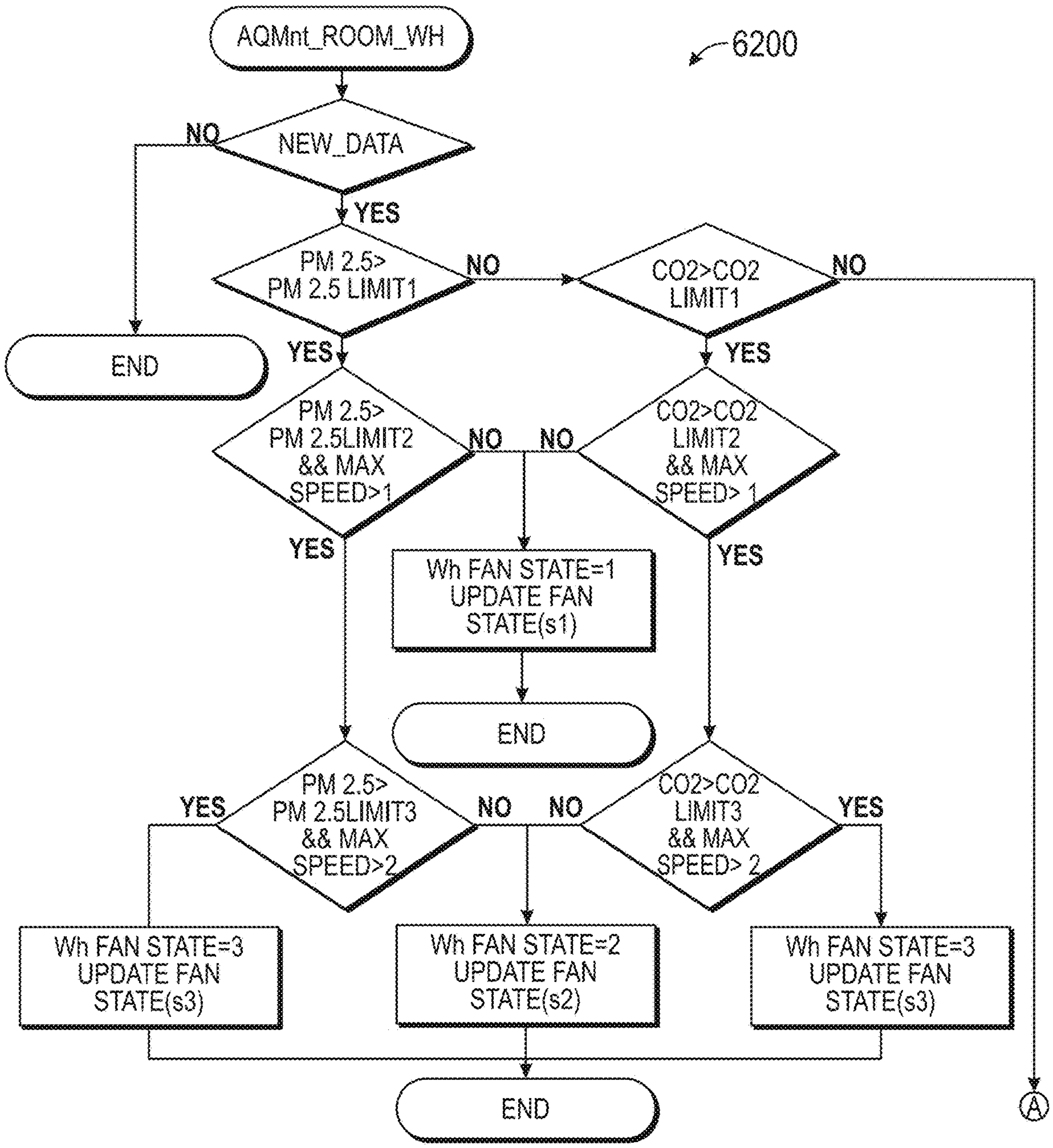
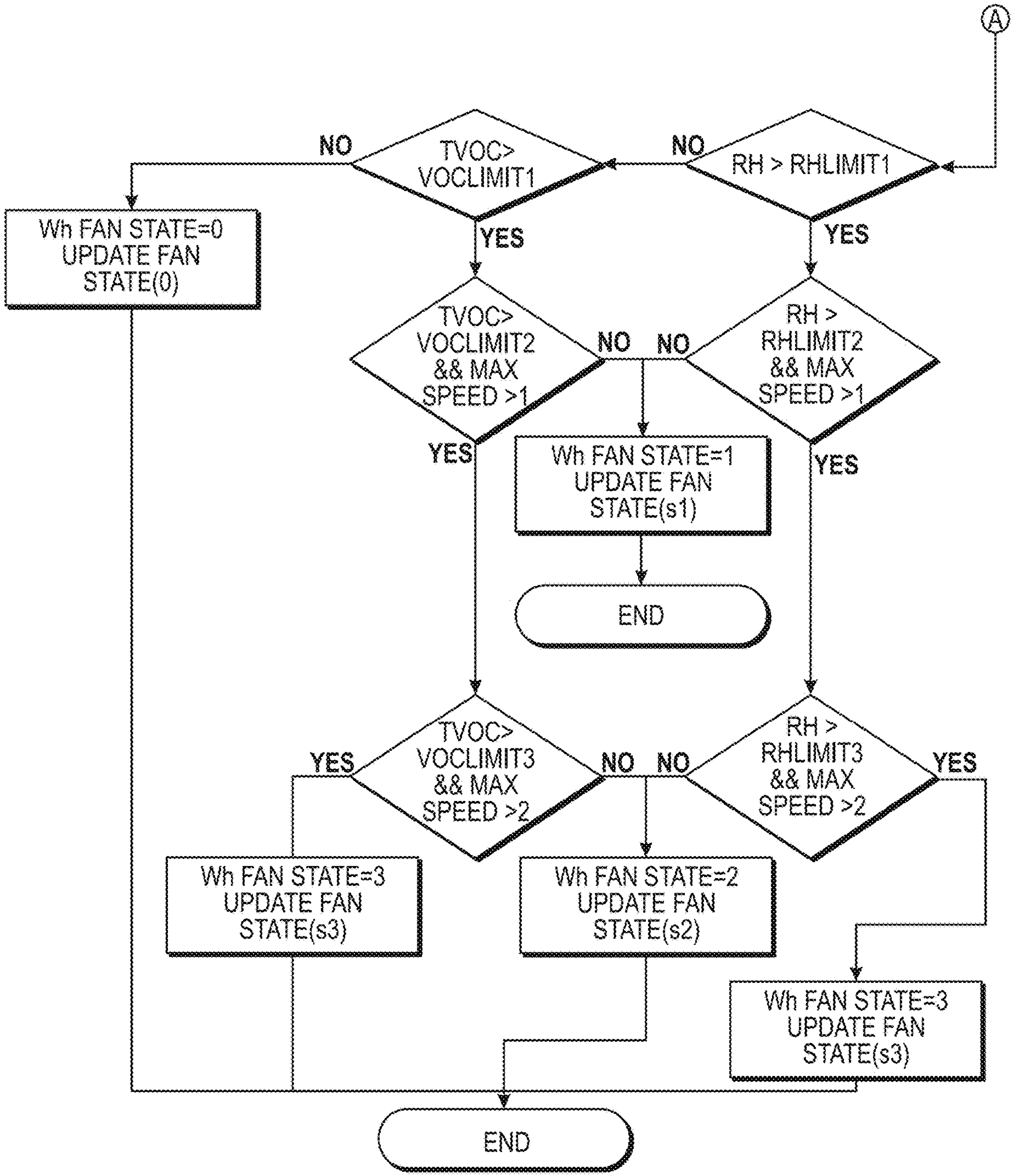

AQMnt_ROOM_WH
6200
NEW_DATA
NO
YES
PM 2.5> PM 2.5 LIMIT1
NO
CO2>CO2 LIMIT1
NO
END
YES
YES
PM 2.5> PM 2.5LIMIT2 && MAX SPEED>1
NO
NO
CO2>CO2 LIMIT2 && MAX SPEED> 1
YES
YES
Wh FAN STATE=1 UPDATE FAN STATE(s1)
END
PM 2.5> PM 2.5LIMIT3 && MAX SPEED>2
YES
NO
NO
CO2>CO2 LIMIT3 && MAX SPEED> 2
YES
Wh FAN STATE=3 UPDATE FAN STATE(s3)
Wh FAN STATE=2 UPDATE FAN STATE(s2)
Wh FAN STATE=3 UPDATE FAN STATE(s3)
END
A

FIG. 68A



A
RH > RHLIMIT1
NO
YES
TVOC> VOCLIMIT1
NO
YES
Wh FAN STATE=0 UPDATE FAN STATE(0)
TVOC> VOCLIMIT2 && MAX SPEED >1
NO
YES
RH > RHLIMIT2 && MAX SPEED >1
NO
YES
Wh FAN STATE=1 UPDATE FAN STATE(s1)
END
TVOC> VOCLIMIT3 && MAX SPEED >2
YES
NO
RH > RHLIMIT3 && MAX SPEED >2
NO
YES
Wh FAN STATE=3 UPDATE FAN STATE(s3)
Wh FAN STATE=2 UPDATE FAN STATE(s2)
Wh FAN STATE=3 UPDATE FAN STATE(s3)
END

FIG. 68B

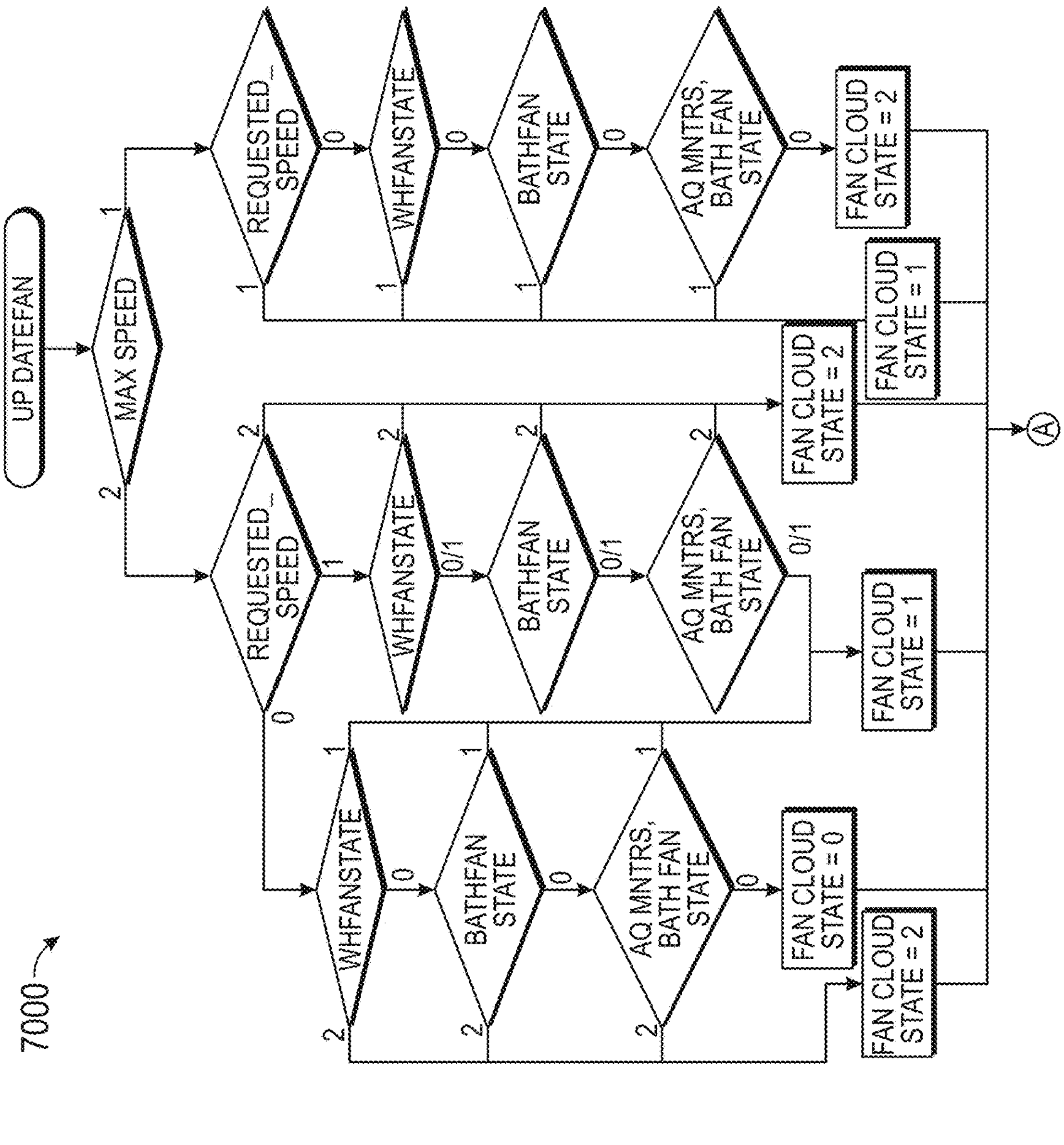


FIG. 69A

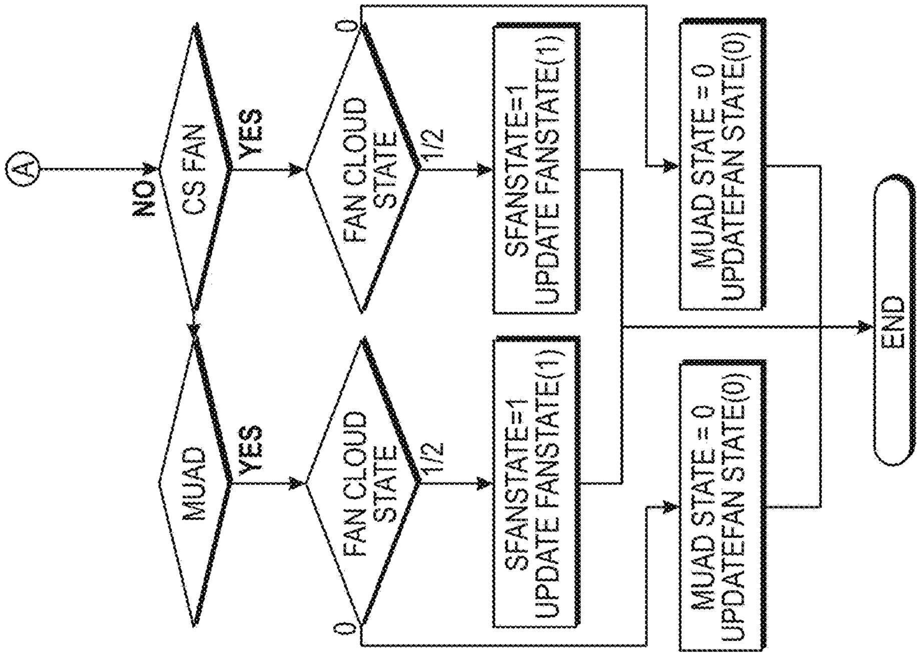


FIG. 69B

7500

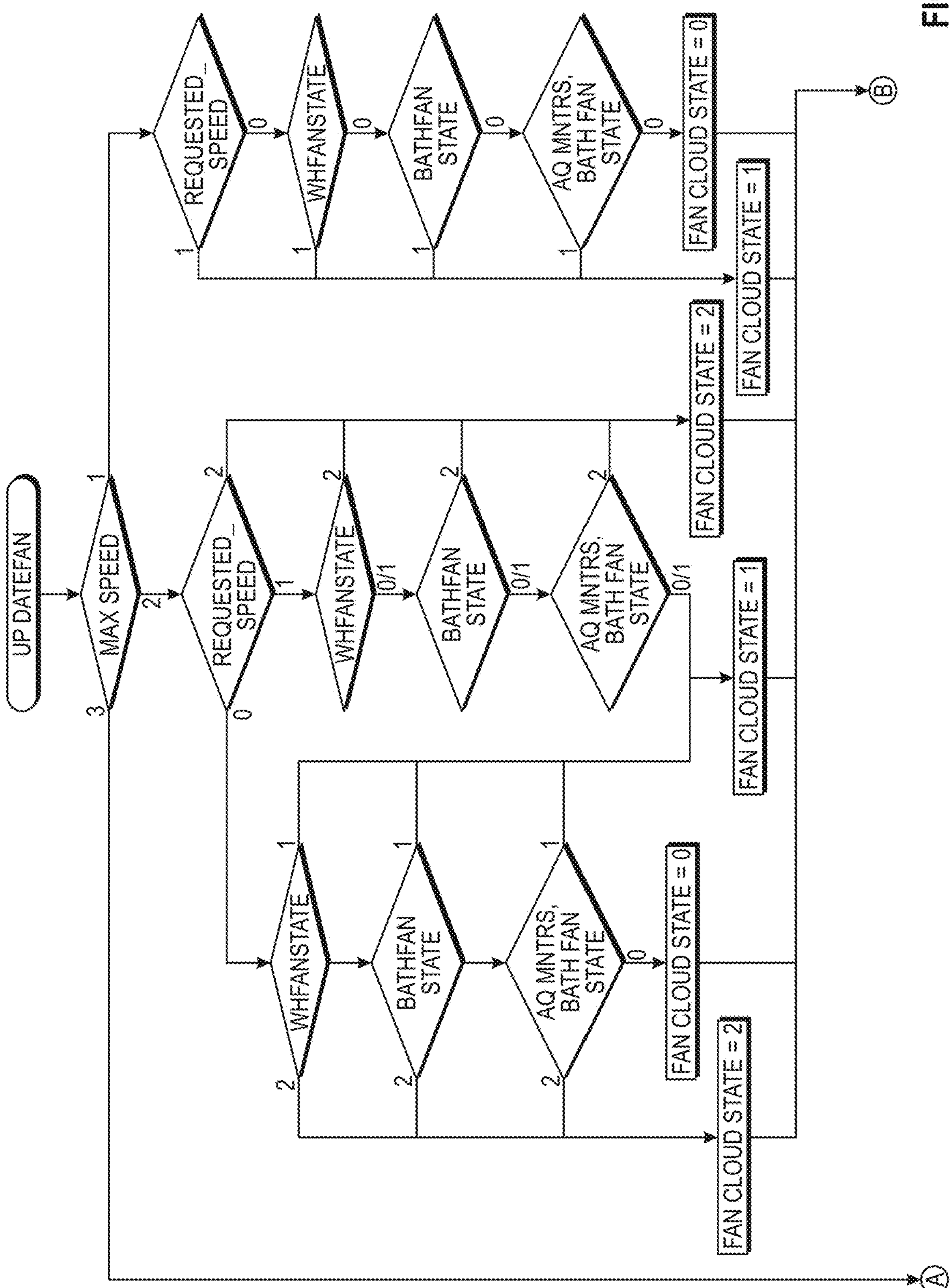

UP DATEFAN
MAX SPEED
REQUESTED_ SPEED
WHFANSTATE
BATHFAN STATE
AQ MNTRS, BATH FAN STATE
FAN CLOUD STATE = 0
FAN CLOUD STATE = 1
FAN CLOUD STATE = 2
REQUESTED_ SPEED
WHFANSTATE
BATHFAN STATE
AQ MNTRS, BATH FAN STATE
FAN CLOUD STATE = 2
FAN CLOUD STATE = 1
WHFANSTATE
BATHFAN STATE
AQ MNTRS, BATH FAN STATE
FAN CLOUD STATE = 0
FAN CLOUD STATE = 2
A
B

FIG. 70A

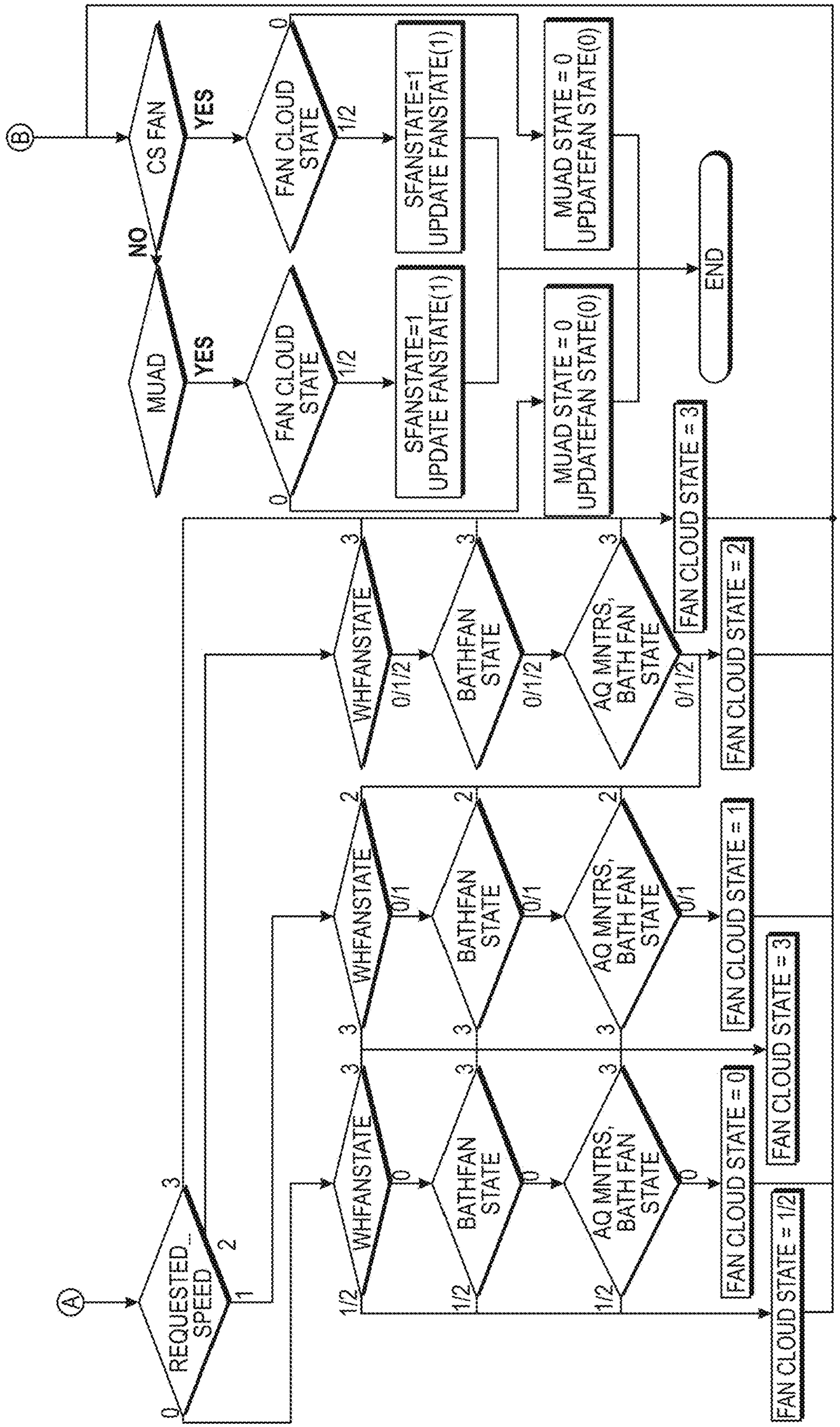

A
REQUESTED SPEED
0
1
2
3
WHFANSTATE
BATHFAN STATE
AQ MNTRS, BATH FAN STATE
1/2
0
0/1
0/1/2
FAN CLOUD STATE = 0
FAN CLOUD STATE = 1/2
FAN CLOUD STATE = 1
FAN CLOUD STATE = 2
FAN CLOUD STATE = 3
B
CS FAN
MUAD
YES
NO
FAN CLOUD STATE
SFANSTATE=1
UPDATE FANSTATE(1)
MUAD STATE = 0
UPDATEFAN STATE(0)
END

FIG. 70B

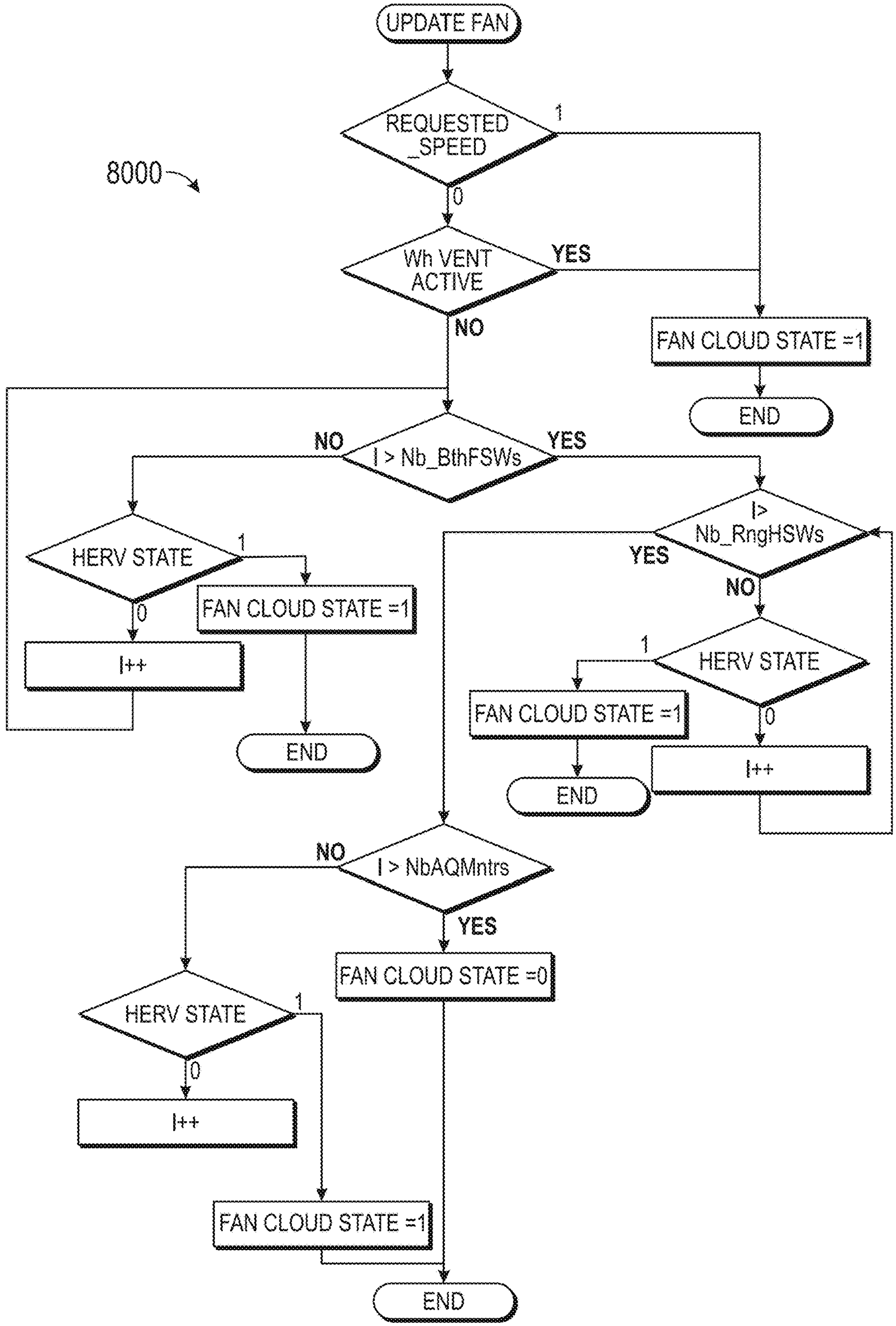

UPDATE FAN
REQUESTED _SPEED
1
0
8000
Wh VENT ACTIVE
YES
NO
FAN CLOUD STATE =1
END
NO
I > Nb_BthFSWs
YES
I> Nb_RngHSWs
YES
NO
HERV STATE
1
0
FAN CLOUD STATE =1
I++
END
HERV STATE
1
0
FAN CLOUD STATE =1
END
I++
NO
I > NbAQMntrs
YES
FAN CLOUD STATE =0
HERV STATE
1
0
I++
FAN CLOUD STATE =1
END

FIG. 71

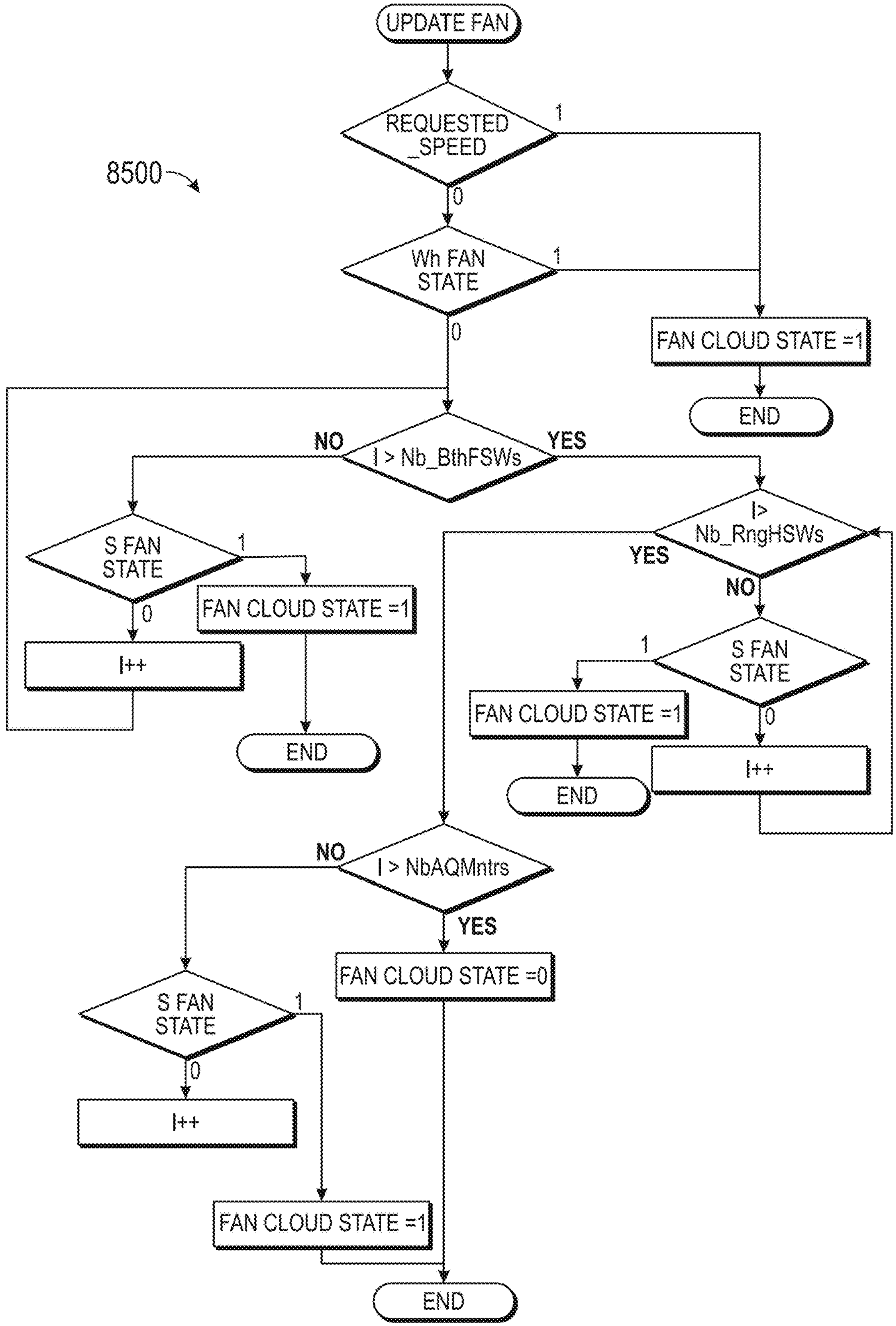

UPDATE FAN
REQUESTED _SPEED
1
0
8500
Wh FAN STATE
1
0
FAN CLOUD STATE =1
END
NO
I > Nb_BthFSWs
YES
I> Nb_RngHSWs
S FAN STATE
1
FAN CLOUD STATE =1
0
I++
END
YES
NO
S FAN STATE
1
FAN CLOUD STATE =1
0
END
I++
NO
I > NbAQMntrs
YES
FAN CLOUD STATE =0
S FAN STATE
1
0
I++
FAN CLOUD STATE =1
END

FIG. 72

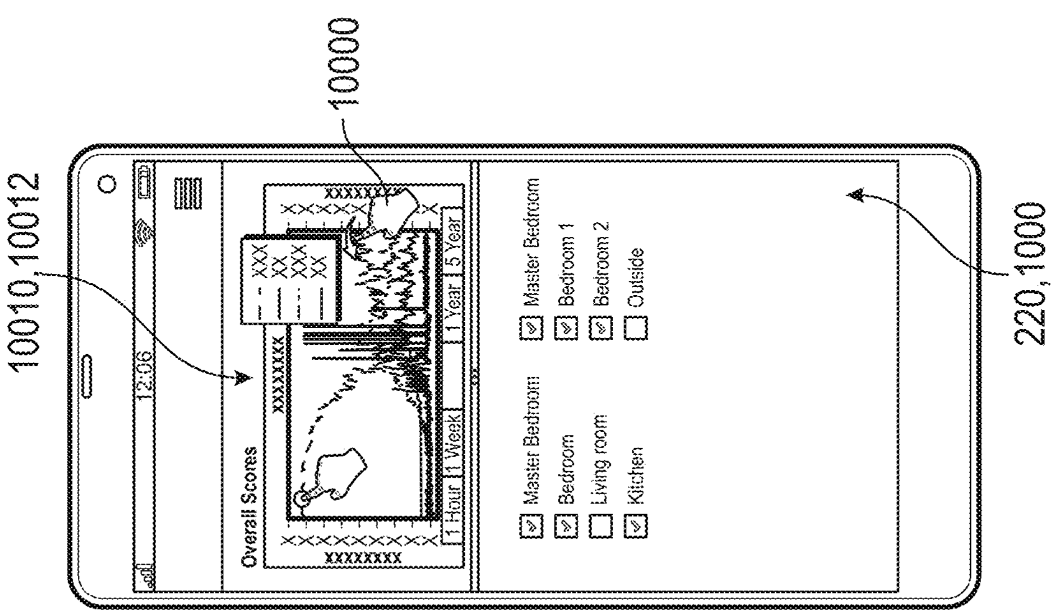


FIG. 73

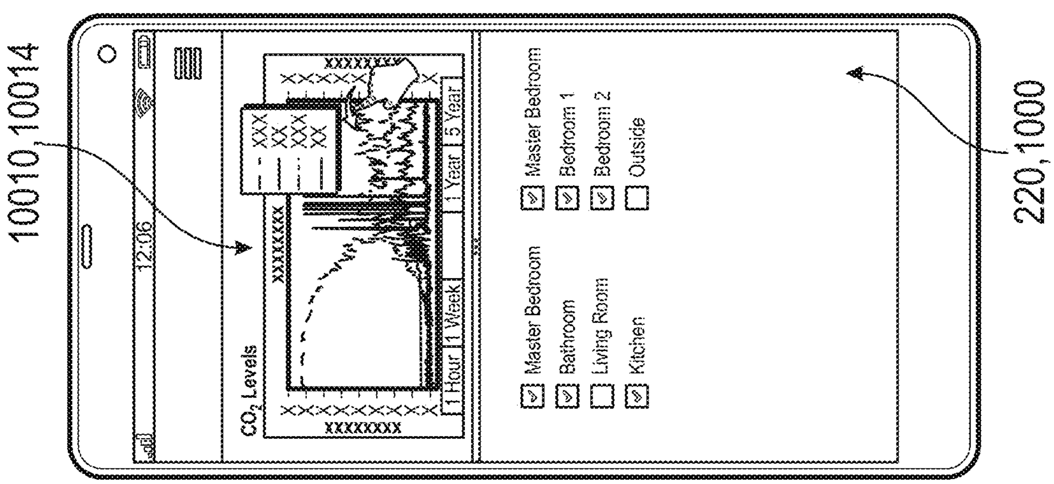


FIG. 74

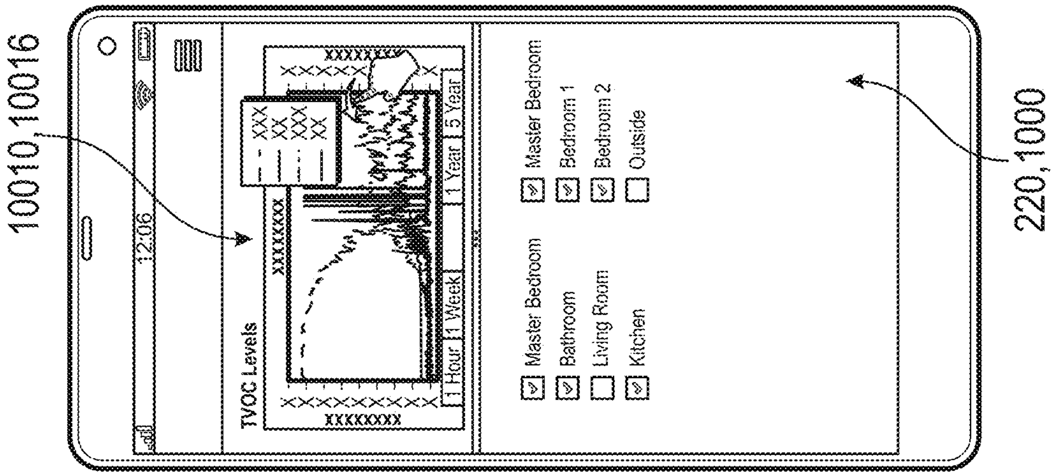


FIG. 75

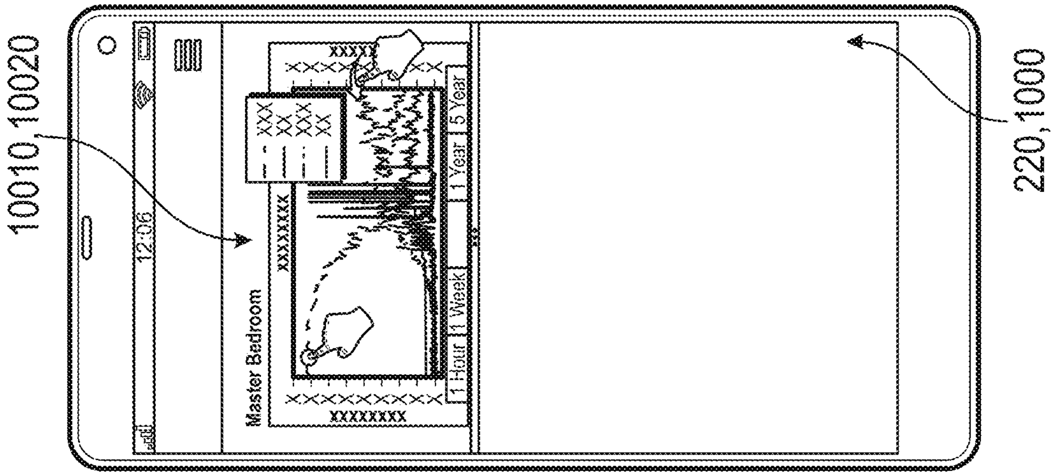

10010,10020
Master Bedroom
12:06
1 Hour
1 Week
1 Year
5 Year
220,1000

FIG. 77

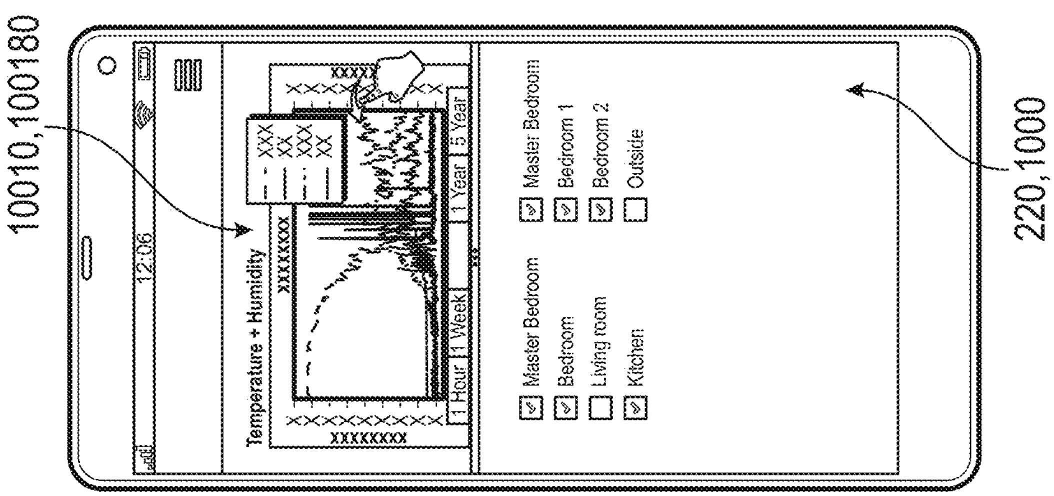

10010,100180
Temperature + Humidity
12:06
1 Hour
1 Week
1 Year
5 Year
Master Bedroom
Bedroom
Living room
Kitchen
Master Bedroom
Bedroom 1
Bedroom 2
Outside
220,1000

FIG. 76

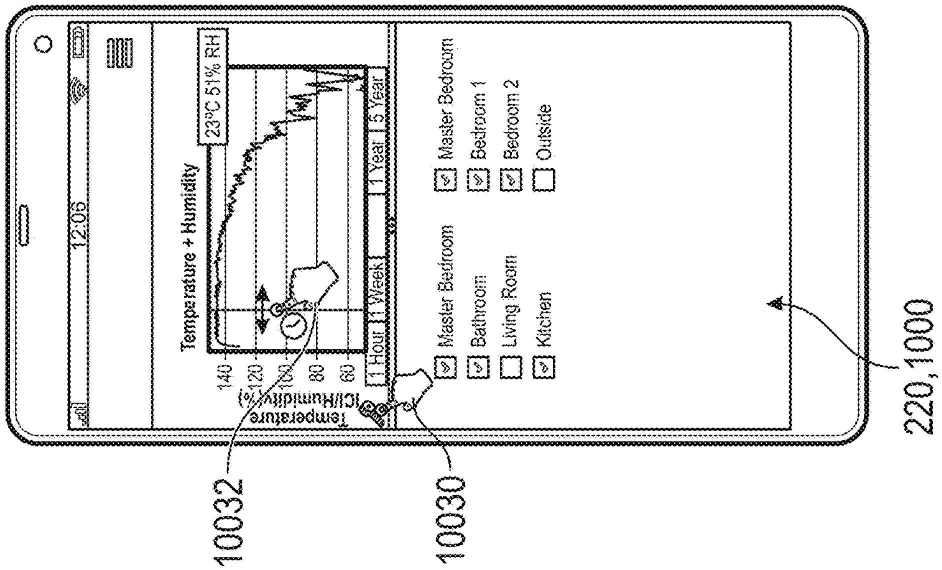


FIG. 80

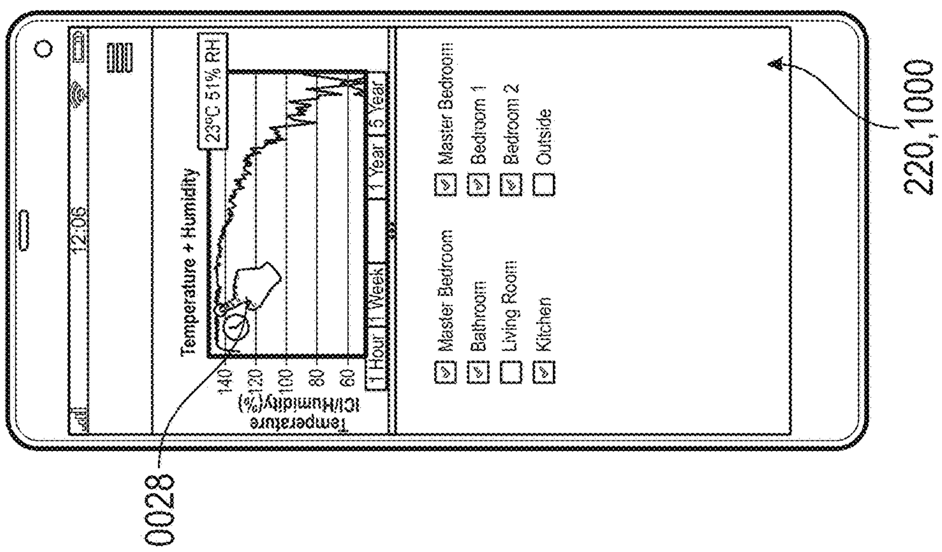


FIG. 79

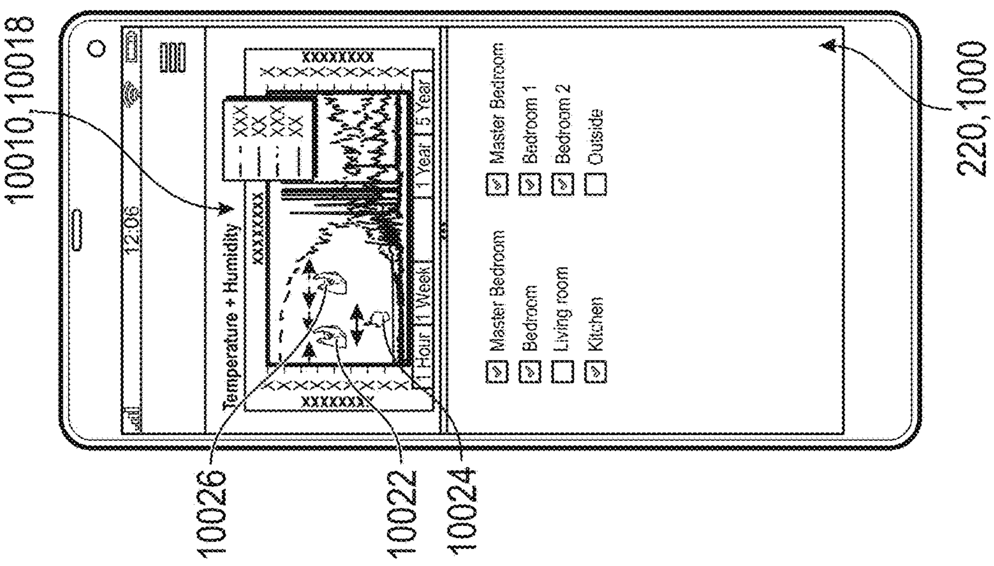


FIG. 78

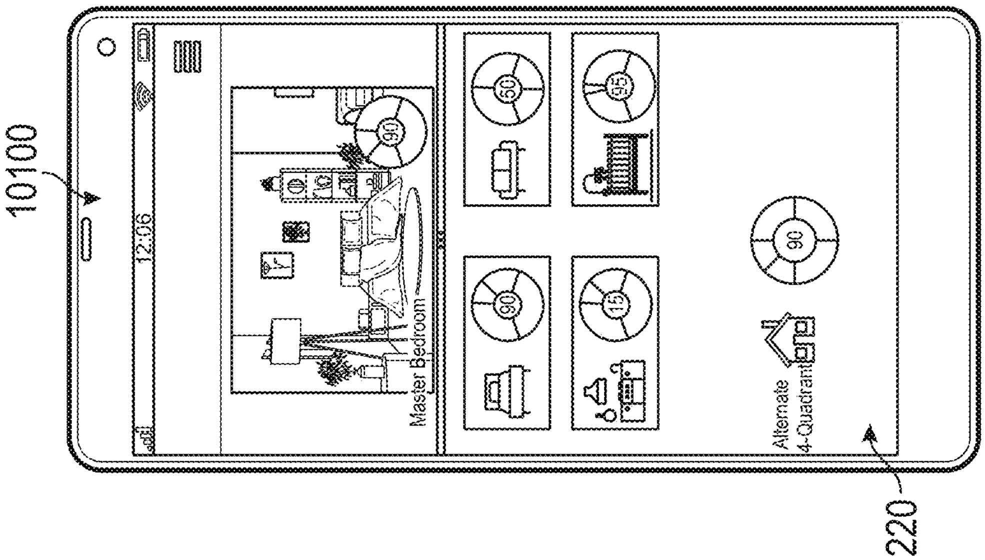

10100
12:06
Master Bedroom
90
60
95
90
15
90
Alternate
4-Quadrant
220

FIG. 82

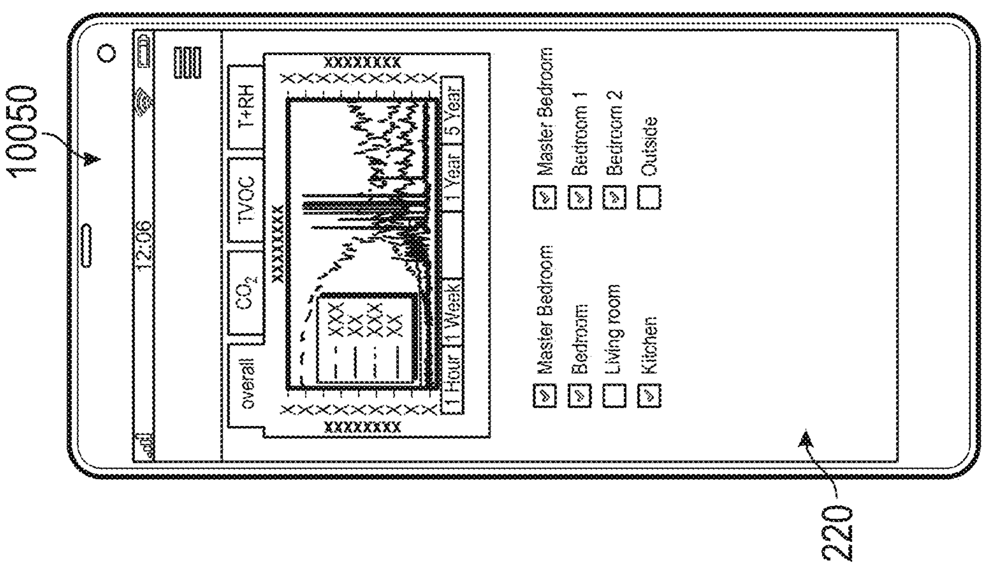

10050
12:06
overall
CO_2
TVOC
T+RH
XXXXXXXX
XXXXXXX
1 Hour
1 Week
1 Year
5 Year
Master Bedroom
Bedroom
Living room
Kitchen
Master Bedroom
Bedroom 1
Bedroom 2
Outside
220

FIG. 81

SYSTEM AND METHOD FOR CONTROLLING INDOOR AIR QUALITY

CROSS-REFERENCE TO OTHER APPLICATIONS

PCT Patent Application No. PCT/US20/12487, filed Jan. 7, 2020, U.S. Provisional Patent Application No. 62/789,501, filed on Jan. 7, 2019, PCT Patent Application No. PCT/US19/63581, filed on Nov. 27, 2019, U.S. patent application Ser. No. 16/243,056, filed on Jan. 8, 2019, U.S. patent application Ser. No. 16/242,498, filed on Jan. 8, 2019, U.S. patent application Ser. No. 15/081,488, filed on Mar. 25, 2016, U.S. patent application Ser. No. 14/593,883, filed on Jan. 9, 2015, U.S. Pat. No. 9,297,540, filed on Aug. 5, 2013, U.S. Pat. No. 10,054,127, filed on Sep. 29, 2017, U.S. Pat. No. 9,816,724, filed on Jan. 29, 2015, U.S. Pat. No. 9,816,699, filed on Sep. 2, 2015, U.S. Pat. No. 9,638,432, filed on Aug. 31, 2010, U.S. Pat. No. 8,100,746, filed on Jan. 4, 2006 and WO 2015/168243, filed on Nov. 5, 2015, all of which are incorporated in their entirety herein by reference and made a part hereof.

TECHNICAL FIELD

The present disclosure relates to indoor air quality ("IAQ") system, and particularly to IAQ system for use with an air venting systems. More particularly, the present disclosure relates to an IAQ system that can control various indoor air ventilation devices in order to regulate the air quality within a structure.

BACKGROUND

Recently researchers have turned their attention to studying the negative effects that poor indoor air quality has on an individual's health because people spend close to 90% of their time indoors and about 65% of their time is in their home. Health condition that appear to be negatively affected by poor indoor air quality include: (i) chronic obstructive pulmonary disease (COPD), asthmatics, heart disease, diabetes, obesity, neurodevelopmental disorders, among many others. Accordingly, a system that can not only monitor and raise awareness about the indoor air quality of a person's home, but can also improve indoor air quality is desirable.

Also, with widespread adoption of smartphones and mobile devices for implementation of smart home and internet of things (IoT) functionality, users are provided with more opportunities to lean about and control their environment. Thus, the ability to control the indoor air quality of a user's home from a remote location is also desirable.

The description provided in the background section should not be assumed to be prior art merely because it is mentioned in or associated with the background section. The background section may include information that describes one or more aspects of the subject technology.

SUMMARY

Described herein is an IAQ system that is capable of obtaining environmental data—namely air quality information—from various devices contained within a structure. In particular, these devices contain sensors that can obtain environmental data. This environmental data is then analyzed by the system to determine if any level of a component within the data is outside of a predefined threshold range. If the system determines that the level of the component is

outside of the predefined threshold range for that given component, the system will carry out certain steps in order to bring the level within the predetermined threshold range. These steps include selecting the appropriate appliance and the proper operating conditions (e.g., turned ON/OFF and/or operating speed) of the selected appliance to most efficiently bring the level back within the predetermined threshold range. Once the system has determined that the level is back within the predetermined threshold range, the system will instruct the selected appliance to turn OFF.

It is understood that other configurations of the subject technology will become readily apparent to those skilled in the art from the following detailed description, wherein various configurations of the subject technology are shown and described by way of illustration. As will be realized, the subject technology is capable of other and different configurations, and its several details are capable of modification in various other respects, all without departing from the scope of the subject technology. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not as restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

The drawing figures depict one or more implementations in accord with the present teachings, by way of example only, not by way of limitation. In the figures, like reference numerals refer to the same or similar elements.

FIG. **1**A is a simplified block diagram of a first exemplary embodiment of an IAQ system;

FIG. **1**B is a simplified block diagram of a second exemplary embodiment of an IAQ system;

FIG. **1**C is a simplified block diagram of a third exemplary embodiment of an IAQ system;

FIG. **1**D is a simplified block diagram of a fourth exemplary embodiment of an IAQ system;

FIG. **1**E is a simplified block diagram of a fifth exemplary embodiment of an IAQ system;

FIG. **1**F is a simplified block diagram of a sixth exemplary embodiment of an IAQ system;

FIG. **1**G is a simplified block diagram of a seventh exemplary embodiment of an IAQ system;

FIG. **2** is a simplified block diagram of a monitoring device that includes sensors, such as environmental sensors;

FIGS. **3**A-**3**B are exemplary in-wall monitoring devices that may be positioned within a switching unit that is designed to control lights;

FIGS. **4**A-**4**D are exemplary monitoring devices that are designed to be plugged into an electrical outlet that is positioned within a wall of the operating environment;

FIG. **5** is a portable exemplary monitoring device;

FIG. **6** is an exemplary monitoring device that is designed to control the operation of an appliance;

FIG. **7** is an exemplary monitoring device that also includes a display that can be utilized to show information to the user;

FIGS. **8**A-**8**C are exemplary connected appliances that are designed to be physically connected to an operating environment and are capable of communicating with the IAQ system without requiring additional hardware;

FIG. **9** is an exemplary non-connected appliance that requires additional hardware to communicate with the IAQ system, such additional hardware includes the monitoring device shown in FIG. **7**;

FIGS. **10**A-**10**B are exemplary non-connected appliances that require additional hardware to communicate with the

IAQ system, such additional hardware includes the monitoring device shown in FIGS. **4**A-4D;

FIG. **11** is an exemplary non-connected appliance that requires additional hardware to communicate with the IAQ system, such additional hardware includes the monitoring device shown in FIG. **7**;

FIG. **12** is an exemplary ventilation grill that can be retrofitted to enable the IAQ system to control air output through the grill;

FIG. **13** is a partial cut-away view of an operating environment, which contains one of the exemplary IAQ systems shown in FIGS. **1**A-**1**G;

FIGS. **14-15** show a graphical user interface ("GUI") that is displayed on a mobile device, wherein the displayed screens, contained within the GUI, allow a user to log into their previously created account;

FIGS. **16-18** show screens, contained within the GUI, that allow an authorized user to enter information about the operating environment;

FIGS. **19-32** show screens, contained within the GUI, that allow the authorized user to connect an appliance and a monitoring device into the IAQ system;

FIG. **33** shows a screen, contained within the GUI, that allows the authorized user to set up alert notifications;

FIG. **34** shows a screen, contained within the GUI, that allows the authorized user to set the IAQ system in a do not disturb mode;

FIGS. **35-40** show a partial cut-away view of an operating environment, which shows the exemplary IAQ system operating under a first set of conditions;

FIGS. **41-45** show a partial cut-away view of an operating environment, which shows the exemplary IAQ system operating under a second set of conditions;

FIGS. **46-50** show a partial cut-away view of an operating environment, which shows the exemplary IAQ system operating under a third set of conditions;

FIGS. **51-55** show a partial cut-away view of an operating environment, which shows the exemplary IAQ system operating under a fourth set of conditions;

FIGS. **56-72** contain flow charts describing how the IAQ system functions;

FIGS. **73-80** show screens, contained within the GUI, that display historical environmental measurements that were recorded by the IAQ system over a predefined amount of time;

FIG. **81** shows a first alternative embodiment of a screen, contained within the GUI, that displays historical environmental measurements that were recorded by the IAQ system over a predefined amount of time; and

FIG. **82** shows a second alternative embodiment of a screen, contained within the GUI, that displays historical environmental measurements that were recorded by the IAQ system over a predefined amount of time.

DETAILED DESCRIPTION

In the following detailed description, numerous specific details are set forth by way of examples in order to provide a thorough understanding of the relevant teachings. However, it should be apparent to those skilled in the art that the present teachings may be practiced without such details. In other instances, well-known methods, procedures, components, and/or circuitry have been described at a relatively high-level, without detail, in order to avoid unnecessarily obscuring aspects of the present disclosure.

While this disclosure includes a number of embodiments in many different forms, there is shown in the drawings and will herein be described in detail particular embodiments with the understanding that the present disclosure is to be considered as an exemplification of the principles of the disclosed methods and systems, and is not intended to limit the broad aspects of the disclosed concepts to the embodiments illustrated. As will be realized, the disclosed methods and systems are capable of other and different configurations and several details are capable of being modified all without departing from the scope of the disclosed methods and systems. For example, one or more of the following embodiments, in part or whole, may be combined consistent with the disclosed methods and systems. As such, one or more steps from the flow charts or components in the Figures may be selectively omitted and/or combined consistent with the disclosed methods and systems. Accordingly, the drawings, flow charts and detailed description are to be regarded as illustrative in nature, not restrictive or limiting.

1) Introduction/Summary

FIGS. **1-82** describe an IAQ system **10** that is capable of obtaining environmental data—namely air quality information, such as pollutant levels—from a monitoring device **102**, a central unit **104**, or a connected appliance **106**, which are contained within an operating environment **98**—namely a structure **100** (e.g., commercial building, a residential building, a single-family home, an apartment, etc.). These devices **102**, **104**, and **106** are configured to record environmental data, which includes various components (e.g., temperature, humidity and/or pollutant levels, such as TVOC, CO_2, PM2.5), and send the recorded levels of the components of the environmental data to a local server/database **110**. This local server/database **110** may: i) analyze the data, ii) determine if all levels contained within environmental data are within predefined threshold ranges, and iii) may recommend that the IAQ system **10** take certain steps (e.g., turn ON/OFF various appliances) to bring certain levels of the components within the predetermined threshold range. The IAQ system **10** can then carry out these steps by controlling the operational mode (e.g., ON/OFF and/or the speed of the fan) of various appliances **106** contained within the operating environment **98**. Once the IAQ system **10** has determined that the levels contained within the environmental data are back within the predetermined threshold ranges, the IAQ system **10** will instruct the appliances **106** to turn OFF.

2) System Configuration

FIGS. **1**A-**1**G illustrate simplified block diagrams of various non-limiting embodiments of an exemplary IAQ system **10** that is designed to operate within a structure **100**. Specifically, the IAQ system **10** may include: i) a CIAQ device **50**, such as a monitoring device **102**, ii) a central unit **104**, iii) one or more appliances **106** (e.g., connected appliances **300** that contain an CIAQ device, monitoring device **102** that can control non-connected appliances **400**, or a controller that can control non-connected appliances **400**), iv) a network **108**, such as any type of wired or wireless communication technology, iv) local server/database **110**, v) national database **112**, vi) data source **113** (e.g., distributed external sensors, weather pattern data, historical air quality databases, air quality prediction databases, and other information about the air that is exterior to the structure **100**), and vii) alerting device **114** (e.g., computer, smartphone, tablet, smartwatch, or similar types of devices). It should be understood in certain embodiments that some of the devices set forth above may be omitted. For example, the IAQ system **10** shown in FIG. **1**G does not contain a central unit **104**. Additionally, in other embodiments, the IAQ system **10** may include additional devices and/or components.

a. First Embodiment of the System

FIG. **1**A illustrates a first exemplary IAQ system **10** that includes the monitoring device **102**, central unit **104**, and the appliance **106**. The monitoring device **102** and the central unit **104** include at least one sensor, which it uses to collect data about the local environment **98**. Some or all of this environment data is then sent to the local server/database **110**, which processes and stores this data. If the local server/database **110** determines that all levels contained within the environmental data are within the predetermined threshold ranges, then the local server/database **110** will simply store the environmental data. However, if the local server/database **110** determines that one level contained within the environmental data is out of the predetermined threshold range, then the IAQ system **10** may be configured to perform one of the following steps from one of the below embodiments:

In a first embodiment, the local server/database **110** will send an alert to the alerting unit **114** via the network **108**. The alert that is sent to the alerting unit **114** informs the user which level was outside of the predetermined threshold range. Along with sending this alert to the alerting unit **114**, the local server/database **110** will send an electronic signal(s) via the router **116** to the appliance(s) **106** (e.g., connected appliances **300** that contain an CIAQ device, monitoring device **102** that can control non-connected appliances **400**, or a controller that can control non-connected appliances **400**) in order to return the level to a state that is within the predetermined threshold range. In this first embodiment, the IAQ system **10** does not ask the user to confirm any steps that the IAQ system **10** has deemed necessary; instead, the IAQ system **10** automatically performs the determined steps. Once the steps have been performed or if the level has been returned to a state that is within the predetermined threshold range, the IAQ system **10**: (i) sends electronic signal(s) via the router **116** to turn OFF the appliance(s) **106** and (ii) sends a signal to the alerting device **114** to inform the authorized user that the alert has been resolved. It should be understood that at any time, including before, during, or after an alert has been received, the user can prevent the system from automatically performing the steps that the IAQ system **10** may or has deemed necessary. It should also be understood that the authorized user may configure the IAQ system **10** such that it automatically performs the steps without sending an alert to the alerting device **114**.

In a second embodiment, the local server/database **110** will send an alert to the alerting unit **114** via the network **108**. This alert informs the user which component was outside of the predetermined threshold range and the steps the IAQ system **10** has deemed necessary to return the component to a state that is within the predetermined threshold range. The IAQ system **10** will then wait for the user to confirm the steps the IAQ system **10** is proposing. In this embodiment, the IAQ system **10** will not perform any steps prior to receiving confirmation from the authorized user. Once the authorized user has confirmed the steps the IAQ system **10** is proposing to implement or has selected an alternate set of steps, the IAQ system **10** sends electronic signal(s) via the router **116** to the appliance(s) **106** in order perform the steps that were approved by the authorized user. Once the steps have been performed or if the level of the component is returned to a state that is within the predetermined threshold range, the IAQ system **10**: (i) sends electronic signal(s) via the router **116** to turn OFF the appliance(s) **106** and (ii) sends a signal to the alerting device **114** to inform the authorized user that the alert has been resolved.

The IAQ system **10** in FIG. **1**A includes a national database **112**. The national database **112** can collect information from other systems **10**, which are deployed in other structures **100**. The national database **112** can compare the current environmental data collected from one specific structure against historical environmental data collected from: i) this specific structure, ii) other structures within the same neighborhood, iii) other structures within the same area or zip code, iv) other structures within the same region, v) other structures within the same country, and vi) all the structures around the world. This may allow the user to understand how their current air quality compares to historical air quality. Thus, this data may suggest that the changes the user made to their structure have improved their air quality. In addition, national database **112** can compare the current environmental data collected from one specific structure against current environmental data collected from: i) other structures within the same neighborhood, ii) other structures within the same area or zip code, iii) other structures within the same region, iv) other structures within the same country, and v) all the structures around the world. This may allow the user to understand how their current air quality compares to the current air quality of others. Thus, this data may suggest that the user needs to make additional changes to their structure to bring their air quality in line with their neighbors. One example of how this data could be utilized in a commercial setting is this data could be used in the marketing of a house. For example, a user that is selling their house may show someone that is interested in buying the house that their air quality is better than their neighbors. Or this environmental data could be used by potential home buyers in order to select a home or a location they desire to live.

The IAQ system **10** shown in FIG. **1**A also includes a data source **113**. This data source **113** may include a prediction table that is based on information derived from current and historical data collected from: i) exterior local/regional/national sensors (e.g. dew point, temperature, air pollutants), ii) sensors installed in other structures, iii) weather information, iv) electricity costs, and v) other similar types of data. This data contained within the data source **113** can be accessed by a combination of the national database **112** and the local server/database **110**. This data can be utilized to help make predictions when levels of the components will be deemed to be out of the predetermined threshold range and to take corrective measures prior to the occurrence of these events. For example, the national database **112** and the local server/database **110** may access the data source **113** and determine that the exterior air quality is predicted to be outside of threshold ranges between the hours of 3:00 pm and 9:00 pm. Thus, the system may try and minimize drawing air into the structure **100** during these times and instead will utilize air purifiers within the structure **100** in order to maximize the quality of the air contained within the structure **100**. In another example, the national database **112** and the local server/database **110** may access the data source **113** and determine that the cost of electricity during a specific month is lower between the hours of 11:00 am and 3:00 pm. Thus, the IAQ system **10** may try and operate devices that use more electricity during these times in order to remove air pollutants from structure **100**. Thus, the national database **112** in conjunction with the data source **113** can be utilized to maximize the quality of the air contained within the structure **100** based on predictions about the exterior environmental conditions.

b. Second-Seventh Embodiments of the System

FIG. **1**B illustrates another exemplary IAQ system **10**, which is similar to FIG. **1**A. However, the IAQ system **10** in FIG. **1**B does not include a national database **112** or a data source **113**. In addition, the monitoring device **103** and the appliance **106** do not report directly to the local server/database **110**. Instead, in this configuration, all data that is generated by the monitoring unit **102** and the signals that are sent to the appliance **106** pass through the central unit **104**. In comparison to FIG. **1**A, this configuration allows the central unit **103** to have more control over the IAQ system **10** and requires that fewer devices connect directly to the local server/database **110**.

FIGS. **1**C and **1**D illustrate exemplary systems **10**, which are similar to FIGS. **1**A and **1**B. However, the systems **10** in FIGS. **1**C and **1**D have the ability to send alerts directly from the router **116** to the alerting unit **114**. For example, these systems **10** may use a Wi-Fi connection or other low powered local area wireless network protocols to send data from the router **116** to the alerting unit **114**. In comparison to FIGS. **1**A and **1**B, this configuration allows the system to reduce the amount of data that travels over the non-local network, which reduces data costs and improves speed. It should be understood that alerts could still be sent over a non-local network in FIGS. **1**C and **1**D, if the alerting unit **114** is outside of the range of the network provided by the router **116**.

FIGS. **1**E and **1**F illustrate exemplary systems **10**, which are similar to FIGS. **1**A and **1**B. However, the systems **10** in FIGS. **1**E and **1**F have the ability to send alerts directly from the central unit **104** to the alerting unit **114**. For example, these systems **10** may use a Bluetooth protocol or other low powered local area wireless network protocols to send data from the central unit **104** to the alerting unit **114**. In comparison to FIGS. **1**A and **1**B, this configuration allows the system to reduce the amount of data that travels over the non-local network and the local network, which reduces data costs and improves speed. It should be understood that alerts can still be sent over a non-local network in FIGS. **1**E and **1**F, if the alerting unit **114** is outside of the range of the network provided by the central unit **104**.

FIG. **1**G illustrates an exemplary IAQ system **10**, which is very similar to FIG. **1**A. However, the IAQ system **10** in FIG. **1**G does not have a central unit **104**. Instead, the monitoring device **102** sends signals either to the alerting unit **114** or to the local server/database **110** via the router **116**. For example, these systems **10** may use a Bluetooth, NFC or other low powered local area wireless network protocols to send data from the monitoring device **102** to the alerting unit **114**. In comparison to FIG. **1**A, this configuration allows the system to reduce the amount of data that travels over the non-local network and the local network, which reduces data costs and improves speed. Also, this IAQ system **10** may be more suitable for smaller installations due to the fact that it does not require a central unit **104**. It should be understood that alerts could still be sent over a non-local network in FIG. **1**G, if the alerting unit **114** is outside of the range of the network provided by the monitoring device **102** or the router **116**.

3) Block Diagram of the Monitoring Device

FIG. **3**A illustrates a block diagram of exemplary monitoring device **102** of the IAQ system **10**. Specifically, the monitoring devices **102** may include the following elements: i) sensors **200**, ii) processor **202**, iii) memory **204**, iv) power control module **206**, v) location module **208**, and vi) connectivity module **210**. In some embodiments, the monitoring devices **102** may include other optional components, which include: i) speaker **212**, ii) microphone **214**, iii) status indicator **216**, or iv) other optional components (e.g., components that can control the operational setting of the device, data inputs, or lights) **218**. Meanwhile, the central unit **104** may be any internet enabled device (e.g., computer, laptop, mobile device, cellular phone, etc.) that includes displaying the current and/or historical data collected by the IAQ system **10**. In alternative embodiments, the central unit **104** may contain all of the same components and features of the monitoring devices **102** along with a display **220** for displaying the current and/or historical data collected by the IAQ system **10**.

a) Sensor(s)

The sensor(s) **200** that are contained within the monitoring device **102** are configured to collect data about the local environment **98**. The sensor(s) **200** may include any one of, or any combination of, the following: (i) air pollutant sensor, (ii) humidity/temperature sensor, (iii) motion sensor, (iv) light/color sensor, (v) camera, (vi) passive infrared (PIR) sensors or (vii) other sensors (e.g., infrared, ultrasonic, microwave, magnetic field sensors). It should be understood that the term environmental data is comprised of measurements taken from these sensors and these measurements are referred to herein as levels of components. In particular, the air pollutant sensor is configured to detect a concentration of one or more air pollutants in the environment within the structure **100**, including: CO, CO_2, NO, NO2, NOX, PM2.5, ultrafine particles, smoke (PM2.5 and PM10), radon, molds and allergens (PM10), volatile organic compounds (VOCs), ozone, dust particulates, lead particles, acrolein, biological pollutants (e.g., bacteria, viruses, animal dander and cat saliva, mites, cockroaches, pollen and etc.), pesticides, and formaldehyde. The humidity/temperature sensor measures the temperature and/or humidity in the environment within the structure **100** to establish an ambient baseline and to detect changes in the conditions of the environment within the structure **100**. The motion sensor, light/color sensors, camera, and other sensors may be used to monitor habits of humans or animals near the monitoring device **102** to establish a baseline trend and to detect changes in the baseline. Changes in this baseline trend may be helpful in determining why changes occurred within the recorded environment data. Alternatively, this baseline may be used by the IAQ system **10** to suggest different or alternative steps to maximize the air quality within the structure **100**.

b) Memory

The memory **204** may be utilized to temporally store the environmental data before this data is sent to the local server/database **110**. Typically, the predetermined threshold range(s) or value(s) may be programmed within the memory contained in the local server/database **110** or the central unit **104**. However, in some embodiments, some or all of the predetermined threshold range(s) or value(s) may be programmed within the memory **204** of the monitoring devices **102**. Regardless of where these predetermined threshold range(s) are stored, the range(s) or value(s) may be preprogramed into the IAQ system **10**. Specifically, there preprogramed range(s) or value(s) may be determined by the system designer based on one or more of the following: regulatory bodies, government agencies, private groups or standard setting bodies, such as the ASHRAE Standard Committee (e.g., ANSI/ASHRAE 62.2-2016, ISSN 1041-2336, which is fully incorporated herein by reference). An example of the range(s) that may be preprogram into the system **10** are shown in the below table, where the system **10** will send the alert or take start to take corrective action when the air quality reaches the "Fair" reference level. It should be understood that the if the air quality reaches the "Poor" reference level or the "Bad" reference level, the system **10** may take additional actions or more aggressive action in order to try and return the air quality within the structure **100** to at least a "Good" reference level within a reasonable amount of time. It should further be understood that these range(s) are only exemplary and should not be construed as limiting.

Reference Level	IAQ Rating	CO_2 (ppm)*	TVOC ($\mu g/m^3$)*	PM2.5 ($\mu g/m_3$)*	RH %*
Excellent	0-20	<600	<300	<25	40-60
Good	21-40	601-1000	301-1000	25-40	<40/>60
Fair	41-60	1001-1500	1001-3000	40-150	<30/>70
Poor	61-80	1501-2000	3001-10000	150-250	<20/>80
Bad	81-100	>2000	>10000	>250	<10/>90

It should be understood that predetermined threshold range (s) or value(s) may be updated by replacing the levels within the local server/database **110** or by using over the air updates in order to update levels that are stored in memory **204** of the monitoring devices **102**.

Instead of preprogramming the predetermined threshold range(s) or value(s) into the IAQ system **10**, the range(s) or value(s) may be determined/modified by calibrating the IAQ system **10** to the structure **100**. In order to provide these range(s) or value(s), the following steps may be undertaken. First, the monitoring unit **102** collects data from the sensors **200** over a predefined time period (e.g., 1 day, 3 days, or 7 days). This environmental data is then compared against recommended levels that are set forth by various regulatory bodies, government agencies, private groups, or standard setting bodies. Based on this comparison, the IAQ system **10** determines the threshold range(s) or value(s). For example, if the measured level of the components are more than one standard deviation below or above the recommended levels, then the system **10** may adjust recommend levels down or up that standard deviation. Performing these steps helps ensure that the IAQ system **10** is calibrated to the specific structure **100**, while being within recommended levels that are provided by the groups. This reduces false alarms and too many alarms, which allows the system **10** to run more efficiently. For example, if the environmental data from the structure **100** suggests that all levels of the components are well within the recommended levels, then set the thresholds at the recommended levels would not provide any useful information and the IAQ system **10** would rarely turn ON, if at all. On the other hand, if the environmental data from the structure **100** suggests that all levels of the components are not within the recommended levels, then set the thresholds based only on the data from the structure **100** would not be very helpful to aid the user in correcting their air quality. Thus, the IAQ system **10** utilizes both the environmental data collected from the structure along with the recommended levels data to provide the most accurate threshold ranges.

In a further alternative, the predetermined threshold range(s) or value(s) may be based on data collected over a predefined amount of time by systems **10** that have been deployed across the country. The collected data can then be analyzed in connection with the recommended levels, which are set forth by various regulatory bodies, government agencies, private groups, or standard setting bodies. Based on this comparison, the system **10** may adjust the predetermined threshold range(s) or value(s). It should be understood that the predetermined threshold range(s) or value(s) may differ on a region, state, city, or neighborhood basis. For example, the analysis of the collected data and the threshold range(s) may suggest that a IAQ system **10** that is located within Downtown, Los Angeles should have different range(s) then system **10** that are installed in: (i) Malibu, California, (ii) Tahoe, California, Oregon, or (iv) within the northwester part of the U.S. Based on this analysis, the system **10** can adjust the range(s) or value(s) to account for these differences. In other words, the system **10** may have one set of range(s) or value(s) for a system **10** located within Downtown, Los Angeles and another set of range(s) or value(s) for a system **10** located within Portland, Oregon. In an even further alternative, the predetermined threshold range(s) or value(s) may be set or modified by the user.

c) Power Control Module

The monitoring devices **102** include a power control module **206**, which controls the power of the monitoring devices **102** and any non-connected appliance **400** that is connected to the monitoring devices **102**. This module **206** allows the user and/or IAQ system **10** to turn ON/OFF the power supplied to an appliance **106**, which is connected to the monitoring devices **102**. In other words, this module **206** allows the IAQ system **10** to control non-connected appliances **400** using the monitoring devices **102**. Examples of non-connected appliances are shown in FIGS. **9**, **10**A and **10**B.

d) Location Module

The monitoring device **102** includes a location module **208** that aids the IAQ system **10** in determining the location of the monitoring device **102** within the structure **100** and what appliances **106** are positioned near or adjacent to the monitoring device **102**. This locational information aids the IAQ system **10** in determining the steps necessary to return a level contained within the environmental data back to the predetermined threshold range. The location module **208** is configured to determine the location of the monitoring devices **102**: (i) based on the information entered by the authorized user, (ii) using an indoor positioning system, (iii) using an absolute locating system, or (iv) a hybrid system. In a first embodiment, the location module **208** may determine the location of the monitoring device **102** and the appliances **106** are positioned nearby based on inputs from the user. Specifically, the IAQ system **10** may utilize an application that is installed on an Internet enabled device to provide the user with a number of questions about the structure **100**. For example, the application may ask generic questions about the structure **100**, which may include: i) number of bedrooms/bathrooms, ii) square footage of the structure, iii) which bathrooms are connected to bedrooms, iv) closest bathroom to the kitchen, v) how many levels does the structure have, vi) rough room dimensions, vii) other questions geared to determining the rough layout of the structure **100**, and viii) other similar questions. Next, the application may ask the user about the location of the devices within the structure **100**. For example, the application may ask generic questions about the location of the monitoring devices **102** and appliances **106**, which may include: i) is the monitoring device **102** located within the master bedroom or kitchen. Next, the application may ask the user for information about the appliances **106**. For example, the application may ask the user the CFM rating of the bathroom fan or the range hood. Once all of this information is inputted into the application by the user, the IAQ system **10** may ask the user which appliance **106** should be turned on when a specific monitoring device **106** measures a level that is outside of a predetermined threshold range.

In an alternative embodiment, the locating module **208** may utilized indoor positioning sensors that are built into each appliance **106** or maybe temporally attached to appliances **106**. For example, upon purchasing the IAQ system **10**, the user may be provided with a number of indoor positioning sensors that can be temporally attached to non-connected appliances **400**. Specifically, indoor positioning sensors may utilize one or a combination of the following technologies: i) magnetic positioning, ii) GPS along with dead reckoning, iii) positioning using visual markers (e.g., use of the camera that is built into the monitoring unit **102**), iv) visible light communication devices, v) infrared systems, vi) wireless technologies (e.g., Wi-Fi positioning system, Bluetooth Low Energy ("BLE"), iBeacon, other beacon technology, received signal strength, ultra wide-band technologies, RFID), or vii) other methods discussed in the papers that were attached to U.S. Provisional Application No. 62/789,501. The user then may be instructed to attach these sensors to these non-connected appliances **400**. Once these sensors are in place and the connected devices and monitoring devices **104** are turned on, the IAQ system **10** can determine which devices are closest to each monitoring device **102** along with the relative positioning of the monitoring devices **104** to one another. Based on this relative location, the IAQ system **10** can then ask the user for additional information about the functionality of each device and additional information about the room layouts. Once this information is entered into the IAQ system **10**, the IAQ system **10** will be able to determine the steps necessary to return a level contained within the environmental data back to the predetermined threshold range.

In a further alternative embodiment, the locating module **208** may utilize sensors that can provide the absolute location of each monitoring unit **102** and appliance **106** within the structure **100**. The absolute location system may require a user to upload a map of the structure **100** to the local server/database **110**. This map of the structure **100** may be generated based on: i) blueprints of the structure **100** or ii) determined by a device that is capable of mapping the structure **100** after the structure **100** was built. Such devices include software programs that can be loaded on a cellular phone or a robotic vacuum. In a particular example, the user may utilize a robotic vacuum to map the structure **100**. Once the structure **100** is mapped, the robotic vacuum can upload the map to the local server/database **110**. The IAQ system **10** can then place the monitoring devices **102** and the appliances **106** within the structure **100** based on the readings from indoor positioning systems. Once the IAQ system **10** has placed the monitoring devices **102** and the appliances **106** within the structure **100**, the user can then login to the local server/database **110** using an internet enabled device and can confirm their position. In an even further embodiment, the locating module **208** may use any combination of the methods described above. For example, the IAQ system **10** may ask the user a number of questions and then use the indoor positioning system in the above described embodiments.

e) Connectivity Module

The connectivity module **210** is a module that enables the monitoring unit **102** to send data to another device, such as the local server/database **110** or the central unit **104**. The connectivity module **210** may use any one, or combination, of the following wireless or wired technologies/communication protocols: Bluetooth (e.g., Bluetooth version 5), ZigBee, Wi-Fi (e.g., 802.11a, b, g, n), Wi-Fi Max (e.g., 802.16e), Digital Enhanced Cordless Telecommunications (DECT), cellular communication technologies (e.g., CDMA-1X, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, or LTE), near field communication (NFC), Ethernet (e.g., 802.3) FireWire, BLE, ZigBee, Z-Wave, 6LoWPAN, Thread, WIFI-ah, RFID, SigFox, LoRaWAN, Ingenu, Weightless, ANT, DigiMesh, MiWi, Dash7, WirelessHART, advanced message queuing, data distribution service, message queue telemetry transport, IFTTT, inter-integrated circuit, serial peripheral interface bus, RS-232, RS485, universal asynchronous receiver transmitter, USB, powerline network protocols, a custom designed wired or wireless communication technology, or any type of technologies/communication protocol listed within the papers that were attached to U.S. Provisional Application No. 62/789,501.

Using any one of the above technologies/communication protocols, the environment data that is collected by the monitoring unit **102** may be sent to a device outside of the monitoring unit **102** in at least three different ways. The first way is where the monitoring device **102** will only send the environment data at a predefined time interval. This predefined time interval (e.g., 30 seconds, 1 minute, 3 minutes, 5 minutes, 10 minutes, 30 minutes, every hour, every 24 hours, or anytime therebetween) may be preprogrammed into the IAQ system **10** or may be set by the user. It should be understood that in this method, the monitoring device **102** does not perform any calculations and instead raw sensor data is simply sent from the monitoring device **102** to the central unit **104** or the local server/database **110** for processing. This method is beneficial because it does not require that the monitoring device **102** perform calculations to determine if a level within the environmental data that is outside of the predefined threshold ranges. However, more data may be transmitted outside of the monitoring device **102** and there may be a lag between when an alert event occurs and when the IAQ system **10** detects the alert event.

A second way of sending environment data to a device that is outside of the monitoring device **102** is where the monitoring device **102** sends data only when an alert event occurs. In this method, the monitoring device **102** must have capabilities sufficient to process the raw data collected by the sensor **200** in order to determine if a level that is within the environmental data is outside of the predefined threshold range(s) or value(s). Upon making a determination that a level within the environmental data that is outside of the predefined threshold ranges, the monitoring device **102** sends this alert data to the central unit **104** or the local server/database **110** for the IAQ system **10** to perform the next steps. This method is beneficial because it requires the least amount of data to be sent from the monitoring device **102** to another device.

The third way of sending environment data to a device that is outside of the monitoring device **102** is a hybrid of the first and second methods. Specifically, the monitoring device **102**: i) sends the environment data at predefined intervals (e.g., 5 minutes, 10 minutes, 30 minutes, every hour, every 24 hours, or anytime therebetween) and ii) sends the environment data when a sensor alert occurs. The hybrid approach requires that the monitoring device **102** send the extra data that is required by the first way and have the additional processing power that is required by the second way. Nevertheless, this hybrid approach avoids the lag time that is described in a first way and allows the user to view historical environmental data that is below the alert level.

f) Other Module(s)

The monitoring devices **102** may include a microphone **214** and other electronic components **218** necessary to allow for voice control of the monitoring devices **102**. In addition, the microphone **214** and other electronic components **218** can be used to allow the monitoring device **102** to be controlled or operate with any virtual assistant (e.g., Amazon Alexa, Microsoft Cortana, Google Assistant, Samsung Bixby, Apple Siri, or any other similar virtual assistant). The monitoring devices **102** may also include a status indicator **216**, which provides a general indication of the indoor air quality at or near the monitoring devices **102**. For example, the monitoring devices **102** may show a red light if the air quality is bad, a green light if the air quality if good, and a yellow light if the air quality is between bad and good.

4) Exemplary Monitoring Devices

FIGS. **3A-5** illustrate exemplary monitoring devices **102** of the IAQ system **10**. Specifically, FIGS. **3A-3B** show two different embodiments of in-wall monitoring devices **502**, **504**. These in-wall monitoring devices **502**, **504** can be installed in the place of a switch or a power outlet. A limited version of this first embodiment of the in-wall monitoring device **502** is described within U.S. patent application Ser. No. 14/593,883, filed on Jan. 9, 2015, which is herein incorporated by reference. This first embodiment may have limited uses because it takes the place of a light switch; thus, the user loses the ability to control a lighting fixture or fan when using this monitoring device **502**. To overcome the limitations associated with the first embodiment **502**, the second embodiment **504** can be utilized without losing the ability to control a lighting fixture or fan. One example of where this second embodiment of the in-wall monitoring device **504** may be utilized is in connection with a non-connected appliance **400** (e.g., range hood/exhaust hood, bathroom fan, supply fan, evaporative cooler, air conditioner, HVAC, HRV, ERV, air cycler, air exchanger, CFIS, garage fan, space heaters, ceiling fans, dehumidifiers, humidifiers, space heaters, air ionizers, or air purifiers) that are affixed to the structure **100**. Specifically, the second embodiment **504** is configured to be wired between an electrical supply for the fan and the fan. This allows the power control module **206** contained within the monitoring device **590** to control whether power is supplied to the fan; thus, controlling when and how long the fan is ON/OFF. Overall, this configuration is desirable because: (i) it allows the IAQ system **10** to control the non-connected appliance **400** (i.e., fan) that are affixed to the structure **100** and (ii) it still allows the user to manually control the fan using the buttons **506**. It should be understood that other configurations of these in-wall monitoring devices **502**, **504** may be utilized. For example, the in-wall monitoring device may span/include multiple light switches and/or plugs.

FIGS. **4A-4D** show four different embodiments of plug-in monitoring devices **540**, **542**, **544**, **546** that are designed to be plugged into an electrical wall outlet **538**. In comparison to the in-wall monitoring devices **502**, **504**, the plug-in monitoring devices **540**, **542**, **544**, **546** are easier to install because they only require a user to plug them into the electrical wall outlet **538** and do not require a user to wire them into an in-wall switching device. Also, in contrast to controlling non-connected appliance **450** (i.e., fan) that are affixed to the structure **100**, these plug-in monitoring devices **540**, **542**, **544**, **546** can control non-connected appliances **400** (i.e., dehumidifier, humidifier, space heater, air ionizer, air purifier, portable fan, and other similar devices that circulate/modify the air) that are not affixed to the structure **100**.

FIG. **5** shows a battery-powered monitoring device **580**. This configuration allows the user to place the monitoring device in the location that they desire without trying to find a plug or light switch. FIG. **6** is an in-line monitoring unit **590**. This in-line monitoring unit **590** is configured to be wired between the non-connected appliance **450** (e.g., range hood/exhaust hood, bathroom fan, supply fan, evaporative cooler, air conditioner, HVAC, HRV, ERV, air cycler, air exchanger, CFIS, garage fan, space heater, ceiling fan, dehumidifier, humidifier, space heater, air ionizer, or air purifier) that is affixed to the structure **100**. This allows the power control module **206** contained within the monitoring device **590** to control whether power is supplied to the non-connected appliance **450**. In some embodiments, the monitoring device **590** includes control wires that tap into the appliances **106** operational centers to enable the monitoring device **590** to control the functionality (e.g., fan speed) of the appliance **106**. It should be understood that these are just a few examples of monitoring devices **102**, where additional monitoring devices **102** may have different shapes, additional functionality, additional features, and etc.

5) Exemplary Central Unit

FIG. **7** shows a table-based central unit **702**. As described above, the central unit **702** contains a display **220**, which can be used by the authorized user to review historical or current environmental data, which is/has been collected by the monitoring units **102**. In some embodiments, the central unit **702** is battery powered and/or can have all the functionality of a monitoring device **102**.

6) Exemplary Connected Appliances

FIGS. **8A-8C** show exemplary connected appliances **300** (e.g., range hood/exhaust hood, bathroom fan, supply fan, evaporative cooler, air conditioner, HVAC, HRV, ERV, air cycler, air exchanger, CFIS, garage fan, space heater, ceiling fan, dehumidifier, humidifier, space heater, air ionizer, or air purifier) that contain a CIAQ **50** device and are typically built into the structure. These connected appliances **300** contain circuits that enable the IAQ system **10** to control the operation of these connected appliances **300** without requiring additional devices. Thus, these connected appliances **300** at least contain a connectivity module **210** and a power control module **206**. In some embodiments, these connected appliances **300** include all of the modules contained within a monitoring device **102**. The inclusion of these additional modules may be beneficial because it provides the local server/database **110** with additional environmental data from other locations within the structure **100**. Specifically, FIG. **8A** shows a connected bathroom fan **304**, while FIG. **8B** shows a ceiling fan **308**. Additionally, FIG. **8C** shows a connected range hood **312**, one example of such is partly discussed in U.S. Provisional Application No. 62/772,724, filed on Nov. 29, 2018, which is hereby incorporated by reference.

FIG. **9** shows exemplary connected appliances **350** that include a CIAQ **50** device and are typically not built into the structure. Like the above described built-in connected appliances **300**, these non-built-in connected appliances **300** contain circuits that enable the IAQ system **10** to control the operation of these connected appliances. Specifically, FIG. **9** shows a connected air ionizer **352**.

7) Non-Exemplary Connected Appliances

FIGS. **10A-10B** show exemplary non-connected appliances **400** that are typically not built-in the structure or are portable. These non-connected appliances **400**, such as a portable humidifier **406** and a fan **410**, cannot communicate with the system **10** and therefore need a device that allows the system **10** to control these non-connected appliances **400**. Examples of CIAQ devices **50** that are designed to control these non-connected appliances **400** have been discussed above in connection with the monitoring devices **102**. Specifically, the in-wall monitoring device **504**, the plug-in monitoring device **540**, **542**, **544**, **546**, and the in-line device **590** can be used to control non-connected appliances **400**. Here, because both of these exemplary non-connected appliances **400** are portable and can be plugged into an electrical outlet, the user would likely utilize one of the plug-in monitoring device **540**, **542**, **544**, **546** to control these devices **406**, **410**.

FIGS. **11-12** show exemplary non-connected appliances **450** that are built into the structure **100**, such as a supply fan **454** and an air vent **458**. Due to the configuration of built-in non-connected appliances **450**, a CIAQ device **50** that is simply a controller may be utilized to connect these devices to the system **10**. The controller is similar to the monitoring device **102** because it can communicate with the IAQ system **10** and be used to control a non-connected appliance **400**. However, unlike the monitoring device **102**, the controller does not contain sensors or most of the modules contained within the monitoring devices **102**. Instead, the controller merely includes a connectivity module **210** and a power control module **206**. By only containing these two modules, the controller can be smaller, may be designed to be retrofitted into existing non-connected devices **450**, and can be utilized in locations where sensor data is not desired.

8) Local Server/Database

Typically, all environmental data that is generated by the IAQ system **10** passes through a wired and/or wireless network to the local server/database **110** that is accessible using an internet enabled device. The local server/database **110** may store the following information: i) maps of the structure **100**, ii) location of the monitoring units **102**, central units **104**, appliances **106** within the structure and their capabilities (e.g., a fan that can move **300** CFM), iii) physical information about each part (e.g., room) of the structure **100**, such as air volume, types of items contained with the part of the structure, ducting and etc., iv) occupant usage information about each part of the structure **100**, such as when that part is most used, by how many people or pets, v) baseline environmental data for each part of the structure **100**, vi) historical environmental data. The information listed above can be obtained by the local server/database **110** through various means. For example, the local server/database **110** may obtain a map of the structure **100** by pulling this information from a robot vacuum, while the occupant usage information may be obtained from the sensors that are housed within the monitoring devices **102** and/or the central unit **104**. It should be understood that the term local server/database refers to a server/database that is local in the terms of its association to the structure **100** and is not local in terms of physical location. In other words, the local server/database **110** is not physically located with the structure **100** and can be physically located anywhere in the world that is accessible via the internet.

Some or all of the above information will be used by the local server/database **110** as inputs to either a basic algorithm or a learning algorithm in order to determine: i) which appliance **106** to turn ON, ii) when to turn the appliance **106** on, and iii) how long to keep the appliance **106** ON. The basic algorithm may utilize a preset table that is contained within the local server/database **110** to make its determinations. For example, if a CO_2 alert is detected, the preset table will instruct the local server/database **110** to avoid circulating air from the basement into the rest of the structure **100**. Instead, the preset table will instruct the IAQ system **10** to turn ON the ventilation devices (e.g., bathroom fan) that are contained within the basement in order to vent the CO_2 outside of the structure **100**. Another example is if the IAQ system **10** determines a localized humidity alert in the bathroom, the preset table will instruct the IAQ system **10** to only turn ON the local bathroom fan and will not turn ON the HVAC system. However, if the humidity alert is not localized to the bathroom, then the preset table will instruct the IAQ system **10** to turn ON a large dehumidifier or run the HVAC system.

Alternatively, the IAQ system **10** may utilize a learning algorithm to make its determinations. Specifically, this learning algorithm will be trained using mock structure **100** setups. This training may be done from the factory or maybe done after the user buys and installs the system within the structure **100**. Training at the factory may be easier to accomplish because a trained algorithm can simply be installed on the IAQ system **10** prior to shipment. However, training at the factory may be less accurate in comparison to training the system after its bought and installed within the structure **100** because training within the structure **100** will be tailored to that structure **100**. Training within the structure **100** may first require that the user set up the system and provide all information about the monitoring devices **102**, central units **104** and the appliances **106**. Once this information is entered into the IAQ system **10**, the local server/database **110** can be trained using a preset algorithm to start from and continue training itself using various mocked up conditions for the specific structure **100**. A person from the factory can oversee the training of the algorithm to ensure that the system **10** is making the proper selections and/or to correct the system's **10** selections.

In other embodiments, the IAQ system **10** may be able to determine that sufficient environmental data is not being collected from certain regions of the structure **100**. In response to this determination, the IAQ system **10** will suggest that the user add more monitoring devices **102** within those locations. In addition, the IAQ system **10** may also suggest relocating various appliances **106** into other locations or adding more appliances **106** within the structure **100** to maximize the air quality. In other embodiments, the IAQ system **10** may be able to determine where the structure **100** lacks proper airflow. The IAQ system **10** then may propose solutions to correct for this lack of proper airflow.

9) Alerting Device

The alerting device **114** is an electronic device that can receive messages from the IAQ system **10** and more particularly the devices shown in FIGS. **1A-1G**. Examples of alerting devices **114** include, but are not limited to: i) cellular phones, ii) computers (e.g., laptops or desktops), iii) tablets,

iv) smartwatches, or v) devices that are designed to be alerting devices **114**. The alerting device **114** includes a software application that is installed thereon, which allows the alerting device **114** to display any information that is generated by the other components of the IAQ system **10** or any information that may be of use to the user. Such information may include, but is not limited to: i) displaying location, time, duration, and type of alert, ii) current levels of the environmental data and their associated ranges, iii) historical levels of the environmental data and their associated ranges, iv) comparisons of one structures current environmental data against historical environmental data collected from: a) this specific structure, b) other structures within the same neighborhood, c) other structures within the same area or zip code, d) other structures within the same region, e) other structures within the same country, and f) all the structures around the world, v) comparisons of one structures **101** current environmental data against current environmental data collected from: a) other structures within the same neighborhood, b) other structures within the same area or zip code, c) other structures within the same region, d) other structures within the same country, and e) all the structures around the world. The alerting device **114** may also display recommend appliances **106** to buy or how to reconfigure a user's current appliances **106** in order to maximize the air quality within the structure **100**. In addition, the alerting device may also display information about environmental conditions outside of the structure **100**.

10) Exemplary System within a Structure

FIG. **13** is a partial cut-away view of an operating environment **98**, which contains one of the exemplary systems **10** shown in FIGS. **1A-1G**. Specifically, this exemplary IAQ system **10** includes: (i) in-wall monitoring device **504**, (ii) plug-in monitoring device **542**, (iii) potable monitoring device **580**, (iv) central unit **702**, (v) connected range hood **312**, (vi) connected air ionizer **352**, (vii) non-connected humidifier **406**, (viii) non-connected supply fan **454**, and (ix) non-connected bathroom fan **460**. It should be understood that this is only exemplary and other configurations of the operating environments **98** are contemplated by this disclosure. FIGS. **14-34** describe logging in/setting up an account for this exemplary structure **100** that is shown in FIG. **13** and connecting an appliance **106** to the system **10**. Once the step up is finished, FIGS. **35-55** shows how this exemplary system **10** reacts to various sets of conditions.

11) Configuration of the System within a Structure

FIG. **14** shows a landing screen **1010** contained within a GUI **1000**. The landing screen **1010** allows a user to: (i) sign into their account by entering their email **1012** an password **1014** and pressing the sign-in button **1016** (shown in FIG. **15**) or (ii) create a new account by pressing the new account button **1018**. If the user signs into their account then screen **1022**, **1030**, **1044** shown in FIGS. **16-18** are skipped and the screen that is shown in FIG. **19** is displayed. Alternatively, if the user presses the new account button **1018**, then the screen **1022**, shown in FIG. **16**, is displayed within the GUI **1000**. Specifically, screen **1022** allows the user to enter their email **1024**, the password **1026** and a confirmation code **1028**. Once this information is entered into the system **10**, the system **10** creates an account for the user. Next, system **10** displays screen **1030**, which includes questions about the structure **100**. Such prompts may include: (i) approximate home square footage **1032**, (ii) number of rooms **1034**, and (iii) zip code **1036**. In response to this information, the GUI **1000** may display a map **1038** of the neighborhood that surrounds the user's structure **100**. After this information is entered into the system **10**, the user presses the next button **1040**.

The system **10** then takes the number of rooms entered by the user on screen **1030** and attempts to estimate the breakdown of the rooms in connection with screen **1044**, which is shown in FIG. **18**. Once the systems **10** estimates is shown in connection with screen **1044**, the user can alter the breakdown of the rooms. For example, the user can inform the system **10** that the structure only has 1 kitchen and not 2 kitchens. Once the user is finished confirming the room breakdown, the user can select the complete button **1048**. The selection of the complete button **1048** finishes the first part of setting up the system **10**. The next part of setting up the system **10** is shown in connection with FIGS. **19-32**, which will be discussed in greater detail below.

FIG. **19** shows the splash screen **1050** that is displayed within the GUI **1000** after: (i) signing into the system **10** and (ii) a structure **1000** has been set up (e.g., screens **1022**, **1030**, and **1044** have been completed). To set up a room within the structure **100**, the user can press the "+" sign **1052** on screen **1050**. Pressing button **1052**, displays screen **1058** that is shown in FIG. **20**. Specifically, screen **1058** is the first device/room setup screen. Here, screen **1058** allows the user to select one of the rooms (e.g., kitchen, living room, bedroom, bathroom) of the structure **100** using the dropdown **1060**, select a device CIAQ device **50** (e.g., in-wall monitoring device **504**, plug-in monitoring device **540**, in-line monitoring device **590**, etc.) using dropdown **1062**, a primary appliance **106** (e.g., non-connected bathroom fan **460**) using dropdown **1064**, a secondary appliance **106** (e.g., supply fan **454**) using dropdown **1066**, and a configuration type using the checkboxes **1068**. Once these selections have been made by the user, then the user can select the "OK" button **1070** to move to the next screen **1076**.

FIG. **21** shows screen **1076**, which displays the second step in configuring the device/room. In particular, this screen **1076** shows multiple steps in the installation of the primary appliance **106** (e.g., non-connected bathroom fan **460**) that was selected in connection with screen **1058**. Here, a few exemplary screens from these steps are shown in FIGS. **21-22**B. Specifically, FIG. **21** shows screen **1076** that includes a warning label **1082**, while FIG. **22**A-**22**B shows screens **1086**, **1088** that include wiring diagrams **1088**. Once the user has finished wiring the primary appliance **106** into the structure **100**, the user may press the complete button **1094** to bring the user to screen **2000** that is shown in FIG. **23**. Next, the GUI **1000** instructs the user to power on the primary appliance **106** to ensure that it is working. The use can confirm that the primary appliance **106** is working by pressing the "OK" button in **2002** in connection with screen **2000**.

Pressing the "OK" button **2002** brings the user to the third step in configuring the device/room, which is shown in connection with FIG. **24**. Specifically, FIG. **24** shows screen **2010**, which requests that the user select the CIAQ device **50** that is connected to the primary appliance **106** by selecting one of the CIAQ devices **50** from a plurality of CIAQ devices **50**. Once the internet enabled device that is running the GUI **1000** is connected to the CIAQ device **50**, the system **10** checks to see if the CIAQ device **50** is updated in connection with screen **2016** shown in FIG. **25**. Once the CIAQ **50** device is updated, the system **10**, will attempt to control the primary appliance **106** in connection with the screen **2020** shown in FIG. **26**. The system **10** then requests that the user confirm the operation of the primary appliance **106** in connection with the screen **2030** shown in FIG. **27**. If the test fails, then screen **2034** shown in FIG. **28** is displayed along with a trouble shooting screen **2038** shown in FIG. **29**. Alternatively, if the system **10** passes the test, screen **2040** is displayed in connection with FIG. **30**. The GUI **1000** finally display the screen **2042** that allows the user to display a tutorial/demo or take a quick tutorial.

Alternatively, if the user sets up a CIAQ device **50** that is located within a room that does not have an appliance **106**, the system **10** will ask the user to select an appliance **106** that should be utilized when an alert event occurs in connection with screen **2050**. The user selects this appliance **106** by selecting one of the radial buttons **2054** that are positioned adjacent to the names of the appliances **106**. It should be understood that alternative methods of determining which appliance **106** should be triggered are discussed in greater detail in other parts of this application. Other screens **2060** and **2064** that show other functionalities that are associated with the GUI are displayed in connection with FIGS. **33-34**. For example, screen **2060** shows that the user can set the frequency and threshold level required for an alert to be sent to the user's alerting device **114**. Additionally, screen **2064** shows how a user can place the system **10** in a do not disturb mode, which instructs the system **10** not to perform any tasks. It should be understood that only exemplary screens contained within the steps of configuring the device/room were displayed in connection with FIGS. **14-34** and as such additional screens are utilized to set up this system **10**.

12) Operation of the System Under Different Sets of Conditions

FIGS. **35-55** illustrate partial cut-away views of operating environments **98**, which contain the exemplary IAQ system **10** and are operating under various sets of conditions. Specifically, these scenarios show how the IAQ system **10** works when the IAQ system **10** is exposed to various conditions. At a high level, these scenarios display that the IAQ system **10** understands the configuration of the structure **100**, the location of the devices **102**, **104**, and **106** within the structure **100**, and that the IAQ system **10** can determine which appliance **106** should be utilized to bring the levels contained within the environmental data back within the predetermined threshold range.

FIGS. **35-40** illustrate the operation of an exemplary IAQ system **10** operating under a first set of conditions. Here, in FIG. **35**, the air that contains a certain level of pollutant is traveling throughout the ventilation system. After a predetermined amount of time, the concentration of the air pollutant exceeds the predetermined threshold value within the bedroom **60**. As shown in FIG. **36**, this environmental data is sent to the local server/database **110** from the monitoring device **102**, **542** contained within the bedroom **60**. After analysis of this data and data from other sensors, the local server/database **110** determines that the air pollutant is above a predetermined threshold value. The local server/database **110** then sends an alert to the alerting unit **114**. Upon further analysis of the collected environmental data, the local server/database **110**, determines that a local air purifier that is contained within the bedroom will not reduce the air pollutant below a predetermined threshold value because certain levels of the air pollutant can be traced throughout the house. Here, the IAQ system **10** determines that all appliances **106** that vent air outside of the house should be turned on. Accordingly, as shown in FIG. **38**, the local server/database **110** sends signals to the connected bathroom fan **304** and the connected range hood **312** to turn ON these devices. After the connected bathroom fan **304** and the connected range hood **312** have been running for a predefined amount of time, the IAQ system **10** determines that air needs to be brought into the house to ensure that the structure **100** is balanced. Thus, as shown in FIG. **39**, the local server/database **110** sends a signal to the CIAQ device **50** that is coupled to the supply fan **454** to turn ON this device. The IAQ system **10** will continue to monitor the levels of the air pollutant and once these levels are brought within their predetermined system thresholds, the system will turn OFF all of the devices **304**, **312**, and **454**, which is shown in FIG. **40**. It should be understood that one device **304** may stay on long then another device **312** within the structure **100** due to the layout of the structure **100**. Specifically, the bathroom fan **304** may be closer and may have a direct access to the bedroom **60** where the alert occurred and as such the bathroom fan **304** may run for a longer amount of time then the range hood **312**.

An alternative description of the scenario shown in FIGS. **35-40** may include the fact that the local server/database **110** may determine that it is best to turn ON all appliances **106** that vent air outside of the house in light of the level of the air pollutant and the fact that the alert occurred within the bedroom, where there is no direct ventilation appliance. In this description, the local server/database **110** only utilized data from the one monitoring device **102** contained within the bedroom and did not consider data from other monitoring devices **102** contained within the structure **100**. Another alternative description of the scenario shown in FIGS. **35-40** may include the fact that the IAQ system **10** may not send the environmental data to the local server/database **110** and instead the monitoring device **102** simply determines that the level of the air pollutant is too high to warrant additional analysis from the local server/database **110**. In this alternative description, the monitoring unit **102** sends an alert to the alerting unit **114** and then sends signals to turn ON the appliances **106**.

FIGS. **41-45** illustrate the operation of an exemplary IAQ system **10** operating under a second set of conditions. Here, in **41**, the air that contains a certain level of pollutant is traveling throughout the ventilation system. After a predetermined amount of time, the concentration of the air pollutant exceeds the predetermined threshold value within the bathroom **62**. As shown in FIG. **42**, this environmental data is sent to the local server/database **110** from the monitoring device **102**, **504** contained within the bathroom **62**. After analysis of this data and data from other sensors, the local server/database **110** determines that the air pollutant is above a predetermined threshold value. The local server/database **110** then sends an alert to the alerting unit **114**. Upon further analysis of the collected environmental data, the local server/database **110**, determines that a local connected bathroom fan **304** can reduce the air pollutant below a predetermined threshold value. Here, the IAQ system **10** turns on only the local connected bathroom fan **304**, as shown in FIG. **43**, and does not turn ON the entire HVAC system or any other systems. In other words, system **10** selectively picks the appliance **106** that can best resolve the issue at hand without turning on all appliances **106**. After the local connected bathroom fan **304** has been running for a predefined amount of time, the IAQ system **10** determines that air needs to be brought into the house to ensure that the structure **100** is balanced. Thus, as shown in FIG. **44**, the local server/database **110** sends a signal to the CIAQ device **50** that is coupled to the non-connected supply fan **454** to turn ON this device. The IAQ system **10** will continue to monitor the levels of the air pollutant and once these levels are brought within their predetermined sensor thresholds, the system will turn OFF the local connected bathroom fan **304**, which is shown in FIG. **45**.

An alternative description of the scenario shown in FIGS. **41-45** may include the fact that the IAQ system **10** may not send the environmental data to the local server/database **110** and instead the monitoring device **102** simply determines that the level of the air pollutant is too high to warrant additional analysis from the local server/database **110**. In this alternative description, the monitoring unit **102** sends an alert to the alerting unit **114** and then sends signals to turn ON the local bathroom fan **304**.

FIGS. **46-50** illustrate the operation of an exemplary IAQ system **10** operating under a third set of conditions. Here, in FIG. **46**, the air that contains a certain level of pollutants is traveling throughout the ventilation system. After a predetermined amount of time, the concentration of the air pollutant exceeds the predetermined threshold value within the living room **64**. As shown in FIG. **47**, this environmental data is sent to the local server/database **110** from the central unit **702** contained within the living room **64**. After analysis of this data and data from other sensors, the local server/database **110** that is contained within the central unit **702** determines that the air pollutant is above a predetermined threshold value. The local server/database **110** then sends an alert to the alerting unit **114**. Upon further analysis of the collected environmental data, the local server/database **110**, determines that a local air purifier that is contained within the living room **64** will not reduce the air pollutant below a predetermined threshold value because certain levels of the air pollutant can be traced throughout the house. Here, the IAQ system **10** determines that all appliances **106** that vent air outside of the house should be turned on. Accordingly, as shown in FIG. **48**, the local server/database **110** sends signals to the connected bathroom fan **304** and the connected range hood **312** to turn ON these devices. After the connected bathroom fan **304** and the connected range hood **312** have been running for a predefined amount of time, the IAQ system **10** determines that air needs to be brought into the house to ensure that the structure **100** is balanced. Thus, as shown in FIG. **49**, the local server/database **110** sends a signal to the CIAQ device **50** that is coupled to the non-connected supply fan **454** to turn ON this device. The IAQ system **10** will continue to monitor the levels of the air pollutant and once these levels are brought within their predetermined sensor thresholds, the system will turn OFF all of the devices, which is shown in FIG. **50**.

An alternative description of the scenario shown in FIGS. **46-50** may include the fact that the local server/database **110** may determine that it is best to turn ON all appliances **106** that vent air outside of the house in light of the level of the air pollutant and the fact that the alert occurred within the living room, where there is no direct ventilation appliance. In this description, the local server/database **110** only utilized data from the one central unit **702** contained within the living room and did not consider data from other monitoring devices contained within the structure **100**.

FIGS. **51-55** illustrate the operation of an exemplary IAQ system **10** operating under a fourth set of conditions. Here, in FIG. **51**, air pollutant travels from the cooktop into the range hood **312**. After a predetermined amount of time, the concentration of the air pollutant exceeds the predetermined threshold value. As shown in FIG. **52**, this environmental data is sent to the local server/database **110** from the connected range hood **312**. After analysis of this data, the local server/database **110** determines that the air pollutant is above a predetermined threshold value. The local server/database **110** then sends an alert to the alerting unit **114**. Upon further analysis of the collected environmental data, the local server/database **110**, determines that a local connected range hood **312** can reduce the air pollutant below a predetermined threshold value. Here, the IAQ system **10** only turns on the local connected range hood **312**, as shown in FIG. **53**. In other words, system **10** selectively turned on the appliance **106** that could efficiently resolve the issue at hand. After the local connected range hood **312** has been running for a predefined amount of time, the IAQ system **10** determines that air needs to be brought into the house to ensure that the structure **100** is balanced. Thus, as shown in FIG. **54**, the local server/database **110** sends a signal to the CIAQ device **50** that is coupled to the non-connected supply fan **454** to turn ON this device. The IAQ system **10** will continue to monitor the levels of the air pollutant and once these levels are brought within their predetermined sensor thresholds, the system will turn OFF the local connected range hood **312**, which is shown in FIG. **55**.

Another alternative description of the scenario shown in FIGS. **51-55** may include the fact that the IAQ system **10** may not send the environmental data to the local server/database **110** and instead the connected range hood **312** simply determines that the level of the air pollutant is too high to warrant additional analysis from the local server/database **110**. In this alternative description, the connected range hood **312** sends an alert to the alerting unit **114** and turns on the connected range hood **312**.

13) Exemplary Flowcharts Showing Operation of the System

FIGS. **56-72** contains flow charts describing how an exemplary embodiment of the IAQ system **10** functions. It should be understood from the start that this is only an exemplary embodiment and as such, the IAQ system **10** may function in other or alternative ways. For example, the flowcharts shown on pages 57-59 could be omitted without affecting the functionality of the IAQ system **10**. Additionally, steps could be added to these flowcharts. For example, if additional monitoring devices **102** and/or appliances **106** were added to the system, then the flowcharts would be altered to account for these changes. Further, it should be understood that a flowchart contained within one figure may reference one or multiple other flowcharts.

FIG. **56** describes the main system **10** algorithms **3000** that references or calls the other IAQ algorithms contained within the IAQ system **10**. This main system algorithm **3000** is preferably called less than every 10 minutes, more preferably less than every minute, and most preferably every 5 seconds. The first step **3002** in this main system algorithm **3000** is to check if the user has enabled the do not disturb mode, which was described above in connection with FIG. **34**. If do not disturb mode is enabled, then the system **10** will perform no additional steps in **3004**. However, if do not disturb mode is not enabled, then the system **10** will proceed down to check all connected bathroom fans **314** and/or in-wall monitoring devices **502**, **504** that are contained within bathrooms in steps **3006**. In other words, the bathroom algorithm **3010** is performed for each and every bathroom that contains a connected bathroom fan **314** and/or in-wall monitoring devices **502**, **504**. Proceeding to FIG. **60** that contains the bathroom algorithm **3010**, this algorithm **3010** again checks to ensure that the do not disturb mode is not enabled in step **3012**. If do not disturb mode is enabled, then no additional steps are performed. However, if do not disturb mode **3012** is not enabled, then the system **10** checks to see if the system **10** has collected new data from the sensors in step **3014**. The system **10** needs to check to see if there is new data because of the frequency that this algorithm **3010** is performed. If there is no new data, then the system **10** does not perform any additional steps within this algorithm. However, if the system did collect new data, then additional steps within this algorithm **3010** will be performed.

Next, the bathroom algorithm **3010** compares the derivative of the relative humidity against a first derivative of the relative humidity threshold in step **3016**. If the derivative of the relative humidity is greater than the first derivative of the relative humidity threshold, then the algorithm **3010** compares the derivative of the relative humidity against a second derivative of the relative humidity threshold in step **3018**. If the derivative of the relative humidity is greater than the second derivative of the relative humidity threshold in step **3018**, then the system **10** turns the connected bathroom fan **314** to level number 2 or the highest level in step **3020**. Alternatively, if the derivative of the relative humidity is less than the second derivative of the relative humidity threshold in step **3018**, then the system **10** turns the connected bathroom fan **314** to level number 1 or the lowest level in step **3022**.

If the derivative of the relative humidity is less than the first derivative of the relative humidity threshold in step **3016**, then the algorithm **3010** compares CO_2 levels from the sensors **200** against a first CO_2 threshold in step **3024**. If the CO_2 level is greater than the first CO_2 threshold, then the algorithm **3010** compares the CO_2 level against a second CO_2 threshold in step **3026**. If the CO_2 level is greater than the second CO_2 threshold in step **3026**, then the system **10** turns the connected bathroom fan **314** to level number 2 or the highest level in step **3028**. Alternatively, if the CO_2 level is less than the second CO_2 threshold in step **3026**, then the system **10** turns the connected bathroom fan **314** to level number 1 or the lowest level in step **3022**.

If the CO_2 level is less than the first CO_2 threshold in step **3024**, then the algorithm **3010** compares the relative humidity levels from sensors **200** against a first relative humidity threshold in step **3032**. If the relative humidity level is greater than the first relative humidity threshold, then the algorithm **3010** compares the relative humidity level against a second relative humidity threshold in step **3034**. If the relative humidity level is greater than the second relative humidity threshold in step **3034**, then the system **10** turns the connected bathroom fan **314** to level number 2 or the highest level in step **3036**. Alternatively, if the relative humidity level is less than the second relative humidity threshold in step **3034**, then the system **10** turns the connected bathroom fan **314** to level number 1 or the lowest level in step **3038**.

If the relative humidity level is less than the first relative humidity threshold in step **3032**, then the algorithm **3010** compares the TVOC levels from sensors **200** against a first TVOC threshold in step **3040**. If the TVOC level is greater than the first TVOC threshold, then the algorithm **3010** compares the TVOC level against a second TVOC threshold in step **3042**. If the TVOC level is greater than the second TVOC threshold in step **3042**, then the system **10** turns the connected bathroom fan **314** to level number 2 or the highest level in step **3046**. Alternatively, if the TVOC level is less than the second TVOC threshold in step **3042**, then the system **10** turns the connected bathroom fan **314** to level number 1 or the lowest level in step **3038**. Last, if the TVOC level is less than the first TVOC threshold in step **3040**, then the algorithm **3010** does not alter the fan speed and the algorithm is finished in step **3050**.

Returning to FIG. **56**, once the bathroom algorithm **3010** is performed for each and every bathroom that contains a connected bathroom fans **314** and/or in-wall monitoring devices **502**, **504**, then the range hood algorithm **3500** is performed for each and every room that contains a connected range hood **312** and/or in-line monitoring device **590** that is connected to a range hood in step **3490**. Proceeding to FIG. **61** that contains the range hood algorithm **3500**, this algorithm **3500** again checks to ensure that the do not disturb mode is not enabled in step **3512**. If do not disturb mode is enabled, then no additional steps are performed. However, if do not disturb mode **3512** is not enabled, then the system **10** checks to see if the system **10** has collected new data from the sensors in step **3514**. The system **10** needs to check to see if there is new data because of the frequency that this algorithm **3500** is performed. If there is no new data, then the system **10** does not perform any additional steps within this algorithm **3500**. However, if the system did collect new data, then additional steps within this algorithm **3500** will be performed.

Next, the range hood algorithm **3500** compares the derivative of the TVOC against a first derivative of the TVOC threshold in step **3516**. If the derivative of the TVOC is greater than the first derivative of the TVOC threshold, then the algorithm **3500** compares the derivative of the TVOC against a second derivative of the TVOC threshold in step **3518**. If the derivative of the TVOC is greater than the second derivative of the TVOC threshold, then the algorithm **3500** compares the derivative of the TVOC against a third derivative of the TVOC threshold in step **3520**. Alternatively, if the derivative of the TVOC is less than the second derivative of the TVOC threshold in step **3518**, then the system **10** turns the connected range hood **312** to level number 1 or the lowest level in step **3522**. If the derivative of the TVOC is greater than the third derivative of the TVOC threshold in step **3520**, then the system **10** turns the connected range hood **312** to level number 3 or the highest level in step **3524**. Alternatively, if the derivative of the TVOC is less than the third derivative of the TVOC threshold in step **3520**, then the system **10** turns the connected range hood **312** to level number 2 or the middle level in step **3526**.

If the derivative of the TVOC is less than the first derivative of the TVOC threshold in step **3516**, then the algorithm **3500** compares CO_2 levels from the sensors **200** against a first CO_2 threshold in step **3530**. If the CO_2 level is greater than the first CO_2 threshold, then the algorithm **3500** compares the CO_2 level against a second CO_2 threshold in step **3532**. If the CO_2 level is greater than the second CO_2 threshold, then the algorithm **3500** compares the CO_2 level against a third CO_2 threshold in step **3534**. Alternatively, if the CO_2 level is less than the second CO_2 threshold in step **3532**, then the system **10** turns the connected range hood **312** to level number 1 or the lowest level in step **3522**. If the CO_2 level is greater than the third CO_2 threshold in step **3534**, then the system **10** turns the connected range hood **312** to level number 3 or the highest level in step **3536**. Alternatively, if the CO_2 level is less than the third CO_2 threshold in step **3534**, then the system **10** turns the connected range hood **312** to level number 2 or the middle level in step **3526**.

If the CO_2 level is less than the first CO_2 threshold in step **3530**, then the algorithm **3500** compares relative humidity levels from the sensors **200** against a first relative humidity threshold in step **3540**. If the relative humidity level is greater than the first relative humidity threshold, then the algorithm **3500** compares the relative humidity level against a second relative humidity threshold in step **3542**. If the relative humidity level is greater than the second relative humidity threshold, then the algorithm **3500** compares the relative humidity level against a third relative humidity threshold in step **3544**. Alternatively, if the relative humidity level is less than the second relative humidity threshold in step **3542**, then the system **10** turns the connected range hood **312** to level number 1 or the lowest level in step **3546**. If the relative humidity level is greater than the third relative humidity threshold in step **3544**, then the system **10** turns the connected range hood **312** to level number 3 or the highest level in step **3548**. Alternatively, if the relative humidity level is less than the third relative humidity threshold in step **3544**, then the system **10** turns the connected range hood **312** to level number 2 or the middle level in step **3550**.

If the relative humidity level is less than the first relative humidity threshold in step **3540**, then the algorithm **3500** compares TVOC levels from the sensors **200** against a first TVOC threshold in step **3552**. If the TVOC level is greater than the first TVOC threshold, then the algorithm **3500** compares the TVOC level against a second TVOC threshold in step **3554**. If the TVOC level is greater than the second TVOC threshold, then the algorithm **3500** compares the TVOC level against a third TVOC threshold in step **3556**. Alternatively, if the TVOC level is less than the second TVOC threshold in step **3554**, then the system **10** turns the connected range hood **312** to level number 1 or the lowest level in step **3546**. If the TVOC level is greater than the third TVOC threshold in step **3556**, then the system **10** turns the connected range hood **312** to level number 3 or the highest level in step **3560**. Alternatively, if the TVOC level is less than the third TVOC threshold in step **3556**, then the system **10** turns the connected range hood **312** to level number 2 or the middle level in step **3550**. Last, if the TVOC level is less than the first TVOC threshold in step **3552**, then the algorithm **3500** does not alter the fan speed and the algorithm is finished in step **3562**.

Returning to FIG. **56**, once the range hood algorithm **3500** is performed for each and every connected range hood **312** and/or in-line monitoring device **590** that is connected to a range hood, then the AQ algorithm **3710** is performed for all room sensors in step **3700**. Proceeding to FIG. **62** that contains kitchen algorithm **4000**, a bathroom algorithm **5000**, and a living room/bedroom algorithm **6000**. Like before, the AQ algorithm **3710** again checks to ensure that the do not disturb mode is not enabled in step **3712**. If do not disturb mode is enabled, then no additional steps are performed. However, if do not disturb mode **3712** is not enabled, then the system **10** checks to see if the system **10** has collected new data from the sensors in step **3714**. The system **10** needs to check to see if there is new data because of the frequency that this algorithm **3700** is performed. If there is no new data, then the system **10** does not perform any additional steps within this algorithm **3700**. However, if the system collects new data, then additional steps within this algorithm **3700** will be performed.

The kitchen algorithm **4000** first determines if a kitchen monitoring device **102** is connected to a range hood in step **4010**. If the kitchen monitoring device **102** is connected to a range hood, then the kitchen range hood algorithm **4100** is performed. This algorithm is almost identical to the range hood algorithm **3500** that is discussed above in connection with FIG. **61**. The only difference is that algorithm **4100** substitutes PM2.5 for the derivative of the TVOC. Thus, for the sake of brevity, algorithm **4100** will be shown in FIG. **63**, but will not be described in additional detail herein. Alternatively, if the kitchen monitoring device **102** is not connected to a range hood, then the kitchen-house algorithm **4200** is performed. The kitchen-house algorithm **4200** is nearly identical to the kitchen range hood algorithm **4100**. The only difference between these algorithms is the fact the kitchen-house algorithm **4200** changes the fan speed of the whole-house ventilation system, while the kitchen range hood algorithm **4100** only changes the fan speed of the range hood. For the same reasons as discussed above and for the sake of brevity, algorithm **4200** will be shown in FIG. **64**, but will not be described in additional detail herein.

The bathroom algorithm **5000** determines if a bathroom monitoring device **102** is connected to a bathroom fan in step **5010**. If the bathroom monitoring device **102** is connected to a bathroom fan, then the bathroom fan algorithm **5100** is performed. This algorithm is almost identical to the range hood algorithm **3010** that is discussed above in connection with FIG. **60**. The only difference is that algorithm **5100** substitutes PM2.5 for the derivative of the relative humidity. Thus, for the sake of brevity, algorithm **5100** will be shown in FIG. **65**, but will not be described in additional detail herein. Alternatively, if the bathroom monitoring device **102** is not connected to a bathroom fan, then the bathroom-house algorithm **4200** is performed. The bathroom-house algorithm **5200** is identical to the kitchen-house algorithm **4200**. For the same reasons as discussed above and for the sake of brevity, algorithm **5200** will be shown in FIG. **66**, but will not be described in additional detail herein.

The living room/bedroom algorithm **6000** first checks to see if the structure **100** has a HERV in step **6010**. If there is a HERV, then the system **10** runs the living room/bedroom-house algorithm **6200**. The living room/bedroom house algorithm **6200** is identical to the kitchen-house algorithm **4200**. For the same reasons as discussed above and for the sake of brevity, algorithm **6200** will be shown in FIG. **68**, but will not be described in additional detail herein. Alternatively, if the structure **100** does not have an entire house ventilation system, then the system **10** will pick the bathroom fan that is closest to the monitoring device **102** in step **6012**. The living room/bedroom fan algorithm **5100** is identical to the bathroom fan algorithm **5100**. Thus, for the sake of brevity, algorithm **6100** will be shown in FIG. **67**, but will not be described in additional detail herein.

While system **10** is performing the above algorithms, the system **10** may receive multiple different requests from different monitoring devices **102**. For example, the connected range hood may be instructing the fan speed to be equal to the max level, while the kitchen-room based sensor may be instructing the fan speed be equal to the lowest level. This is possible due to the concentration of pollutants in a single location. Thus, the algorithms shown in FIGS. **69**-**72** ensure that the highest fan speed is chosen for the selected device. Specifically, the algorithms shown in these figures compare all inputs from the monitoring devices **102** and then select the highest level. Specifically, the algorithm **7000** is shown in FIG. **69** is related to the bathroom fans, algorithm **7500** that is shown in FIG. **70** is related to the range hoods, algorithm **8000** that is shown in FIG. **71** is related to a HERV. Also, to balance the air exhaust from these devices, the system **10** utilizes a supply fan or make-up air damper to help ensure that the house stays balanced. Similar to how the monitoring devices **102** can have different levels, algorithms **7000**, **7500**, and **8000** can also have different requested levels from the supply fan. Thus, algorithm **8500** is utilized to ensure that the highest requested supply fan setting is utilized.

FIG. **57** contains the building code algorithm **9000** that ensures that more than a predefined amount of air is vented out of the structure **100** over a predefined amount of time. Similar to the other algorithms that are discussed above, building code algorithm **9000** first checks to see if the do not disturb setting is on in step **9010**. If the do not disturb is on, then the building code algorithm **9000** does not perform any additional steps. However, if the do not disturb is off, then the building code algorithm **9000** checks to see if the system **10** utilizes the basic ventilation algorithm **9100** or the advanced algorithm **9500**. Referring to FIG. **58**, if the system **10** is set up to use the basic ventilation algorithm **9100**, the system **10** first determines a fan speed and time that the whole house ventilation fan needs to run in order to meet the code requirements. For example, a lower fan speed will require additional running time. Next, the algorithm **9100** splits the total run time of the fan over four hour blocks and then runs the fan for that set amount of time. For example, if the daily run time of the fan was calculated to be 8 hours, then the fan would need to run for 1.33 hours out of every 4 hours. Unlike the basic algorithm **9100** shown in FIG. **59**, the advanced algorithm **9500** shown in FIG. **59** gives credit for all ventilation devices (bathroom fans, range hoods, etc.) contained within the structure **100**.

In an alternative embodiment, the building code algorithm **9000** may be replaced by the algorithms described within U.S. patent application Ser. No. 16/243,056 or 16/242,498, both of which are fully incorporated herein by reference.

Below is a list of the acronyms that are used in connection with FIGS. **56-72**:

Acronym	Description
	FIG. 56
IAQ_Algo	Indoor air quality main algorithm 3000
DnD	Do not disturb mode for the entire system 3002
NB_BthFSWs	Number of connected bath fan switches
BthF_Algo	Bath fan control algorithm 3010
Nb_RngHSWs	Number of connected range hood switches
RngH_Algo	Range hood control algorithm 3490
Nb_AQMntrs	Number of connected air quality monitors (Room Monitoring Devices 102)
AQMntr_Algo	Air quality monitor algorithm 3710
	FIG. 57
CodeVentHandler	Building code Algorithm 9000
DnD	Do not disturb mode for the entire system 3010
CodeMode	User preference on code mode
WhHBasicCodeVent	Algorithm for basic whole house ventilation mode 9100
WhHAdvCodeVent	Algorithm for advanced whole house ventilation mode 9500
	FIG. 58
WhHBasicCodeVent	Algorithm for basic whole house ventilation mode 9100
maxSpeed	Maximum speed available on the ventilation device
WhHReq	The ventilation requirement for the whole house as calculated by the building code's equations
RunTime	The required portion of time to run the ventilation device
WhVentcfm	Airflow rate of the whole house ventilation device
CfmRateS1	Airflow rate of the whole house ventilation device at speed 1
CfmRateS2	Airflow rate of the whole house ventilation device at speed 2
CfmRateS3	Airflow rate of the whole house ventilation device at speed 3
S1	set device to first speed
S2	set device to second speed
S3	set device to third speed
T1	The required portion of time based on the air flow rate for speed 1
T2	The required portion of time based on the air flow rate for speed 2
T3	The required portion of time based on the air flow rate for speed 3
Timer	A running timer for reference
WhFanState	The recommended whole house ventilation device's state based on the whole house ventilation algorithm
UpdateFanState	A call of central algorithm to handle updating the fan state for a the whole house ventilation device
Rst	Reset the reference timer
	FIG. 59
WhHAdvCodeVent	Algorithm for advanced whole house ventilation mode
VentCounter	A counter of the accumalated airflow rate from the whole house ventilation device as well as other connected ventilation devices
ReqCFM	The ventilation requirement for the whole house as calculated by the building code's equations
WhHouseDevice	The ventilation device set as the whole house ventilation device
Max	Maximum speed available on the whole houseventilation device
CFMLimit	The limit needed for the counter to reach to allow the whole house ventilation device to be turned off
WhVentActive	Is the whole house ventilation device running
WhFanState	The recommended whole house ventilation device's state based on the whole house ventilation algorithm
UpdateFanState	A call of central algorithm to handle updating the fan state for a the whole house ventilation device
Nb_BthFSWs	Number of connected bath fan switches
Nb_RngHSWs	Number of connected range hood switches

-continued

Acronym	Description
NbSmartPlugs	Number of devices connected to a smart plug
FanRealState	The current state of the device's fan as reported by the hardware
DeviceSiCFM	The device's airflow rate
NegLimit	The negative limit before reseting the counter
	FIG. 60
BathFanAlgo	Air quality algorithm for a bath fan switch 3010
DnD	Do not disturb mode for this device 3012
New_Data	was the sensor data updated since the last execution 3014
dRH	Derivative of the relative humidity
dRHLIMIT1	First threshold for the derivative of the relative humidity
dRHLIMIT2	Second threshold for the derivative of the relative humidity
CO2	CO2 level from the sensor data
CO2LIMIT1	First threshold for the CO2 level
CO2LIMIT2	Second threshold for the CO2 level
TVOC	TVOC level from the sensor data
VOCLIMIT1	First threshold for the TVOC level
VOCLIMIT2	Second threshold for the TVOC level
RH	Relative Humidity level from the sensor data
RHLIMIT1	First threshold for the relative humidity
RHLIMIT2	Second threshold for the relative humidity
BathFanState	The recommended bath fan state based on the air quality algorithm
UpdateFanState	A call of central algorithm to handle updating the fan state for a bathfan switch
s1	set device to first speed
s2	set device to second speed
s3	set device to third speed
maxSpeed	Maximum speed available on the ventilation device
	FIG. 61
RngH_Algo	Air quality algorithm for a range hood switch 3500
DnD	Do not disturb mode for this device 3512
New_Data	was the sensor data updated since the last execution 3514
dVOC	Derivative of the relative humidity
dVOCLIMIT1	First threshold for the derivative of the TVOC
dVOCLIMIT2	Second threshold for the derivative of TVOC
dVOCLIMIT3	Third threshold for the derivative of TVOC
CO2	CO2 level from the sensor data
CO2LIMIT1	First threshold for the CO2 level
CO2LIMIT2	Second threshold for the CO2 level
CO2LIMIT3	Third threshold for the CO2 level
TVOC	TVOC level from the sensor data
VOCLIMIT1	First threshold for the TVOC level
VOCLIMIT2	Second threshold for the TVOC level
VOCLIMIT3	Third threshold for the TVOC level
RH	Relative Humidity level from the sensor data
RHLIMIT1	First threshold for the relative humidity
RHLIMIT2	Second threshold for the relative humidity
RHLIMIT3	Third threshold for the relative humidity
RngHoodState	The recommended range hood state based on the air quality algorithm
UpdateFanState	A call of central algorithm to handle updating the fan state for a rangehood switch
s1	set device to first speed
s2	set device to second speed
s3	set device to third speed
maxSpeed	Maximum speed available on the ventilation device
	FIG. 62
AQMntrAlgo	Air quality algorithm for an AQ Monitor 3710
DnD	Do not disturb mode for this device 3712
NewData	was the sensor data updated since the last execution 2714
Location	The current location of the AQ monitor
C_RngH	Is there a connected range hood in this kitchen 4010
C_BthF	Is there a connected bath fan in this bathroom 5010
WhVentDevice	The ventilation device set as the whole house ventilation device 6010
C_BthFNear	Is there a linked bath fan set by the user 6012
AQMnt_Ktchn_RH	Air quality algorithm for an AQ Monitor in a Kitchen with range hood 4100
AQMnt_Ktchn_WH	Air quality algorithm for an AQ Monitor in a Kitchen without range hood 4200
AQMnt_BthR_BF	Air quality algorithm for an AQ Monitor in a bathroom with bath fan 5100

-continued

Acronym	Description
AQMnt_BthR_WH	Air quality algorithm for an AQ Monitor in a bathroom without bath fan 5200
AQMnt_Room_WH	Air quality algorithm for an AQ Monitor in a room without user linked bath fan 6200
AQMnt_Room_BF	Air quality algorithm for an AQ Monitor in a room with user linked bath fan 6100

FIG. 63

AQMnt_Ktchn_RH	Air quality algorithm for an AQ Monitor in a Kitchen with range hood 4100
NewData	Do not disturb mode for this device
PM2.5	PM2.5 level from the sensor data
PM2.5LIMIT1	First threshold for the PM2.5 level
PM2.5LIMIT2	Second threshold for the PM2.5 level
PM2.5LIMIT3	Third threshold for the PM2.5 level
CO2	CO2 level from the sensor data
CO2LIMIT1	First threshold for the CO2 level
CO2LIMIT2	Second threshold for the CO2 level
CO2LIMIT3	Third threshold for the CO2 level
TVOC	TVOC level from the sensor data
VOCLIMIT1	First threshold for the TVOC level
VOCLIMIT2	Second threshold for the TVOC level
VOCLIMIT3	Third threshold for the TVOC level
RH	Relative Humidity level from the sensor data
RHLIMIT1	First threshold for the relative humidity
RHLIMIT2	Second threshold for the relative humidity
RHLIMIT3	Third threshold for the relative humidity
RngHoodState	The recommended range hood state based on the air quality algorithm
UpdateFanState	A call of central algorithm to handle updating the fan state for a rangehood switch
s1	set device to first speed
s2	set device to second speed
s3	set device to third speed
maxSpeed	Maximum speed available on the ventilation device

FIG. 64

AQMnt_Ktchn_WH	Air quality algorithm for an AQ Monitor in a Kitchen without range hood 4200
NewData	Do not disturb mode for this device
PM2.5	PM2.5 level from the sensor data
PM2.5LIMIT1	First threshold for the PM2.5 level
PM2.5LIMIT2	Second threshold for the PM2.5 level
PM2.5LIMIT3	Third threshold for the PM2.5 level
CO2	CO2 level from the sensor data
CO2LIMIT1	First threshold for the CO2 level
CO2LIMIT2	Second threshold for the CO2 level
CO2LIMIT3	Third threshold for the CO2 level
TVOC	TVOC level from the sensor data
VOCLIMIT1	First threshold for the TVOC level
VOCLIMIT2	Second threshold for the TVOC level
VOCLIMIT3	Third threshold for the TVOC level
RH	Relative Humidity level from the sensor data
RHLIMIT1	First threshold for the relative humidity
RHLIMIT2	Second threshold for the relative humidity
RHLIMIT3	Third threshold for the relative humidity
WhFanState	The recommended whole house ventilation device's fan state based on the air quality algorithm
UpdateFanState	A call of central algorithm to handle updating the fan state for a the whole house ventilation device
s1	set device to first speed
s2	set device to second speed
s3	set device to third speed
maxSpeed	Maximum speed available on the ventilation device

FIG. 65

AQMnt_BthR_BF	Air quality algorithm for an AQ Monitor in a bathroom with bath fan 5100
NewData	Do not disturb mode for this device
PM2.5	PM2.5 level from the sensor data
PM2.5LIMIT1	First threshold for the PM2.5 level
PM2.5LIMIT2	Second threshold for the PM2.5 level
PM2.5LIMIT3	Third threshold for the PM2.5 level
CO2	CO2 level from the sensor data
CO2LIMIT1	First threshold for the CO2 level
CO2LIMIT2	Second threshold for the CO2 level
CO2LIMIT3	Third threshold for the CO2 level

-continued

Acronym	Description
TVOC	TVOC level from the sensor data
VOCLIMIT1	First threshold for the TVOC level
VOCLIMIT2	Second threshold for the TVOC level
VOCLIMIT3	Third threshold for the TVOC level
RH	Relative Humidity level from the sensor data
RHLIMIT1	First threshold for the relative humidity
RHLIMIT2	Second threshold for the relative humidity
RHLIMIT3	Third threshold for the relative humidity
BathFanState	The recommended bath fan state based on the air quality algorithm
UpdateFanState	A call of central algorithm to handle updating the fan state for a bathfan switch
s1	set device to first speed
s2	set device to second speed
s3	set device to third speed
maxSpeed	Maximum speed available on the ventilation device
	FIG. 66
AQMnt_BthR_WH	Air quality algorithm for an AQ Monitor in a bathroom without bath fan 5200
NewData	Do not disturb mode for this device
PM2.5	PM2.5 level from the sensor data
PM2.5LIMIT1	First threshold for the PM2.5 level
PM2.5LIMIT2	Second threshold for the PM2.5 level
PM2.5LIMIT3	Third threshold for the PM2.5 level
CO2	CO2 level from the sensor data
CO2LIMIT1	First threshold for the CO2 level
CO2LIMIT2	Second threshold for the CO2 level
CO2LIMIT3	Third threshold for the CO2 level
TVOC	TVOC level from the sensor data
VOCLIMIT1	First threshold for the TVOC level
VOCLIMIT2	Second threshold for the TVOC level
VOCLIMIT3	Third threshold for the TVOC level
RH	Relative Humidity level from the sensor data
RHLIMIT1	First threshold for the relative humidity
RHLIMIT2	Second threshold for the relative humidity
RHLIMIT3	Third threshold for the relative humidity
WhFanState	The recommended whole house ventilation device's fan state based on the air quality algorithm
UpdateFanState	A call of central algorithm to handle updating the fan state for a the whole house ventilation device
s1	set device to first speed
s2	set device to second speed
s3	set device to third speed
maxSpeed	Maximum speed available on the ventilation device
	FIG. 67
AQMnt_Room_BF	Air quality algorithm for an AQ Monitor in a room with user linked bath fan 6100
NewData	Do not disturb mode for this device
PM2.5	PM2.5 level from the sensor data
PM2.5LIMIT1	First threshold for the PM2.5 level
PM2.5LIMIT2	Second threshold for the PM2.5 level
PM2.5LIMIT3	Third threshold for the PM2.5 level
CO2	CO2 level from the sensor data
CO2LIMIT1	First threshold for the CO2 level
CO2LIMIT2	Second threshold for the CO2 level
CO2LIMIT3	Third threshold for the CO2 level
TVOC	TVOC level from the sensor data
VOCLIMIT1	First threshold for the TVOC level
VOCLIMIT2	Second threshold for the TVOC level
VOCLIMIT3	Third threshold for the TVOC level
RH	Relative Humidity level from the sensor data
RHLIMIT1	First threshold for the relative humidity
RHLIMIT2	Second threshold for the relative humidity
RHLIMIT3	Third threshold for the relative humidity
BathFanState	The recommended bath fan state based on the air quality algorithm
UpdateFanState	A call of central algorithm to handle updating the fan state for a bathfan switch
s1	set device to first speed
s2	set device to second speed
s3	set device to third speed
maxSpeed	Maximum speed available on the ventilation device
	FIG. 68
AQMnt_Room_WH	Air quality algorithm for an AQ Monitor in a room without user linked bath fan 6200

-continued

Acronym	Description
NewData	Do not disturb mode for this device
PM2.5	PM2.5 level from the sensor data
PM2.5LIMIT1	First threshold for the PM2.5 level
PM2.5LIMIT2	Second threshold for the PM2.5 level
PM2.5LIMIT3	Third threshold for the PM2.5 level
CO2	CO2 level from the sensor data
CO2LIMIT1	First threshold for the CO2 level
CO2LIMIT2	Second threshold for the CO2 level
CO2LIMIT3	Third threshold for the CO2 level
TVOC	TVOC level from the sensor data
VOCLIMIT1	First threshold for the TVOC level
VOCLIMIT2	Second threshold for the TVOC level
VOCLIMIT3	Third threshold for the TVOC level
RH	Relative Humidity level from the sensor data
RHLIMIT1	First threshold for the relative humidity
RHLIMIT2	Second threshold for the relative humidity
RHLIMIT3	Third threshold for the relative humidity
WhFanState	The recommended whole house ventilation device's fan state based on the air quality algorithm
UpdateFanState	A call of central algorithm to handle updating the fan state for a the whole house ventilation device
s1	set device to first speed
s2	set device to second speed
s3	set device to third speed
maxSpeed	Maximum speed available on the ventilation device
	FIG. 69
UpdateFan	Central algorithm to handle updating the fan state for a switch controlled bathfan 7000
MaxSpeed	Maximum speed available on the ventilation device
RequestedSpeed	The speed to update the ventilation device's fan to
WhFanState	The recommended whole house ventilation device's state based on the whole house ventilation algorithm (if applicable)
BathFanState	The recommended bath fan state based on the smart switch's air quality algorithm
AQMntrs.BathFanState	The recommended bath fan state based on the AQ Monitor's air quality algorithm
FanCloudState	The status of the fan's ventilation device on the cloud
CSFan	Is there a connected supply fan in this smart home system
MUAD	Is there a connected Make up Air Damper in this smart home system
SFanState	The recommended supply fan state by the exhaust ventilation device
MUADState	The recommended MUAD state by the exhaust ventilation device
UpdateFanState	Central algorithm to handle updating the device's (Sfan or MUAD) fan state
	FIG. 70
UpdateFan	Central algorithm to handle updating the fan state for a switch controlled rangehood 7500
MaxSpeed	Maximum speed available on the ventilation device
RequestedSpeed	The speed to update the ventilation device's fan to
WhFanState	The recommended whole house ventilation device's state based on the whole house ventilation algorithm (if applicable)
RangeHoodState	The recommended range hood state based on the smart switch's air quality algorithm
AQMntrs.RangeHoodState	The recommended range hood state based on the AQ Monitor's air quality algorithm
FanCloudState	The status, on the cloud, of the fan's ventilation device
CSFan	Is there a connected supply fan in this smart home system
MUAD	Is there a connected Make up Air Damper in this smart home system
SFanState	The recommended supply fan state by the exhaust ventilation device
MUADState	The recommended MUAD state by the exhaust ventilation device
UpdateFanState	Central algorithm to handle updating the device's (Sfan or MUAD) fan state
	FIG. 71
UpdateFan	Central algorithm to handle updating the fan state for a HERV 8000
RequestedSpeed	The speed to update the ventilation device's fan to
WhFanState	The recommended whole house ventilation device's state based on the whole house ventilation algorithm (if applicable)

-continued

Acronym	Description
NB_BthFSWs	Number of connected bath fan switches
Nb_RngHSWs	Number of connected range hood switches
Nb_AQMntrs	Number of connected air quality monitors (Room sensors)
HERVState	The state of the HERV as recommended by each device
FanCloudState	The status, on the cloud, of the fan's ventilation device
	FIG. 72
UpdateFan	Central algorithm to handle updating the fan state for a supply fan 8500
RequestedSpeed	The speed to update the ventilation device's fan to
WhFanState	The recommended whole house ventilation device's state based on the whole house ventilation algorithm (if applicable)
NB_BthFSWs	Number of connected bath fan switches
Nb_RngHSWs	Number of connected range hood switches
Nb_AQMntrs	Number of connected air quality monitors (Room sensors)
SFanState	The state of the supply fan as recommended by each device
FanCloudState	The status, on the cloud, of the fan's ventilation device

14) Exemplary Historical Measurements

FIGS. **73**-**80** show screens, contained within the GUI **1000**, that display historical environmental measurements that were recorded by the IAQ system **10** over a predefined amount of time. Specifically, FIGS. **73**-**76** show swiping left **10000** or right on: (i) the display **220** of the central unit **104** or (ii) on a display contained within an Internet enabled device that is signed in to the system as an authorized user can show graphs **10010** that contain different information about the environmental measurements that were recorded by the IAQ system **10**. Specifically, FIG. **73** shows an overall score **10012**, FIG. **74** shows CO_2 levels **10014**, FIG. **75** shows TVOC levels **10016**, and FIG. **76** shows Temperature+Humidity levels **10018**. FIG. **77** then shows how the authorized user can display the environmental measurements **10020** that were recorded by the IAQ system **10** for a single room (e.g., master bedroom) within the structure **100**. Next, FIGS. **78**-**80** shows how a user can zoom in to a specific time period or can scroll through different time periods by using various guessers **10022**, **10024**, **10026**, **10028**, **10030**, **10032**. It should be understood that these historical environmental measurements that were recorded by the IAQ system **10** may be displayed to the user in different formats or may be emailed to the user after a predefined amount of time. For example, two different embodiments **10050**, **10100** of how the environmental measurements that were recorded by the IAQ system **10** can be displayed to the user are shown in FIGS. **81** and **82**.

It should be understood that this GUI **1000** provides a significant improvement in the efficiency of using the system **10** by bringing together and effectively visually presenting a limited list of high priority information without requiring the user to navigate through multiple screens in order to obtain this information. This in turn improves the efficiency of using the system **10** because it saves the user form navigating to a selected screen, manipulating the data associated with that screen, and then trying to interpret the resulting data. These factors tangibly improve the functionality of the system **10**, particularly the user interface, and more particularly effectively displaying the user interface on a central device **104** that has a small screen (e.g., mobile phone).

15) Industrial Design

While the foregoing has described what are considered to be the best mode and/or other examples, it is understood that various modifications may be made therein and that the subject matter disclosed herein may be implemented in various forms and examples, and that the teachings may be applied in numerous applications, only some of which have been described herein. It is intended by the following claims to claim any and all applications, modifications and variations that fall within the true scope of the present teachings. Other implementations are also contemplated.

While some implementations have been illustrated and described, numerous modifications come to mind without significantly departing from the spirit of the disclosure; and the scope of protection is only limited by the scope of the accompanying claims. Headings and subheadings, if any, are used for convenience only and are not limiting. The word exemplary is used to mean serving as an example or illustration. To the extent that the term includes, have, or the like is used, such term is intended to be inclusive in a manner similar to the term comprise as comprise is interpreted when employed as a transitional word in a claim. Relational terms such as first and second and the like may be used to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions.

Phrases such as an aspect, the aspect, another aspect, some aspects, one or more aspects, an implementation, the implementation, another implementation, some implementations, one or more implementations, an embodiment, the embodiment, another embodiment, some embodiments, one or more embodiments, a configuration, the configuration, another configuration, some configurations, one or more configurations, the subject technology, the disclosure, the present disclosure, other variations thereof and alike are for convenience and do not imply that a disclosure relating to such phrase(s) is essential to the subject technology or that such disclosure applies to all configurations of the subject technology. A disclosure relating to such phrase(s) may apply to all configurations, or one or more configurations. A disclosure relating to such phrase(s) may provide one or more examples. A phrase such as an aspect or some aspects may refer to one or more aspects and vice versa, and this applies similarly to other foregoing phrases.

Numerous modifications to the present disclosure will be apparent to those skilled in the art in view of the foregoing description. Preferred embodiments of this disclosure are described herein, including the best mode known to the inventors for carrying out the disclosure. It should be understood that the illustrated embodiments are exemplary only, and should not be taken as limiting the scope of the disclosure.

The invention claimed is:

1. An indoor air quality ("IAQ") system for operating an exhaust fan in a structure having a first room and a second room adjacent to the first room, the IAQ system comprising:

a monitoring device for being located in the first room, the monitoring device includes: (i) an identity, (ii) a sensor and (iii) a connectivity module, wherein the sensor is configured to record environment data;

a server/database and a connectivity module configured to receive: (i) the monitoring device's identity and (ii) environmental data that has been recorded by the monitoring device;

a first exhaust fan for being located in the structure and a second exhaust fan for being located in the structure, each of the first and second exhaust fans selectively controlled by the server/database, wherein one of the first and second exhaust fans is located in one of the first and second rooms;

wherein the server/database is configured to (i) assign the monitoring device to whichever of the first and second exhaust fans is located in the first room, and (ii) if neither the first or second exhaust fan is in the first room, then assign the monitoring device to whichever of the first and second exhaust fans is located in the second room; and

wherein the server/database is configured to: (i) analyze the received environmental data from the monitoring device and (ii) selectively control, based on the received environmental data, whichever of the first and second exhaust fans is assigned to the monitoring device.

2. The IAQ system of claim **1**, wherein the monitoring device is not electrically or directly connected to one of the first and second exhaust fans.

3. The IAQ system of claim **1**, wherein the server/database turn ON the assigned one of the first and second exhaust fans when a level of an air component contained within the environmental data is over a predefined threshold value.

4. The IAQ system of claim **3**, wherein the predefined threshold value is set by a regulatory body, government agency, private group or standard setting body.

5. The IAQ system of claim **4**, wherein the predefined threshold value is set: (i) using the sensor to record environmental data over a predefined amount of time and (ii) adjusting the predefined threshold value in light of the recorded environmental data.

6. The IAQ system of claim **1**, wherein the sensor measures levels of at least one of the following: CO, CO2, NO, NO2, NOX, PM2.5, ultrafine particles, radon, volatile organic compounds, ozone, dust particulates, lead particles, acrolein, biological pollutants, pesticides, or formaldehyde.

7. The IAQ system of claim **1**, further comprising a plurality of monitoring devices, wherein each monitoring device within the plurality of monitoring devices is assigned to at least one of the first and second exhaust fans.

8. The IAQ system of claim **1**, wherein controlling the assigned one of the first and second exhaust fans includes: (i) turning the assigned one of the first and second exhaust fans to a first setting if a level contained within the environmental data is over a first predetermined threshold, (ii) turning the assigned one of the first and second exhaust fans to a second setting if the level contained within the environmental data is over a second predetermine threshold, and

(ii) turning the assigned one of the first and second exhaust fans to a third setting if the level contained within the environmental data is over a third predetermine threshold.

9. The IAQ system of claim **1**, further includes an internet enabled device that is configured to display environmental data that has been collected over a predefined amount of time.

10. The IAQ system of claim **1**, wherein the assigned one of the first and second exhaust fans is an exhaust ventilation device, a supply fan or an air exchanger.

11. The IAQ system of claim **1**, wherein the assigned one of the first and second exhaust fans is one of the following: a range hood, a bathroom fan, or a supply fan.

12. The IAQ system of claim **1**, wherein the server/database uses an indoor positioning system to determine whether one of the first and second exhaust fans is positioned in the first room.

13. The IAQ system of claim **1**, wherein the server/database uses user input to determine whether one of the first and second exhaust fans is positioned in the first room.

14. A method for operating an exhaust fan within a structure, the method comprising:

providing a monitoring device for being located in a first room of the structure, the monitoring device includes: (i) an identity and (ii) a sensor that records environment data;

receiving, at a server/database: (i) the monitoring device's identity and (ii) environmental data that has been recorded by the monitoring device;

providing a first exhaust fan for being located in the structure and a second exhaust fan for being located in the structure, wherein at least one of the first and second exhaust fans is located in either the first room or a room adjacent to the first room, each of the first and second exhaust fans having a plurality of operation modes;

determining whether or not one of the first and second exhaust fans is in the first room, and

if one of the first and second exhaust fans is in the first room, then assigning to the monitoring device that one of the first and second exhaust fans,

but if neither of the first and second exhaust fans is in the first room, then assigning to the monitoring device whichever of the first and second exhaust fans is the room adjacent to the first room; and

using the server/database to control the operation mode of the one of the first and second exhaust fans assigned to the monitoring device, wherein the operation mode is selected based on a comparison of the environment data with predetermined threshold values.

15. The method of claim **14**, wherein the selected operation mode is not directly determined by the monitoring device.

16. The method of claim **14**, wherein the operation mode is set to ON, when a level contained within the environmental data is over a predefined threshold value.

17. The method of claim **16**, wherein the predefined threshold value is set by a regulatory body, government agency, private group or standard setting body.

18. The method of claim **14**, further comprising the following steps:

receiving a first set of environmental data that includes one level that is above a predefined threshold value;

selecting one of the first and second exhaust fans that can bring the level within the environmental data below the predefined threshold value;

sending a signal from the server/database to turn ON the selected one of the first and second exhaust fans;

receiving a second set of environmental data that includes one level that is below a predefined threshold value;

sending a signal from the server/database to turn OFF the selected one of the first and second exhaust fans.

19. The method of claim **18**, wherein the step of selecting the one of the first and second exhaust fans that can bring the level of the level within the environmental data below the predefined threshold value includes selecting the one of the first and second exhaust fans that is assigned to the monitoring device.

20. The method of claim **14**, further comprising the step of displaying the recorded environmental data on an internet enabled device.

21. The method of claim **14**, wherein an indoor positioning system is used to determine whether one of the first and second exhaust fans is positioned in the first room.

22. The method of claim **14**, wherein user input is used to determine whether one of the first and second exhaust fans is positioned in the first room.

23. A method for operating one of a plurality of exhaust ventilation devices within a structure having a first room and a second room adjacent to the first room arranged so that one of the plurality of exhaust ventilation devices is located in one of the first and second rooms, the method comprising:

monitoring levels of components of air within the structure using a monitoring device having at least one sensor in the first room of the structure;

determining that a level of an air component is above a predefined threshold range for that air component;

analyzing data from other sensors to determine if the other sensors measured the level for that air component is above a predefined threshold range;

determining whether one of a plurality of exhaust ventilation devices located in the structure is in the first room and

if one of the plurality of exhaust ventilation devices is in the first room, then assigning to the monitoring device that one of the plurality of exhaust ventilation devices,

but if none of the plurality of exhaust ventilation devices is in the first room, then assigning to the monitoring device any of the plurality of exhaust ventilation devices in the second room;

generating a plan designed to return the level of the air component within the predefined threshold range, wherein the plan comprises instructing the one of the plurality of exhaust ventilation devices assigned to the at least one monitoring device to turn ON;

informing a user of the generate plan; and

performing the generated plan.

24. The method of claim **23**, wherein the step of performing the generated plan includes: (i) instructing the one of exhaust ventilation devices assigned to the at least one monitoring device to turn ON and (ii) instructing the one of exhaust ventilation devices assigned to the at least one monitoring device to turn OFF, when the level is within the predefined threshold range.

25. The method of claim **23**, wherein an indoor positioning system is used to determine whether one one of the plurality of exhaust ventilation devices is positioned in the first room.

26. The method of claim **23**, wherein user input is used to determine whether one one of the plurality of exhaust ventilation devices is positioned in the first room.

27. An indoor air quality ("IAQ") system for operating an exhaust ventilation device in a structure having a first room and a second room adjacent to the first room, the IAQ system comprising:

a monitoring device for being located in the first room, the monitoring device includes: (i) an identity, (ii) a sensor and (iii) a connectivity module, wherein the sensor is configured to record environment data;

a server/database and a connectivity module configured to receive: (i) the monitoring device's identity and (ii) environmental data that has been recorded by the monitoring device;

a plurality of exhaust ventilation devices that are selectively controlled by the server/database, wherein at least one of the plurality of exhaust ventilation devices is located in one of the first and second rooms; and

wherein the server/database is configured to: (i) analyze the received environmental data (ii) identify which one of the plurality of exhaust ventilation devices is located in the first room and assign that one of the plurality of exhaust ventilation devices to the monitoring device, and if none of the plurality of exhaust ventilation devices located in the first room, then assign the monitoring device to whichever of the plurality of exhaust ventilation devices located in the second room and (iii) selectively control the one of the plurality of exhaust ventilation devices assigned to the monitoring device based on the received environmental data.

28. The IAQ system of claim **27**, wherein the exhaust ventilation device assigned to the monitoring device is a range hood.

29. The IAQ system of claim **27**, wherein an indoor positioning system is used to determine whether one of the plurality of exhaust ventilation devices is positioned in the first room.

30. The IAQ system of claim **27**, wherein user input is used to determine whether one of the plurality of exhaust ventilation devices is positioned in the first room.

* * * * *